QUANTITATIVE EQUITY PORTFOLIO MANAGEMENT

SECOND EDITION

AN ACTIVE APPROACH TO PORTFOLIO CONSTRUCTION AND MANAGEMENT

LUDWIG B. CHINCARINI & DAEHWAN KIM

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FOREWORD

This is an ambitious book that both develops the broad range of artillery employed in quantitative equity investment management and also provides the reader with a host of relevant practical examples. While the authors firmly take the view that active management can be rewarded, their book offers a solid practitioner's guide to quantitative passive management in the efficient-market/index-fund world on the one hand and to active managers on the other. Far too often quantitative management is equated with passive management, and active quantitative management is thought to be an oxymoron. The authors' rigorous dispelling of this tired canard opens the way for the traditional active manager to make use of quantitative tools. Readers of this book soon learn that quantitative management is a technology that they can use to hone their evaluation of their fundamental ideals, implement them successfully, and finally, control the risk of the portfolios they manage while doing so.

Often investment ideas are anomalies that fit uncomfortably, if at all, with neoclassical efficient-market theories. Typically, such anomalies are observations formed from studying historical stock market returns. This book surveys the rich collection of anomalies that form the basis for much of what is current practice in the active quantitative arena. The student of portfolio management who is interested in learning what academics and empiricists have discovered will find this of particular interest. What about the small-firm effect or momentum? This book is an excellent place to introduce the manager to these important anomalies.

But where this book excels is in melding theory with practice. Issues of liquidity, leverage, market neutrality, transactions costs, and the pitfalls and virtues of backtesting that are so often skirted in other treatments take center stage here. So, too, there is an extensive analysis of optimal after-tax portfolio management, a topic often not even mentioned in other books.

Throughout this book, the authors expend much useful effort on ridding their analysis of naive reliance on mathematics at the expense of practicality.

While the mathematics can at times be demanding, it is about at the level of the CFA requirements and should be well within the command of the analytic abilities of most portfolio managers.

Helpfully, the prose is lively and far removed from the usual pedantry that surrounds mathematics in finance. Extensive questions test the reader's understanding and make this book perfectly suited to the needs of an advanced course in investment management at the MBA or PhD level.

Stephen A. Ross
The late Franco Modigliani Professor of
Finance and Economics
Massachusetts Institute of Technology

PREFACE TO THE FIRST EDITION

The world of active portfolio management has been changing over the last few years to become more quantitative in nature. This trend is inspiring because it lends itself to a more controlled approach to asset management, which ultimately benefits individual and institutional investors. In some ways quantitative asset management is an old field, but in many ways it's a very new field that has bundled together lots of old concepts. It is a vast and diverse field because quantitative managers use a variety of different techniques to manage their portfolios. Despite this diversity, though, there are central themes that remain at the core of the work of most quantitative asset management firms.

When we were first introduced to the field of quantitative portfolio management, we sensed that a lot of the issues it covered were unclear not only to us but also to our colleagues. In fact, there was no formal, authoritative source on the topic. Naturally, through our years at Seoul National University, UC Berkeley, Harvard, and MIT, we learned many of the concepts related to quantitative portfolio management, such as statistics and basic financial theory. And through our years working for portfolio management companies, we learned some of the real-world aspects of the trade. Still, we never really had a comprehensive reference to turn to for an understanding of the nuts and bolts of quantitative equity portfolio management. We also would sometimes see practitioners approach this kind of management with holes in their analytics or listen to academics speak about the theory with no attention to the details of real-world portfolio management. We began teaching the concepts to students at our respective universities and realized that it would be useful to write a book on the subject that attempted to cover the whole spectrum of quantitative equity portfolio management, including the theoretical side and the practical side.

Since the field of quantitative equity portfolio management (QEPM) is vast, we chose to focus on its core concepts. We chose to write a book that goes step by step through the entire process of building a quantitative equity portfolio. At times we explain concepts in much detail for those who are new to the field, yet we use mathematical rigor and develop new concepts for those already quite proficient in the field. Our main goal was to write a book that would be useful as a reference to professional money managers but also very useful for teaching an advanced investments course to undergraduate, MBA, or PhD students. Although our topic is QEPM, parts of this book can be used to teach the fundamental concepts of most advanced financial economics programs. We also wrote some parts of the book in such a way that other departments within a portfolio management operation can understand the main drivers of QEPM. Thus, marketing, business development, and salespeople should feel comfortable using this book.

We would like to thank Emilie Striker, a graduate of Georgetown University, for doing an excellent job of editing our writing. She pursued her work on this book with an enthusiasm and effort that is rare among people today. Not only did she improve the prose, but she also did a thorough job of catching errors and made a number of substantial improvements. We thank Joe Abboud, Stela Hristova, and Mariya Mitova for excellent research assistance. Furthermore, we would like to thank Steve Ross, Dan diBartolomeo, Mark Holowesko, and William Seale for extensive comments and suggestions, as well as Eric Rosenfeld, David Blitzer, Lawrence Pohlman, Prem Jain, Laurens Leerink, Ron Kahn, Wayne Wagner, Wolfgang Chincarini, Neer Asherie, Kevin Garrow, Mark Schroeder, and Mark Esposito for their comments. We thank David Bieri for supplying us with macroeconomic data and KLD for supplying us with social-responsibility data.

We would also like to thank everyone at McGraw-Hill who made this book possible, including James K. Madru, Cheryl Hudson, and especially Stephen Isaacs and Daina Penikas.

We are delighted with the positive response to our book. Since publication, we have received many comments and suggestions from readers, quantitative practitioners, and professors. We wish to thank all of you; including Ravi Agarwal, Mark Bradshaw, Yuan Chumming, Levon Goukasian, Allen Huang, Gergana Jostova, Doug Martin, Natalie

Michelson, Masaka Mori, Marco Navone, Merav Ozair, Angel Samaniego, Lucio Sarno, Robert Schumaker, Felipe Vargas, Michael Verhofen, Bovorn Vichiansin, Russ Wermers, Chun Xia, Arthur Young, Giovanna Zanotti, Guofu Zhou, and the students of quantitative investment management at the University of San Francisco. If you are using this book in the classroom, we would appreciate hearing from you. Please contact us so that we can include your name in our acknowledgments. Additional teaching materials for the book are available at www.ludwigbc.com.

Ludwig Chincarini would like to thank the undergraduate students of Georgetown who patiently sat through his derivatives, investment, and global financial markets courses (in particular an anonymous student who wrote, "Write the book, man!") and the MBA students of 2005, 2006 and 2007 who took his course on investment analysis and special topics in investments. Thanks as well to Mellissa Cobb, Emma Curtis, John Carpenter, Juanita Arrington, David Walker, Keith Ord, Rohan Williamson, George Daly, Lee Pinkowitz, Gary Blemaster, and Reena Aggarwal and others of the faculty and staff of Georgetown University for making his life at Georgetown more enjoyable. Ludwig also would like to thank James Angel, William Droms, Joseph Mazzola, and Georgetown University for their initial encouragement for him to come back to teaching. Without that encouragement, this book may never have been written. Finally, he thanks the late Fischer Black and the late Rudiger Dornbusch for the inspirational influence they had on him.

Ludwig would like to dedicate this book to his family: to his brother, for being his best friend and always encouraging him; to his mother, for her love and encouragement, for wanting the best for him, and for opening a window for him when the door would close, in particular for giving him the opportunity to attend a top-rated university; to his father, for showing him, by example, the importance of thinking and of questioning; to his sister, the promise of tomorrow; and to everyone in his family, for allowing him the freedom to think or say anything, however outrageous, yet making him realize that it was he who would bear the burden of any rational or irrational thoughts.

Daehwan Kim would like to thank the American University in Bulgaria and Korea University, which provided a stable cash flow while the book was being written. Students in the two universities were the source of constant inspiration, whether they were taking a finance class or something else. Taeyoon Sung, a colleague at KAIST Graduate School of Management, provided various supports to the book writing. Daehwan also would like to thank his lovely wife, Ksenia Chizhova. Not only did she dutifully perform the traditional role of a book writer's wife (i.e., spending weekend after weekend without her husband at home), but she also was an excellent student of languages and literature who undertook the difficult task of trying to improve his writing style.

In the movie *Wall Street*, Gordon Gekko gives a famous speech about greed, a speech that is based on one given by Ivan Boesky years earlier at a University of California commencement. In his speech, Gekko states that "Greed—for lack of a better word—is good. Greed is right. Greed works. Greed clarifies, cuts through, and captures the essence of the evolutionary spirit … and has marked the upward surge of mankind." Unfortunately, both Gekko and Boesky got it wrong. There is a better word than greed, and it is *truth*. It is the search for truth that clarifies, works, and marks the upward surge of humankind. Speaking the truth eliminates inefficiency in all its forms; seeking the truth leads to great discoveries; passing the truth between loved ones provides strength over oceans, over time, and over death. Honesty would have prevented the corporate disasters that occurred in the last several years, and the truth is finally bringing the guilty, who were motivated by greed, to justice. Truth is indeed the highest summit we can achieve, whether we are managing a portfolio, managing a corporation, involved in a relationship, or taking part in any other aspect of everyday life.

If you would like to send us suggestions or comments on this book, please send them to our e-mail addresses with the subject line "Book comments."

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PREFACE TO THE SECOND EDITION

When we wrote the first edition of *Quantitative Equity Portfolio Management* in 2006, quantitative equity portfolio management was still a new field mainly reserved for elite hedge funds. Since the publication of our book, the world of quantitative management has exploded. The most revolutionary changes have been smart beta and the launch by mutual funds, ETFs, and many companies of their own quantitative factor indices. In fact, the amount of assets under management following quantitative strategies has increased from approximately $238 billion when we published the first edition of *QEPM* to $2.3 trillion in 2020.[1] There are now all kinds of companies providing smart beta or factor indices, including Charles Schwab, Vanguard, BlackRock's iShares, Invesco, Goldman Sachs, and many others. In fact, the enormous growth in quantitative strategies has raised concerns among investors that there may be too much crowding in factor strategies, crowding that might be linked to crises like the quant crisis of 2007 [Chincarini (2012)]. Despite the concern about crowding and the efforts to understand it, we are extremely excited about the growth of this form of portfolio management.

Our book was also among the earliest to discuss the use of socially responsible factors in quantitative portfolio management, factors that have since become more commonly called ESG (environmental, social, and governance) factors. Since the first edition of our book was published, the amount of funds managed by ESG factors has increased from $2.5 billion to $249 billion worldwide.[2] Although we do not advocate for or against the use of ESG factors for portfolio construction, we are still amazed by the growth in this quantifiable rule-based method of socially responsible investing.

For the first edition of *QEPM*, we were fortunate enough to have Professor Stephen Ross of MIT write the Foreword to our book. He was one of the big thinkers in finance, being widely credited with the invention of the APT, the binomial option pricing formula, and agency theory.[3] Stephen Ross was not only successful in his career but also generous in writing the Foreword to our book and helping two young, talented researchers get a push. In the words of Anton Ego in the movie *Ratatouille*, “But there are times when a critic truly risks something, and that is in the discovery and defense of the new. The world is often unkind to new talent, new creations. The new needs friends.” In 2018, Stephen Ross suddenly died. Although he is no longer here to dazzle us, he remains in our thoughts and in our prayers. Thank you, Steve.

In this new edition of *QEPM*, we have added some additional anomalies, a new list of behavioral explanations for some of the anomalies, new discoveries in transactions costs optimization, and new techniques for optimizing market-neutral portfolios. In addition, we present the backtests for many factors so that readers can get a peek at all of them. We also update all the data in the book through the end of 2020. Finally, we provide, on the book’s website, teaching materials for instructors and practice materials for practitioners, including solutions to all the end-of-the-chapter questions, a folder with additional appendices for each chapter, a folder with general appendices (Appendices A—E), a folder with programs for every chapter written in MATLAB, R, and Stata for learning or for classroom labs, a folder with PowerPoint slides for every chapter, a folder with sample stock data, and a downloadable file of historical factor returns for 164 factors.[4] We hope you enjoy the website. If you are an instructor using our book, let us know, and we will thank you in subsequent printings of the book.

Along the road to writing this new edition, we benefited from the generous help of many people. We would especially like to thank Steve Glass, Michael Chigirinsky, Richard Sloan, Scott Esser, Rob Arnott, Fabio Moneta, Dan diBartolomeo, Snehil Misha, Brandon Lee Gipper, Jeffrey Pontiff, Lubos Pastor, Pete Kyle, Jim Angel, Andrei Bolshakov, Jeff Silk, Ryan Losson, Susan Leake, Fernando Comiran, Michael Scanlon, Chris Murray, Natasha Grasso, Jamie Comer, Phil Ellis, and William Ding. Daehwan would like to thank Martin Brückner, Tobias Klein, Walter Levy and other colleagues at First Private, Lewis Chan at Maunakai Capital, and

Taegyeong Han, Yunho Heo, and Hyunjai Shin at Samsung Hedge for sharing their perspectives on quantitative investing with him and keeping him interested in this field in the years after publication of the first edition. At McGraw Hill, we thank Judith Newlin, Stephen Isaacs, Jim “Hawk Eyes” Madru, and the man who makes the magic happen Peter Mccurdy.

It’s never easy to write a book or even modify a book. It takes hours of endless patience and solitude. During my rewriting (Ludwig), my wife was pregnant with our second child, and we had our beautiful two-year-old daughter causing us all kinds of chaos, while I was diagnosed with two types of cancer. I strongly encourage all people to inform themselves about the important checkups all men and women should be scheduling with their doctors and to insist that your doctors perform those tests yearly. For example, all of us should check our skin every year, women should check for breast cancer, and men should do a PSA test every year. I also remind people to stop and enjoy the beautiful things in life and worry less about the rat race. With help from my mother and my in-laws, and the touch of God, somehow this new edition was created. I pray every day that I can continue to enjoy this beautiful life as my body is now on the mend. I would like to dedicate this new edition of the book to the most important people in my life: my wife, Gina, my daughter, Gemma, and my son, Angelo Ludovico. They make everything worthwhile. I am also eager to watch Angelo and Gemma achieve much more than I have—I only hope that the first steps that I build for them launch them onto a path of happiness, success, and fulfillment.

In the Preface to the first edition of *QEPM*, we wrote that truth is one of the purest objectives we can pursue as humans. In recent times, while free speech and freedom to disagree with the crowd are seemingly under attack, we often wake up feeling as though it is “Opposite Day.” The danger is that these social fads seemingly designed to conceal the truth may ultimately lead to inefficiency and unhappiness for everyone. Keep fighting for liberty, my friends!

If you would like to send us suggestions or comments on the book, please send them to our email addresses with the subject line “Book Comments.”

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June 1, 2022
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[1] *Source*: Morgan Stanley and author's interpolation.

[2] *Source*: Morningstar's Passive Sustainable Funds: Global Landscape 2020.

[3] See Chincarini and Fabozzi (2018) for a detailed description of his contributions.

[4] Available at www.ludwigbc.com. Once you reach the website, choose Books, then Quantitative Equity Portfolio Management, then QEPM Exclusive Content. Or just go to this link https://ludwigbc.com/books/qepm/exclusive_qepm_content_2020/ The password is qepm2020rig179.

NOTATIONS AND ABBREVIATIONS

MATHEMATICAL SYMBOLS

$d_{i,t}$	the dividend paid by stock i during period t
$f_{k,t}$	the premium of factor k for period t
$\mathbf{f}_t$	the premium vector of many factors for period t
F_t	the price of futures contracts
$\mathbf{I}$	the identity matrix
$I_{t,t+k}$	the gross interest of a cash position whose portions may be paid differing interest
l	the net leverage rate
m_f	the required margin for futures contracts
N_f	the number of futures contracts
N_o	the number of call options
$p_{i,t}$	the price of stock i at the end of period t
q	the futures contract multiplier
q_s	the single-stock-futures contract multiplier
$r_{i,t}$	the return of stock i for period t
$r_{B,t}$	the benchmark return for period t
$r_{P,t}$	the portfolio return for period t
R^2	the goodness-of-fit from a regression

$\bar{R}^2$	the adjusted goodness-of-fit from a regression
s_i	the number of shares of stock i holding
S_t	the value of the equity index
V_t	the dollar value of a portfolio at time t
w_i^B	the benchmark weight of stock i
$\mathbf{w}^B$	the benchmark weight vector
w_i^P	the portfolio weight of stock i
$\mathbf{w}^P$	the portfolio weight vector
x_t	the portfolio return in excess of the benchmark at time t $(= r_{P,t} - r_{B,t})$
$x_{i,t}$	the contribution of stock return i to the portfolio excess return at time t $[= r_{i,t}(w_{i,t}^P - w_{i,t}^B)]$
$z_{i,k,t}$	the Z-score of stock i to factor k for period t
$\mathbf{z}_{i,t}$	the Z-score vector of stock i to many factors for period t
$\bar{z}_{i,t}$	the aggregate Z-score of stock i at time t
α_i^B	the benchmark alpha of stock i
α_i^{CAPM}	the CAPM alpha of stock i
α_i^{MF}	the multifactor alpha of stock i
$\beta_{i,j,t}$	the exposure of stock i to factor j for period t
$\boldsymbol{\beta}_{i,t}$	the exposure vector of stock i to many factors for period t
Δ	the delta of an option
ϵ_i	the residual return of stock i
ι	the vector of 1
μ_B	the expected return of the benchmark
μ_i	the expected return of stock i
μ_P	the expected return of the portfolio

$\boldsymbol{\mu}$	the expected return vector of stocks
ξ	the percentage of equity capital invested in cash
$\rho_{i,j}$	the correlation of the returns of stock *i* and stock *j*
σ_B	the standard deviation of the benchmark return
σ_i	the standard deviation of the return on stock *i*
σ_P	the standard deviation of the portfolio return
σ_x	the standard deviation of the portfolio excess return over the benchmark
$\boldsymbol{\Sigma}$	the variance-covariance matrix of stock returns
τ	the tax rate
ω_i	the residual risk of stock *i*
ω_i^2	the residual variance of stock *i*

MATHEMATICAL FUNCTIONS

$C(\cdot)$	the covariance function
$E(\cdot)$	the expectation function (returns the mean of a random variable)
$I(\cdot)$	the indicator function (1 if true and 0 otherwise)
$M(\cdot)$	the median function
$P(\cdot)$	the probability function
$p(\cdot)$	the probability density function
$S(\cdot)$	the standard deviation function
$V(\cdot)$	the variance function
$\varphi(\cdot)$	the standard normal probability density function
$\Phi(\cdot)$	the standard normal cumulative distribution function
$\rho(\cdot)$	the correlation function

ABBREVIATIONS

ADTV	the average daily trading volume
ADDTV	the average daily dollar trading volume
AE	the allocation effect
APT	arbitrage pricing theory
BR	breadth
CAPM	the capital asset pricing model
DVOL	the dollar trading volume
GLS	the generalized least squares method
IC	the information coefficient
IL	information loss
IR	the information ratio
MAD	the minium absolute deviations method
MVO	mean variance optimization
OLS	the ordinary least squares method
QEPM	quantitative equity portfolio management
SD	standard deviation
SR	the Sharpe ratio
SSE	the security selection effect
SVOL	the share trading volume
TC	transactions costs
TE	tracking error
TWR	time-weighted return
VAR	vector autoregression
VaR	value-at-risk

ABBREVIATION OF FACTOR NAMES

ADR	the advance–decline ratio
AILIQ	the Amihud illiquidity
ARMD	the median change in analyst ratings
ARMR	the median analyst rating
ARMS	the Arms index
ARPB	the percent of analyst buy recommendations
ARPD	the percent of analyst recommendation downgrades
ARPS	the percent of analyst sell recommendations
ARPU	the percent of analyst recommendation upgrades
ARSD	the standard deviation in analyst ratings
BB	the Bollinger band
BBBS	the BBB spread
BBBSD	the change in BBB spread
BCG	the growth rate of the business confidence index
BETA	the beta premium
B/P	the book-to-price ratio
CACC	the current accruals
CB	the channel breakout
CC	the community contribution index
CCC	the cash conversion cycle
CCCD	the change in the cash conversion cycle
CCCX	the cash conversion cycle in excess of the industry average
CCG	the growth rate of the consumer confidence index
CCX	the community contribution index in excess of the industry average
CE	the natural log of common equity
CEG	the growth rate in the natural log of common equity
CFOR	the cash-flow-from-operations ratio

CFORD	the change in the cash-flow-from-operations ratio
CF/P	the cash-flow-to-price
CF/TA	the cash-flow-to-total assets
CF/TS	the cash-flow-to-total sales
CG	the corporate governance index
CGX	the corporate governance index in excess of the industry average
CIG	the growth rate of the commodity price index
CMI	the cost management index
CMID	the change in the cost management index
CMIX	the cost management index in excess of the industry average
CPG	the gross profit growth
CPIG	the growth rate of the consumer price index (i.e., inflation)
CR	the cash ratio
CRD	the change in cash ratio
CROWDA	the crowding measure based on analysts
CROWDV	the crowding measure based on valuation
CSG	the growth rate of the consumer sentiment index
CUR	the current ratio
CURD	the change in current ratio
DE (D/E)	the debt-to-equity ratio
DED	the change in debt-to-equity ratio
DEX	the debt-to-equity ratio in excess of the industry average
DIV	the diversity index
DIVX	the diversity index in excess of the industry average
DY (D/P)	the dividend yield
E	the environment index

EBIT	earnings before interest and taxes
EBITDA	earnings before interest, taxes, depreciation, and amortization
EBITDA/EV (EBITIDAEV)	earnings before interest, taxes, depreciation, and amortization divided by enterprise value
EBITDA/P (EBITDAP)	earnings before interest, taxes, depreciation, and amortization divided by stock price
E/P	earnings-to-price ratio
EPS	earnings-per-share
EPSFPD	the percent of downward revisions in earnings-per-share forecasts
EPSFPU	the percent of upward revisions in earnings-per-share forecasts
EPSFSD	the standard deviation of earnings-per-share forecasts
EPSSD	the earnings-per-share forecasts divided by the standard deviation of analyst forecasts
ER	the employee relation index
ERX	the employee relation index in excess of the industry average
ET	equity turnover
ETD	the change in equity turnover
ETX	equity turnover in excess of the industry average
EV/EBITDA (EVEBITDA)	enterprise value divided by earnings before interest, taxes, depreciation, and amortization
EX	the environment index in excess of the industry average
FAT	fixed-asset turnover
FATD	change in fixed-asset turnover
FATX	fixed-asset turnover in excess of the industry average
FLR	financial leverage ratio

FLRD	change in financial leverage ratio
FLRX	financial leverage ratio in excess of the industry average
GDPR	the revision in forecast of gross domestic product
GDPS	the surprise in forecast of gross domestic product
GDPG	the growth rate of real gross domestic product
GPG	gross profit growth
GPM	the gross profit margin
GPMD	the change in gross profit margin
GPTA	gross profits to total assets
HR	the human rights index
HRX	the human rights index in excess of the industry average
ICFCR	the inverse cash-flow coverage ratio
ICFCRD	the change in the inverse cash-flow coverage ratio
ICFCRX	the inverse cash-flow coverage ratio in excess of the industry average
ICR	the interest coverage ratio
IICR	the inverse interest coverage ratio
IICRD	the change in the inverse interest coverage ratio
IICRX	the inverse interest coverage ratio in excess of the industry avareage
INVIL	the invariance illiquidity measure
IPEGF	the earnings-to-price ratio multiplied by forecasted earnings growth
IPEGH	the earnings-to-price ratio multiplied by historical earnings growth
IPEGY	the earnings-to-price ratio multiplied by earnings growth plus dividend yield
IPG	the growth rate of industrial production
IPR	the revision in industrial production

IT	inventory turnover
ITD	the change in inventory turnover
ITX	the inventory turnover in excess of the industry average
LGPG	long-term gross profit growth
LNPG	long-term net profit growth
LOGCE	log of common equity
LOGCEG	equity growth
LLOGCEG	long-term equity growth
LOGSIZE	log of market capitalization
LOGTA	log of total assets
LOGTAG	total asset growth
LLOGTAG	long-term total asset growth
LOPG	long-term operating profit growth
M12M	12-month momentum
M1M	one-month momentum
M3M	three-month momentum
M6M	six-month momentum
M9M	ninth-month momentum
M9ML1	nine-month momentum lagged one month
MA	the moving-average indicator
MACD	the moving-average convergence divergence indicator
MAILIQ	the modified Amihud illiquidity measure
NPG	net profit growth
NPM	the net profit margin
NPMD	the change in net profit margin
NPTA	net profit to total assets
OBV	on-balance volume
OLBI	the odd-lot balance index

OPG	operating profit growth
OPM	the operating profit margin
OPMD	the change in operating profit margin
OPTA	operating profit to total assets
P	the price of the stock
P/B	the price-to-book ratio
P/CF	the price-to-cash-flow ratio
P/E	the price-to-earnings ratio
P/EBITDA	the price divided by the earnings before interest, taxes, depreciation, and amortization
P/S	the price-to-sales ratio
PCG	the growth rate of personal consumption
PCR	the revision in personal consumption growth
PEG	the price-to-earnings-to-growth ratio
PEGF	the forecasted price-to-earnings growth ratio
PEGH	the historical price-to-earnings-growth ratio
PEGY	the price-to-earnings-to-growth-to-dividend-yield ratio
PRO	the product index
PROX	the product index in excess of the industry average
PSL	the Pastor and Stambaugh liquidity measure
QR	the quick ratio
QRD	the change in quick ratio
RNDS	the research-and-development-to-sales ratio
RNDSD	the change in the research-and-development-to-sales ratio
ROA	the return on assets
ROCE	the return on common equity
ROE	the return on owner's equity
ROTC	the return on total capital

RSI	the relative strengh index
RT	receivables turnover
RTD	the change in receivables turnover
RTX	the receivables turnover in excess of the industry average
SB	stock buybacks
SIR	the short interest ratio
SIRX	the short interest ratio minus its one-year average
SIZE	the market capitalization of a company
S/P	the sales-to-price ratio
SR	the support and resistance indicator
SS	the swap spread
SSD	the change in the swap spread
SUE	standardized unanticipated earnings
TACC	the total accruals
TAT	total asset turnover
TATD	the change in total asset turnover
TATX	the total asset turnover in excess of the industry average
TDR	the total debt ratio
TDRD	the change in the total debt ratio
TDRX	the total debt ratio in excess of the industry average
TP2Y	the term premium (10-year rate over 2-year rate)
TP3M	the term premium (10-year rate over 3-month rate)
TT	trading turnover
UDR	the upside–downside ratio
UDV	the upside–downside volume
UR	the unemployment rate
URD	the change in the unemployment rate
VL	the volume lead indicator

VOL	the volatility of a stock

PART I

An Overview of QEPM

Modern financial theory has long described the stock market as a place that rewards investors who take calculated risks over the long run. Today's understanding of the market, however, argues for a different kind of approach to risk-taking from the kind that was popular just a couple of decades ago. The conventional wisdom at that time was that stock returns related only to stocks' correlations with the total market and that the best investment strategy was simply to follow the market. More recent insights show that other sources of risk fuel stock returns and that the market rewards investors who seek them.

More specifically, recent work in finance finds that stock returns, over time periods of a year or more, are fairly predictable with certain groups of factors. Prices no longer seem to zigzag randomly in Brownian motion. Rather, when viewed through the right prism of risk factors, they follow decipherable patterns. These additional insights in financial theory not only reveal the possibility of profiting from active investment strategies, but they also make the case for a specifically quantitative approach. If it takes multiple factors to predict stock returns most accurately, then quantitative models of stock returns are needed to identify and combine factors efficiently. If returns are somewhat predictable over the long run, then stable quantitative models should work more reliably than picking individual stocks intermittently on qualitative information. The current state of technology, which supports data-heavy quantitative research and complex trading strategies, makes it possible to put these ideas into practice.

Quantitative equity portfolio management (QEPM) is what we call the approach to portfolio management that takes full advantage of today's better understanding of the markets and greater technological capacity for sophisticated investing. QEPM is a broad and flexible umbrella that encompasses as many individual strategies as managers can develop with the set of quantitative methods that we explain in this book. The commonality of all the various QEPM applications is the discipline and accuracy that mathematics lend to the pursuit of returns and the control of risk. In the first three chapters we will introduce you to QEPM, why and how it works, and the essential framework on which it operates.

CHAPTER 1

The Power of QEPM

The first duty of intelligent men is the restatement of the obvious.

—George Orwell

1.1 INTRODUCTION

Personal investors place their savings in the hands of professional money managers in the belief that the professionals, with their specialized skills, will make the best investment decisions for them. In fact, more than 106 million Americans, the equivalent of about half of all U.S. households, entrust their money to mutual funds. They and other investors are the reason why there are more than 11,000 U.S. stock mutual funds and exchange-traded funds (ETFs) managing $24.9 trillion and more than 3,000 U.S. hedge funds.[1] Yet, while assets in these types of investment funds stand in the trillions of dollars, many people have begun to question whether the professionals really do have an edge on amateur investors. There is evidence that only 14% of equity mutual funds managed to beat the Standard and Poor's (S&P) 1500 from 2000 to 2020.[2] Despite poor performance by some funds, however, we firmly believe that professional managers *do* invest better than the average investor when they use certain tools available to them to quantify and truly understand the risks they are taking. Superior portfolio returns are possible through *quantitative equity portfolio management* (QEPM).

In this book, we use *QEPM* to refer mostly to an active, quantitative style of equity portfolio management, although the quantitative tools we describe can be applied easily to passive management strategies. Broadly speaking, equity portfolio management styles can be defined in two dimensions: *passive* versus *active* and *qualitative* versus *quantitative*. The passive-versus-active dimension reflects whether the portfolio is being managed simply to match the return of the benchmark or exceed the return of the benchmark. Passive management, also referred to as *indexing*, involves following and trying to match the returns of an equity index (e.g., the S&P 500) or other benchmark as closely as possible. The "passive" portfolio manager only initiates trades in order to mimic changes in the composition of the index, to reinvest dividends, to deal with the portfolio's cash inflows and outflows, or to respond to corporate actions that affect stocks that make up the index. Index portfolio managers typically are rewarded for their ability to replicate the index. In the U.S., passive management grew from 8.69% in 2000 to 40% in 2020 of all managed mutual funds and ETFs, primarily owing to the poor performance of many active fund managers versus standard equity indices. Passive management implicitly assumes that portfolio managers cannot beat the market.

Active management takes the view that it is possible to choose stocks that will outperform an equity index or other benchmark. "Active" managers also sometimes aim for some absolute level of performance without any reference to an index or benchmark. Trading takes place when the manager wants to buy stocks expected to have superior returns, when there are dividends to reinvest, or when cash flows into or out of the portfolio. In many cases, actively managed portfolios have higher turnover than passively managed ones because active portfolio managers tend to trade more frequently than passive managers do. Active managers usually are rewarded for the portfolio's absolute return or risk-adjusted return over a benchmark.

The second way to define portfolio management styles is to look at whether the manager bases decisions mainly on qualitative or quantitative analysis. Of the two general styles, perhaps the easiest for the average investor to understand is *qualitative management*, which is sometimes called *fundamental management* (although that term can be misleading because quantitative managers look at stock fundamentals as well). What makes the style qualitative is the fact that the research focuses on intangibles and generally does not involve using mathematics or computer programs specifically to identify “good” and “bad” stocks. Qualitative management is almost always a kind of active management because qualitative managers handpick stocks that they expect to outperform the market. Their selections are based on information from income statements and balance sheets, financial ratios, phone interviews with company personnel, research reports, and ad hoc methods of analysis. They also rely on their own gut reactions. For the most part, qualitative managers use their own judgment and informal calculations to filter the information that they and their analysts gather.

Peter Lynch, who led the Fidelity Magellan Fund to a compounded return of more than 2,700% during his tenure as fund manager from 1977 to 1990, is one of the best-known practitioners of the qualitative style. One of Lynch’s largest holdings at Magellan was inspired by his wife’s enthusiasm for L’eggs, Hanes Corporation’s brand of women’s hosiery packaged in egg-shaped containers and sold at local drugstores and supermarkets. Magellan’s position in Hanes prospered when L’eggs became a huge hit with consumers. Subsequently, a competitor of Hanes, the Kayser-Roth Corporation, tried to copy the success of L’eggs by selling its own brand of panty hose. Concerned about possible erosion of Hanes’ market share, Lynch undertook what he has termed “fundamental research” on the matter. He bought 48 pairs of Kayser-Roth’s No Nonsense panty hose and asked a group of female coworkers to try them for a few weeks. Based on their assessment that the No Nonsense product was not nearly as good as L’eggs, Lynch decided to hold onto Hanes’ stock. He was richly rewarded for his (somewhat unconventional) qualitative methods when Hanes’ stock continued to rise, and the company eventually was acquired by what is currently known as the Sara Lee Corporation.[3]

Quantitative management, unlike the rather intuitive process of qualitative management, is rooted in mathematics and statistics and less concerned with intangibles. Quantitative portfolio managers use any numerical data or quantifiable information relevant to the investment decision. This could include stock fundamentals from the income statement and balance sheet, technical data (e.g., stock prices and trading volumes), macroeconomic data, survey data, analyst recommendations, and any other data collected and stored in a database. Quantitative managers, unlike their counterparts in the qualitative tradition, use their data to build quantitative models of security returns. These models, along with advanced statistics, mathematics, and computer software, are used to identify "good" and "bad" stocks. Essentially, quantitative managers filter information mathematically rather than intuitively.

The particular field of quantitative management that we refer to as QEPM is, like most forms of qualitative management, an active approach to investing. With QEPM, the manager aims for returns that exceed a benchmark or market index. QEPM's tools for measuring and controlling risk do allow for highly accurate passive management. QEPM can accomplish much more than pure indexing, though, so our focus is on exploiting quantitative methods for outperformance.

Quantitative management is associated less with great individuals than with great institutions, and many successful mutual funds practice QEPM. There is a strong quantitative management presence at Acadian Asset Management, AQR Capital, Blackrock, Goldman Sachs Asset Management, Parametric Portfolio Associates, Putnam Investments, Quantitative Management Associates, State Street Global Advisors, and Two Sigma, among others. Many hedge funds' portfolios are based on QEPM as well. And even many enhanced index managers (more passive style) who manage portfolios with respect to a benchmark with the goal of modest excess performances also practice QEPM.

Over the years, quantitative management gradually has gained prevalence as even self-described qualitative managers have adopted some quantitative methods. A number of forces have propelled this shift toward the quantitative, the first being the advancement of technology since the 1990s. Complicated computer models of stock returns that once took days to run are now generated in a matter of minutes. Computing speed also allows computer programs to dig through great amounts of data in order to uncover buried treasures. The internet, meanwhile, makes it easier to access a wealth of data to analyze. With so much information at their fingertips, though, investors sometimes can become overconfident and make poor investment choices. The near glut of data only increases the need for quantitative analysis, which imposes discipline on the decision-making process.

In some ways, quantitative approaches also fare better in the post-Enron regulatory environment than qualitative styles do. Now that companies are required to give *fair disclosure* of events, portfolio managers and analysts can no longer get company news ahead of the rest of the market by chatting with the CFO, for instance. Fair disclosure means that all information must be uniformly distributed and available through public data resources—a boon for quantitative managers, who typically use software programs to access large quantities of data, but a blow to qualitative managers, who traditionally have gathered a great deal of their information through informal, one-on-one conversations with company executives.

Quantitative methods help managers to respond to calls for greater transparency as well. Average folks are becoming savvier investors and demanding more information from the people who manage their money. Employees want to know exactly how pension funds are investing their retirement savings. When questioned about their investment strategies, quantitative fund managers can point to clear, objective methods as the basis for their decisions.

Finally, the stability of quantitatively managed portfolios is becoming a selling point with investors. Quantitative strategies can control risk precisely. Precise risk control helps the portfolio avoid large swings in value and instead earn reliable, if modest, returns, which are what many investors seek. Although there have been a number of star managers who have consistently figured out how to beat the market, many qualitative managers have failed to beat the S&P 500 on average over time. Their portfolios have sporadically earned extremely high returns only to sink subsequently into years of underperformance. Portfolios that employ the quantitative risk controls that keep volatility low offer an attractive alternative to such roller-coaster rides.

1.2 THE ADVANTAGES OF QEPM

Quantitative equity portfolio managers gain numerous advantages over their traditional, qualitatively oriented counterparts by organizing and filtering great amounts of data with advanced statistics and mathematics. The disadvantages of QEPM mainly have to do with the possibility of relying the wrong way on quantitative models and historical data. Table 1.1 lists QEPM's advantages and disadvantages in comparison with qualitative management.

TABLE 1.1

The Advantages and Disadvantages of QEPM versus Qualitative Management

Advantages		
Criteria	**QEPM**	**Qualitative**
Objectivity	High	Low
Breadth	High	Low
Behavioral errors	Low	High
Replicability	High	Low
Costs	Low	High
Controlled risk	High	Low
Disadvantages		
Qualitative inputs	Low	High
Historical data reliance	High	Low
Data mining	High	Low
Reactivity	Low	High

One of the greatest advantages of QEPM is that when a model of stock returns is in place, construction of the portfolio is a highly objective process. The quantitative manager creates quantitative models of returns from underlying financial and other data, and these models tell the manager how to construct an optimal portfolio of stocks. The actual buy and sell decisions of the quantitative portfolio manager come directly from the model. This significantly lessens the impact of one person's biases on the portfolio. In contrast, a qualitative manager's buy and sell decisions often are based solely on the manager's opinion and thus are relatively more susceptible to the influence of behavioral biases. The objectivity of QEPM boosts the portfolio's returns and also supports management transparency. There is a clearly defined process for selecting stocks that can be presented to investors.

Another major advantage of QEPM is that computerized quantitative models can analyze large amounts of data and a large volume of stocks in a short amount of time. We call this the *advantage of breadth*. The same breadth is practically impossible with qualitative management because, to use Peter Lynch's metaphor, there are simply too many rocks to turn over one by one.[4] In the course of analyzing literally thousands of stocks with computer programs, the quantitative manager may unearth some diamonds in the rough that the qualitative manager would never find. Some qualitative portfolio managers do use stock screens and elements of quantitative management to help them sort through the stock universe. Ultimately, though, a total QEPM approach is a more complete analysis of the entire stock universe than this sort of mixed analysis.

As we mentioned earlier, QEPM inoculates the portfolio, to some extent, against behavioral biases and errors. The area of behavioral finance has grown in recent years as economists have identified the types of impulses that lead to irrational investment decisions. Portfolio managers may be better than the average investor at controlling these impulses, but they have them just the same. One such impulse is the *disposition effect*, the desire to hold onto "loser," or poorly performing, stocks too long. Investors often hope that losers will rebound despite all evidence to the contrary. Strict adherence to QEPM procedures helps to prevent a portfolio manager from trading on this sort of wishful thinking because it takes the final decision out of the manager's hands to some extent. If the quantitative model of stock returns recognizes a "bad" stock, that is the trigger for selling the stock. (It is easy for the model to make the tough calls about selling; after all, it lacks any emotional attachment to the stocks.) *Overconfidence*, another common behavioral bias, leads to too much trading, raising transactions costs. QEPM curbs overconfidence because the optimization model specifically controls trading costs. *Confirmation bias* leads some investors to block out relevant bad news on stocks they like. Again, with QEPM, the quantitative model processes all relevant information objectively.

QEPM strategies also have the benefit of being replicable. Portfolio managers can pass their models on to their successors when they leave a firm. The firm is not completely dependent, therefore, on the presence of a star manager. Replicability also makes it possible to backtest investment strategies on historical data over different time periods, in different markets, and with alternative specifications. Unlike quantitative methods, qualitative interpretations of market events are largely in the eye of the beholder, making them difficult to replicate in the absence of the manager who comes up with them, difficult to backtest, and difficult to articulate to investors as a methodology.

The cost of portfolio management is generally lower with QEPM than it is with qualitative management. The Ph.D.'s and other "quants" who must be hired to build the stock models generally demand high salaries, but, after the models are implemented, computers do a big share of the work. This keeps the QEPM department's head count relatively low compared with departments that delve into extensive qualitative research.

One of the most important advantages of QEPM is that it provides precise measurements of risk. A good understanding of stocks' exposure to risk factors is essential to the entire construction of the portfolio. More specifically, the ability to measure the risk of the portfolio versus a benchmark has opened the gate for controlled enhanced index management. By quantifying the tracking error of the portfolio, managers can select stocks that both earn high returns and keep the risk of the portfolio within very specific boundaries. This is difficult to do without using quantitative risk-control mechanisms.

There are some minor disadvantages to QEPM, the most prominent being the problem of translating qualitative inputs into quantitative data for use in a quantitative model. Despite the problems inherent in investing in subjective perceptions, there are valuable insights to be gained from, for instance, visiting a store and evaluating its level of customer service. Incorporating this sort of evaluation into QEPM is not simple. Numerical customer satisfaction ratings of the store might be a useful stand-in for a manager's first-hand observations, but such information probably would be difficult to obtain. Even if it were available, how should it be added to a model already built on other data? Later in this book we will show how qualitative inputs from fundamental analysts can be translated into data suitable for quantitative models.[5]

QEPM's heavy reliance on historical data has drawbacks. Historical relationships may not continue in the future, throwing off stock-return forecasts. New types of companies and new market environments, such as the internet bubble of the late nineties, diminish the relevance of inferences and expectations based on past patterns. QEPM is not unique in its reliance on historical information, however, and the statistical tests that quantitative managers apply to a set of data may help them to determine what portion of it is no longer useful. Statistical tests resolve some, but not all, questions about the continuity of trends in the data.

There is the potential for misuse of statistical tests. Data mining, a highly inappropriate practice, involves testing many statistical relationships in historical data and picking the one that apparently explains past stock returns most accurately.[6] The "mined" strategy will have almost no relation to current market conditions and therefore very little ability to predict future stock returns.[7] Unfortunately, many quantitative managers and analysts nonetheless are tempted to do data mining because it is so easy to keep testing and discarding models until finding one that works well on historical data. It takes integrity and discipline to resist the temptation.[8] Qualitative managers are susceptible to a form of data mining known as *data snooping*,[9] so the misapplication of historical relationships is endemic to active portfolio management as a whole.

The last disadvantage we associate with QEPM is its reaction time.[10] QEPM strategies may be slow to react to a shift in the economic paradigm or a change in the investment environment because they are drawn from historical data. Advanced statistical analysis, research, and ingenuity can improve the reaction time. Delayed reaction to new conditions is a problem shared, though to a lesser extent, by qualitative managers. Qualitative strategies can be modified quickly in the face of changing conditions. How well the modifications work depends, of course, on how accurately the manager and analysts interpret events.

Overall, we believe that QEPM's advantages far outdistance its disadvantages. Many of the disadvantages of QEPM are common to all types of active portfolio management. Its particular benefits, however, make it especially well suited to this age of information overload and ever-growing competition among investment funds to find good, unexploited opportunities.

1.3 QUANTITATIVE AND QUALITATIVE APPROACHES TO SIMILAR INVESTMENT SITUATIONS

We have spoken in general about the advantages of QEPM, but you may be wondering how a quantitative manager's response to specific market conditions differs from a qualitative manager's. In this section we discuss quantitative versus qualitative strategies for real-world investment problems.[11]

The Federal Reserve of the United States sets interest rates through the Fed funds target rate, which is typically announced at Federal Open Market Committee (FOMC) meetings. At its meeting on March 15, 2020, at the outbreak of the Covid-19 pandemic, for instance, the Fed lowered the Fed funds target rate to a range between 0% and 0.25%. Such changes in the target rate reverberate throughout the bond and equity markets, so investors try to anticipate them and also gauge what the rest of the market expects them to be. The market trades on these expectations via Fed futures that are based on the effective Fed funds rate, the daily average of the Fed funds rate over the previous month.

With an FOMC meeting on the horizon, a qualitative manager might say, "It's highly likely that Fed Chairman Powell will raise rates by 25 basis points. We should reduce the β exposure of the portfolio." This assessment may be right on the nose, especially given the qualitative manager's years of experience watching the markets. A quantitative manager, however, is more likely to use market data, such as Fed fund futures prices, to specify exactly the implied probability that the Fed will raise rates. In this way, the quantitative manager can state with confidence, "The market has already priced in a 98% probability of a 25 basis point raise in rates." The quantitative manager also has the tools to quantify the effect of a rise in rates on various types of equities, including cases in which the market expects the increase.[12] This is a much more precise analysis, for investment purposes, than the qualitative manager's gut-feeling approach.

Recently, the government of a major country, certain that interest rates were on the rise, wanted to invest in a portfolio of stocks inversely related to interest rates. There are many types of fixed-income investments that would have been appropriate for protecting against a rise in interest rates, but the government wanted an all-equity portfolio. A qualitative analyst might have begun constructing the portfolio using some rule of thumb, such as screening for companies with low debt-to-equity ratios or for companies in the utilities industry, which is known to hold up well in the face of high interest rates. The screen would have yielded a list of stocks that ought to do well in the upcoming high-rate environment.

By contrast, a quantitative analyst probably would have started work on this portfolio by creating an economic factor model that explicitly modeled stock returns in relation to the macroeconomic factor of concern, interest rates.[13] From the results of the model, the analyst then would have constructed a portfolio of equities with low or negative exposures to interest rates. As opposed to the merely directional prediction of the qualitative manager's rule of thumb, the quantitative model would have shown *how much* individual stocks and the entire portfolio likely would react to higher interest rates, with an estimate of the degree of uncertainty of the expected behaviors. Being able to anticipate not only the direction but also the amount and uncertainty of movement in stock prices, the quantitative manager would have formulated a more precise interest-rate hedge.

Sometimes companies are provided with windfall revenue streams, such as when another company or the government awards them a contract. This happened during the Covid-19 pandemic to AstraZeneca when it won a contract with the U.S. government for its Covid-19 product AZD7442. On October 9, 2020, the U.S. government announced that it would provide AstraZeneca with approximately $486 million for two phase 3 clinical trials and related development activities. A qualitative manager might have seen this announcement and said, "The stock has only gone up about 62 cents per share. I have a feeling this deal is worth a lot more than that. Let's make it a short-term buy." This intuition may have merit, but it is not very precise. Given the available information, it would have been possible to do a much more precise analysis quantitatively.

The manager could have used a modified discounted cash-flow model to evaluate the impact of this deal on the stock price of the company. The manager then could have observed the actual change in the stock price, compared it with the predicted change in the stock price from the model, and made an informed decision about whether to go long or short the stock as a short-term trade.[14]

Performance attribution is another type of analysis that benefits from quantitative techniques. With classic performance attribution, a quantitative performance analyst can split the portfolio's excess returns over the benchmark into their underlying sources, making it easier to pinpoint the investment decisions that augmented or diminished the portfolio's performance. Calculations of risk-adjusted performance tell the analyst whether a portfolio's excess performance was due to extra risk (pseudo-outperformance) or to additional return without additional risk (true outperformance). For analyzing the performance of a competitor's portfolio, techniques such as style analysis help the analyst to get an idea of the competitor's investment strategy even if the individual securities in the portfolio are unknown. QEPM approaches to performance measurement analyze performance rigorously and provide a great deal of information and feedback to portfolio managers and investment committees.

Ultimately, investors care about the after-tax returns of their portfolios. Both qualitative and quantitative managers try various ways to reduce the tax burden. Qualitative managers use some very good rules of thumb, such as selling the oldest tax lots first and selling the tax lots with the lowest gains. They also take futures positions that can be traded in the short term with relatively little tax burden. Quantitative methods incorporate these techniques and also generate many more tax-reduction strategies. Moreover, it is possible to integrate tax considerations directly into a quantitative investment model.[15] With QEPM, the manager can look at the tax effects of a transaction before deciding whether to buy or sell. QEPM also offers a solution to instances in which trades made to reduce the tax burden end up disturbing the balance of stocks in the portfolio. For instance, selling poorly performing stocks at year end to generate capital losses that offset capital gains may make the portfolio less than optimal versus the benchmark or increase its tracking error. A method known as *characteristic matching* can be used to find stocks similar to those that were sold to generate capital losses. Purchasing the matching stocks restores some of the overall characteristics of the portfolio even though some of the original stocks were sold temporarily for tax purposes. Such quantitative tax management strategies protect and often significantly increase after-tax returns.

The financial markets often stray from efficiency. For example, when companies report higher-than-expected earnings, their stocks often earn higher-than-normal returns during the few weeks following the report. Qualitative managers may or may not trade on this sort of market anomaly. If they do—for instance, by purchasing stocks with higher-than-expected earnings announcements—it is often in an ad hoc fashion. QEPM equips quantitative managers to study anomalies in detail and exploit them in a calculated fashion. Quantitative methods help to determine the underlying source of an anomaly, which (if any) time period or industry it is specific to, and the expected excess return of a strategy that centers on it. Quantifying the risks, the gains, and the idiosyncrasies of the anomaly makes for a well-informed trading decision.

Managing a portfolio involves buying and selling stocks repeatedly. Buying and selling generates transactions costs in the form of commissions, price impact, and delay. Studies of equity mutual funds show that most mutual fund managers fail to beat the S&P 500 after accounting for transactions costs.[16] Qualitative managers typically only consider transactions costs implicitly. Quantitative managers can use optimization algorithms to focus directly on the effect of transactions costs on the portfolio's return. Now that some commercial research vendors gather detailed data on the costs of trading, including commissions, price impact, and delay, it is possible to determine the effect of the costs on returns quite precisely. Some investment funds also do their own in-house estimates of transactions costs. From either in-house or vendor-provided data, the quantitative manager can find out whether certain transactions are worthwhile and can avoid *churning*, or excessive turnover.[17]

Many equity portfolio managers use leverage to increase the returns of their portfolio. The typical way to leverage the portfolio—through index futures such as the S&P 500 futures—achieves levered exposure to the overall market but is not always the optimal route because it dilutes a manager's excess performance. Qualitative managers, for the most part, ignore the dilution effect and leverage with index futures. QEPM makes it possible to design a plan that does not dilute the performance. Single-stock futures or equity-swap baskets lever the excess returns of the portfolio in addition to its market exposure, thereby multiplying the excess return rather than diminishing it. To the extent that a manager is able to generate excess returns, the quantitative approach to leverage produces better results.[18]

The QEPM method of *factor tilting* highlights another difference between qualitative and quantitative approaches. Qualitative managers typically purchase stocks that they believe will outperform, accepting all risks associated with each stock. Although this is not necessarily a bad decision, and quantitative managers sometimes do the same, it is possible, using factor tilting, to calibrate the portfolio's exposures to different types of risks. Suppose that a manager is very good at forecasting the future value premiums of stocks but very bad at forecasting the other variables that compromise his or her stock-return model. Factor tilting lets the manager create a portfolio with zero exposure, relative to the benchmark, to all variables except the future-value factor. The result is a model that is relatively more exposed to the factor that the manager is fairly capable of forecasting and relatively less exposed to the ones that he or she cannot forecast well. Factor tilting can be a very effective way of managing an equity portfolio and is one of the many powerful tools of QEPM.

These are only a sample of the types of decisions that might differentiate portfolio managers who use QEPM from those who do not avail themselves of quantitative methods. Clearly, responses to specific investment situations vary from manager to manager. Along the spectrum of management styles, some qualitative managers use quantitative methods frequently, and some quantitative managers draw significantly on qualitative information. As we see it, the more consistently a manager uses quantitative methods, the more consistent and precise are the portfolio's results. QEPM structures the decision-making process, and it is a structure adaptable to practically all types of investment scenarios.

1.4 A TOUR OF THE BOOK

The field of QEPM is vast, and there are literally thousands of ways to go about building quantitative models to select stocks for a portfolio. In this book we focus on the most prevalent QEPM methods. Our goal is to cover the entire QEPM process, from modeling stock returns, to building the actual portfolio, to assessing the performance of the portfolio.

The book is divided into five parts. Part I introduces the concept of quantitative equity portfolio management. Chapter 2 discusses the fundamental principles of QEPM, as well as the concept of market efficiency and why QEPM works in mostly efficient markets. Chapter 3 describes the typical QEPM process and introduces the most common models of stock returns.

Part II of the book is about portfolio construction and maintenance. The first step in building a model of stock returns is to choose a mix of explanatory variables for the model. Chapter 4 defines the most commonly used factors and provides simple methods for selecting ones for the model. Factors also can be used for preliminary stock screening and ranking.

Chapter 5 explains the basics of stock screening and introduces the aggregate Z-score model, a simple model for ranking stocks. We also describe the investment philosophies of famous portfolio managers and suggest stock screens that emulate their strategies.

Chapters 6 and 7 discuss in detail how to build fundamental and economic factor models, the two types of models that quantitative managers use to estimate and explain stock returns and volatility. Chapter 8 discusses how to forecast future factor premiums, which, in the framework of the factor models, can be used to forecast future stock returns and risks. The forecasts are the basis for including or excluding stocks from the portfolio.

Chapter 9 ties together Chapters 4 through 8 by showing how to use stock-return models and concepts from optimization theory to determine the optimal weights for the stocks in the portfolio while abiding by any investment constraints.

Chapters 10 and 11 discuss refinements to the basic construction and maintenance process. Chapter 10 explains ways to improve the performance of the portfolio by paying particular attention to transactions costs. Managers can avoid excessively high turnover and high transactions costs by incorporating the costs explicitly into the model of stock returns. Chapter 11 discusses ways to improve performance by managing the effect of taxes. By taking taxes into account in the model itself, the manager can be clever about the timing and composition of trades.

In Part III of the book we step outside the core set of QEPM procedures to explore methods for increasing the portfolio's performance that are not related to actual stock picking. We refer to these methods collectively as α *mojo* ("alpha mojo") because they boost α, which is the portion of a portfolio's returns that goes above and beyond the return of the benchmark or reference portfolio.

Chapter 12 looks at the first form of α mojo, leverage. Leverage is used by many portfolio managers to increase the exposure of the portfolio to the market. The chapter shows various methods for leveraging an equity portfolio to boost its excess returns.

Chapter 13 discusses creation of the market-neutral portfolio. Market-neutral and long-short portfolios are widely used structures for quantitative portfolios, especially among hedge funds. They eliminate or reduce market exposure and increase α, improving the risk-return profile of the portfolio.

A market-neutral portfolio is ideal for a quantitative equity portfolio manager who wants to focus on his or her specialty of picking good stocks.

In Chapter 14 we use some advanced statistical concepts, known collectively as *Bayesian analysis*, to show how a quantitative portfolio manager can take advantage of qualitative insights. We also show how scenario analysis fits into the quantitative stock-return model.

Part IV of the book brings the portfolio management cycle to its last stage, performance analysis. Though often neglected, the cultivation of an excellent performance measurement department is one of the keys to an investment firm's success. Chapter 15 discuses various standard practices, as well as some new concepts related to quantitative performance measurement and attribution.

Part V implements all the ideas in the first four parts of the book. With an example of the practical application of QEPM, we discuss everything related to building a quantitative portfolio, including data issues, estimation problems, portfolio construction with real data, transactions costs, tax issues, leverage, and performance analysis.

Five appendices can be found online.[19] Appendix A is a history of financial theory for readers who need a quick overview of the subject. Appendix B explains three well-known models of stock returns: the dividend discount model (DDM), the capital asset pricing model (CAPM), and the arbitrage pricing theory (APT) model. Appendix C provides a basic review of mathematical and statistical concepts useful to readers of this book. Appendix D includes a discussion of organizational considerations for investment research departments, followed by a summary of commercial databases and software modeling programs useful for QEPM. Finally, Appendix E looks at the game of craps as an entertaining example of the importance of using quantitative techniques.

1.5 CONCLUSION

Quantitative equity portfolio management (QEPM) is growing in popularity among professional portfolio managers. This style of management offers plenty of advantages over traditional qualitative portfolio management. A combination of advanced statistics, mathematics, and a disciplined approach, QEPM makes it possible to quantify stocks' expected returns and risks with a good degree of precision, as well as quantify the uncertainty of the forecasts themselves. To begin the book, we described what we see as the major advantages of QEPM over qualitative approaches to asset management, and we gave examples of typical differences between the two styles of management in the face of specific investment decisions. In the rest of the book we will attempt to guide you through the very broad field of QEPM. We will take you step by step through every aspect of the QEPM process, from stock-return forecasting, to portfolio building, to performance measurement.

QUESTIONS

1.1. Name three advantages of QEPM versus qualitative equity management.

1.2. Name three disadvantages of QEPM versus qualitative equity management.

1.3. Name four realistic investment situations in which a quantitative portfolio manager differs from a qualitative portfolio manager in his or her approach.

1.4. In some ways, a qualitative portfolio manager could never really be an index portfolio manager, whereas a quantitative portfolio manager could be. Explain why.

1.5. What is factor tilting? What type of portfolio manager engages in it?

1.6. Fed fund futures are actively traded instruments on the Chicago Board of Trade. The contract is cash settled to the simple average of the daily effective Fed funds rate for the delivery month. Although the daily effective Fed funds rate is not perfectly equal to the Fed funds target rate, it is very close. Market practitioners use the Fed fund futures rate to gauge the market's assessment that the Fed will change the Fed fund's rate. Suppose that there will be an FOMC meeting 10 days into the current month. Use the following variables, $i^f{}_t$ as the Fed fund futures rate, $i^{pre}{}_t$ as the target rate prevailing before the FOMC meeting, $i^{post}{}_t$ as the target rate expected to prevail *after* the FOMC meeting, p as the probability of a target rate change, d_1 as the number of days between previous month end and the FOMC meeting, d_2 as the number of days between the FOMC meeting at the current month end, and B as the number of days in the month. (*Note*: In practice, the Fed attempts to achieve a certain target rate or target range, but it varies from day to day. For i^{pre}_t you can think of it as the midpoint of the previous target range or use the average Fed funds effective rate of the prior month. For i^{post}_t you can use the midpoint of the next jump in the target range [e.g., 25 basis points] or some other value of your choice.)

(a) Write down a general formula for the probability of an FOMC target-rate change.

(b) Given that the current target rate is 3.5%, the expected rate after the meeting is 3.75%, the Fed futures implied rate is 3.60%, there are 30 days in the month, and the FOMC meeting will take place on the tenth day of the month, what is the probability implied by the market prices that the Federal Reserve will raise interest rates?

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1 In 2021, there were approximately 11,323 stock mutual funds and 3,508 U.S. hedge funds. There were also about 126,457 open-end funds in the world managing $63.1 trillion in assets and 8,042 hedge funds worldwide managing $3.6 trillion. In 2021, there were also approximately 1,974 quantitative hedge funds in the world managing $1.004 trillion. *Source:* Collins et al. (2021) and Hedge Fund Research (www.hfr.com).

2 See Liu and Sinha (2021). In addition, over this 20-year period, only 6% of large-cap funds beat the S&P 500, 12% of mid-cap funds beat the S&P 400, and 12% of small-cap funds beat the S&P 600.

3 See "Interview with Peter Lynch," https://www.pbs.org/wgbh/pages/frontline/shows/betting/pros/lynch.html. Warren Buffett is also another well-known qualitative-style manager.

4 See Morgenson (1997).

5 Chapter 14 explains how Bayesian statistics serve as a powerful method for converting qualitative data into quantitative data.

6 In general, QEPM requires a good understanding of statistics. Not using statistical methods appropriately may create a number of problems. Data mining is just one of them. Not accounting for parameter uncertainty and overinterpreting the estimation from a small sample are other serious problems. We discuss some of these issues in Chapter 2. Nonetheless, the abuse and misunderstanding of statistics, not QEPM itself, are mainly to be blamed.

7 We discuss the issue of data mining in more detail in Chapter 2.

8 We have noticed that data mining occurs with particular frequency in investment departments dominated by office politics.

9 See Chapter 2 for an explanation of this concept as well.

10 This is not the same as *skid*, a term used in the industry to describe a movement in price that occurs in the time between the recognition of an idea and its implementation in the portfolio. Skid is also a problem for portfolio managers because a price movement may eliminate an investment opportunity.

11 On a lighter note, in Appendix E found at www.ludwigbc.com under QEPM Exclusive Content, we also hypothesize about how a quantitative manager and a qualitative manager might differ in their approaches to the game of craps.

12 For more details, the reader is referred to Bieri and Chincarini (2005).

13 We introduce economic factor models in Chapter 3 and discuss them in greater detail in Chapter 7.

14 In Appendix B at www.ludwigbc.com under QEPM Exclusive Content, we discuss additional realistic uses of discounted cash-flow models.

15 See Chapter 11 on tax management.

16 See Bogle (1999).

17 We discuss transactions costs further in Chapter 10.

18 For more on leverage, see Chapter 12.

19 Go to the website www.ludwigbc.com. Once you reach the website, choose Books, then Quantitative Equity Portfolio Management, then QEPM Exclusive Content. The password is qepm2020rig179. You will find a folder with all the solutions to the end-of-chapter questions in the book, a folder with additional appendices for each chapter, a folder with General Appendices (Appendices A–E), a folder with programs and examples written in MATLAB, R, and Stata for learning and for classroom labs, a folder with PowerPoint slides for every chapter, a folder with sample stock data, and a downloadable file of historical factor returns.

CHAPTER 2

The Fundamentals of QEPM

Some people recognize beauty before others have recognized it, some people see beauty that no one else can see, and some people see no beauty.

2.1 INTRODUCTION

Quantitative equity portfolio managers cannot simply go through the motions of quantitative analysis. Implementing quantitative equity portfolio management (QEPM) the right way requires a firm grasp of the underlying concepts that make the analysis work. Since quantitative equity portfolio managers generally aim to outperform a benchmark or index, we start our conceptual discussion with alpha (α), the measurement of a portfolio's risk-adjusted returns over and above a reference instrument. There are a number of variations of alpha, but no matter what type of alpha a manager strives for, his or her work should be guided by the seven tenets of QEPM. The seven tenets encompass ideas necessary to the very existence of QEPM. One of these essential ideas is that financial markets do not operate entirely efficiently. Neither QEPM nor any other form of active management would succeed in a perfectly efficient market. We examine the evidence for market inefficiency, which is found mainly in studies of market anomalies, and we explore possible explanations for these anomalies. The seven tenets of QEPM also lay down important principles for the practice of QEPM, including the efficient use of information in quantitative models. We discuss the popular but frequently misinterpreted fundamental law of active management as an important framework for understanding and achieving efficiency. Finally, the seven tenets demand the consideration of statistical issues that have direct bearing on the statistics-intensive methods of QEPM. We reflect on these issues, which include data mining, parameter instability, and parameter uncertainty, at the end of the chapter.

2.2 QEPM α

The word *alpha* is constantly bandied about in the world of active portfolio management. Active managers use the term in many contexts, and sometimes it is not clear exactly what they are referring to when they use it. The simplest, colloquial meaning of α is outperformance. You might hear a portfolio manager say, "I'm trying to generate positive α." He or she probably means that he or she wants the portfolio to outperform some other instrument. In general, α represents the excess return of the portfolio over the return of a reference instrument. There are different ways to define *excess performance*, though. In QEPM, α is a measure of the *risk-adjusted* excess return, which is the portfolio's performance after accounting for its risk relative to the reference instrument. Increasing the α therefore means increasing the portfolio's return without increasing its direct exposure risk to the reference instrument. When the reference instrument is the portfolio manager's benchmark, we will refer to α as the *benchmark* α or α^B.[1] When the reference instrument is a series of multifactor benchmarks, we will refer to α as *multifactor* α or α^{MF}. When the reference instrument is the market portfolio, we will refer to α as the *CAPM* α or α^{CAPM}.[2]

Given the returns of the reference instrument, it is possible to use statistical techniques to decompose the portfolio's return into two parts—one that is related to the reference instrument and one that is not. The related part is typically called *expected return* or *consensus return*, and the second part is known as *residual return* or *the return not explained by the model*. For each of the three types of α, there is a slightly different method for separating these two components of the portfolio's return.

2.2.1 Benchmark α

Given the portfolio return r_P and the benchmark return r_B, we can estimate the following equation[3]:

$$r_P = \alpha + \beta r_B + \epsilon \tag{2.1}$$

In this equation, βr_B is the expected or consensus return, which is the part of the portfolio's return related to the benchmark. The remaining $\alpha + \epsilon$ is the residual return.[4]

The residual return is all that matters to the quantitative portfolio manager, because his or her goal is to increase the risk-adjusted return. If the benchmark return is positive, it is easy enough to generate higher returns simply by increasing the portfolio's exposure to the benchmark, but the portfolio manager has not added value. The part of the portfolio return that represents an increase in return independent of increased direct benchmark exposure is the residual return. Benchmark α is the expected value of the residual return, and the second component of the residual return, ϵ, is the deviation of the residual return from its mean. Equation (2.1) is constructed so that ϵ averages zero. The benchmark α, also known as α^B, therefore has special significance: It is the risk-adjusted excess return over the benchmark.

2.2.2 CAPM α

Given the portfolio return r_P and the market return r_M, we can estimate the following equation:

$$r_P = \alpha + \beta r_M + \epsilon \tag{2.2}$$

In this equation, βr_M is the expected or consensus return, which is the part of the portfolio's return related to the market. The remaining $\alpha + \epsilon$ is the residual return.

The reader will notice that this version of α is very similar to the one created using the benchmark return. In fact, since the market return is typically the Standard and Poor's (S&P) 500 return, if the benchmark is the S&P 500, then those two measures of α are the same. In this equation, however, α is called the *CAPM α* because this decomposition of a portfolio's returns follows from the capital asset pricing model (CAPM).[5] The equation says that the returns of all portfolios are related to the market portfolio. According to CAPM theory, the CAPM α should equal zero. When it is significantly positive, the portfolio manager is providing excess risk-adjusted returns. The CAPM α, or α^{CAPM}, is the risk-adjusted excess return over the market.

2.2.3 Multifactor α

Given the portfolio return r_P and a series of factor returns $f_1, \ldots, f_K$, we can estimate the following equation:

$$r_P = \alpha + \beta_1 f_1 + \cdots + \beta_K f_K + \epsilon \tag{2.3}$$

where $\beta_1 f_1 + \ldots + \beta_K f_K$ is the expected or consensus return from a multifactor model of stock returns (i.e., the part of the portfolio's return related to the underlying risk factors in the economy). The remaining $\alpha + \epsilon$ is the residual return, as before.

Note that this version of α is created using a model with many factors that influence stock returns. There will be more discussion of this framework in subsequent chapters. The α in this model is called the *multifactor* α, or a^{MF}, and it is a measurement of risk-adjusted excess return given multiple explanatory variables.

2.2.4 A Variety of α's

The α's in Eqs. (2.1), (2.2), and (2.3) are interrelated. In special cases, they are equivalent to each other.

1. The multifactor α of a stock or a portfolio is the same as the CAPM α if the market return is the only factor in the model.
2. The benchmark α of a stock or a portfolio is the same as the CAPM α if the market portfolio is the benchmark.
3. The multifactor α and the benchmark α of a stock or portfolio are the same if the market return is the only factor in the model and is also the benchmark.

In other cases, the three α's are not equivalent. Suppose that the manager uses an inefficient benchmark, which is evidenced by the fact that the α of the benchmark against the market return is negative. As a result, the benchmark α of the portfolio will be higher than the market α of the portfolio. To use our notation, $\alpha^B > \alpha^{\text{CAPM}}$. Using an inefficient benchmark therefore will make the portfolio's performance look better than it would if it were measured against the entire market.[6]

Many practitioners and academics may ask whether or not positive benchmark α is consistent with the arbitrage pricing theory (APT).[7] In fact, the answer is yes. Although it is practically difficult to test the APT because the theory does not specify the underlying factors of security returns, let's assume for a moment that we know the true factors of stock returns. Suppose that the portfolio's return is determined according to APT and can be expressed as

$$r_P = \alpha + \beta_1 f_1 + \cdots + \beta_K f_K + \epsilon \tag{2.4}$$

Let us also suppose that the first factor, f_1, is the benchmark (e.g., S&P 500). What would the benchmark α be in this case? Could it be positive?

Recall that the benchmark α is the part of the average return that is not related to the benchmark. If all the other factors in the APT model are uncorrelated with the first factor (the benchmark), then the benchmark α is given by

$$\alpha^B = E(\alpha + \beta_2 f_2 + \cdots + \beta_K f_K) \tag{2.5}$$

The benchmark α includes the effect of all the other factors. Thus it is quite likely that the benchmark α is positive. In general, if we allow the benchmark to be correlated with all other factors, the benchmark α is given by

$$\alpha^B = E(\alpha + \beta_2 f_2 + \cdots + \beta_K f_K) - (\beta_2 \gamma_2 + \cdots + \beta_K \gamma_K) E(f_1) \tag{2.6}$$

where γ_j is the coefficient in the regression of f_j on f_1 [i.e., $C(f_1, f_j)/V(f_1)$]. While the formula is somewhat more complicated, the conclusion is the same. The benchmark α includes the effect of all factors aside from the benchmark, so it is very possible that the benchmark α is positive.

This analysis shows that when measuring the performance of a manager with benchmark α, benchmark α can be positive even if multifactor α equals zero. Although multifactor α equals zero, benchmark α still can be positive if the portfolio manager loads his or her portfolio with positive exposures to factors that have positive premiums according to the APT. A positive benchmark α therefore is consistent with the APT model, in which managers are rewarded for statistical arbitrages.

2.2.5 Ex-Ante and Ex-Post α

Whichever type of α we use, there is usually a difference between the value we expect it to reach in the future and the value that it ultimately realizes. The *ex-ante* α is the expected α, whereas the *ex-post* α is the realized one. The ex-ante α is of interest to the quantitative portfolio manager when he or she is constructing his or her portfolio. Clearly, he or she sets out trying to achieve the highest possible α when he or she builds the portfolio subject to the portfolio constraints. Once the portfolio has been active for some period of time, the ex-post α shows whether the actual risk-adjusted performance lived up to expectations. The manager's bonus is based on the ex-post α.[8] The best a manager can realistically hope for is that the ex-post and ex-ante α's will be highly correlated.

2.2.6 Ex-Ante and Ex-Post Information Ratio

Though α by itself is the primary measure of a portfolio's excess return, there is also a key metric, the *information ratio*, that adjusts α for the portfolio's residual risk. As with α, there are ex-ante and ex-post information ratios. We will discuss only the ex-ante information ratio at this stage because the ex-post information ratio is discussed in detail in Chapter 15 on performance measurement. The information ratio is directly related to the benchmark α, α^B. The ex-ante information ratio (IR) is given by

$$IR = \frac{\alpha^B}{\omega} \tag{2.7}$$

where α^B is the expected or predicted α of the portfolio manager, and ω is the predicted standard deviation of the residual [i.e., $S(\epsilon)$]. The ex-ante information ratio measures the expected excess return over the benchmark per unit of excess risk over the benchmark. The higher the ex-ante information ratio, the better we expect the portfolio to perform. All else equal, a higher forecasted ratio is better.

2.3 THE SEVEN TENETS OF QEPM

A quantitative equity portfolio manager aims for the twin goalposts of a high alpha and a high information ratio, and getting there requires knowing the rules of the game. QEPM is organized around certain principles. We call these principles the *seven tenets of QEPM*.

Tenet 1: *Markets are mostly efficient*.
Tenet 2: *Pure arbitrage opportunities do not exist*.
Tenet 3: *Quantitative analysis creates statistical arbitrage opportunities*.
Tenet 4: *Quantitative analysis combines all the available information in an efficient way*.
Tenet 5: *Quantitative models should be based on sound economic theories*.
Tenet 6: *Quantitative models should reflect persistent and stable patterns*.
Tenet 7: *Deviations of a portfolio from the benchmark are justified only if the uncertainty is small enough*.

Tenets 1 and 2 set boundaries on QEPM's reach. Markets are mostly efficient, which means that it is not possible to make profits without taking risk. Risk-free opportunities, or pure arbitrage opportunities, do not exist. Still, the markets are not perfectly efficient. There is room to make profits by taking relatively small amounts of additional risk. QEPM is the process of searching for such "statistical arbitrage opportunities," as stated in tenet 3. Statistical arbitrage opportunities exist because stock prices do not always properly reflect all the available information. QEPM provides statistical methods for identifying information that the market has overlooked. As tenet 4 says, the quantitative manager must combine all the available information into an efficient model in order to identify the pieces that are key to earning higher returns.

Tenets 5, 6, and 7 establish standards for applying econometric techniques to choices involving the portfolio. Tenet 5 is absolutely fundamental to creating quantitative models of stock returns. All models used to pick stocks should be based on sound economic theory. Data mining violates this tenet. Factors dredged up from the data through data mining may seem to relate to stock returns on their surface, but they are unlikely to reveal real opportunities for statistical arbitrage. The manager has to have a good reason for choosing each factor that he or she includes in the model. There should be reason to believe that stock prices do not yet, but eventually will, reflect the information contained in each factor or in the group of factors as combined in the model. The model is not meaningful without a strong theoretical underpinning.

Likewise, the relationship between the factors and stock returns needs to be one that holds up over time. As tenet 6 says, the model should only make use of data patterns that are persistent and stable. Parameter stability is essential to the generation of precise estimates and reliable forecasts with the model.

Tenet 7 cautions that QEPM does not always involve differentiating the portfolio from the benchmark. Many quantitative portfolio managers mistakenly believe that deviations from the benchmark are necessary in QEPM. In fact, a deviation from the benchmark portfolio is justified only if the portfolio's estimation error (a measure of uncertainty) is small enough. Quantitative managers always should take parameter uncertainty into account before steering the portfolio away from the benchmark onto another path dictated by the model.[9]

In the following sections we discuss the seven tenets of QEPM in greater detail. We start with the concept of market efficiency and its implications for QEPM. Then we discuss how to use information efficiently, adopting the framework of the fundamental law of active management. Finally, we examine issues arising from the application of econometric techniques to QEPM.

2.4 TENETS 1 AND 2: MARKET EFFICIENCY AND QEPM

Tenets 1 and 2 of QEPM are based on the belief that markets are mostly, but not completely, efficient. If markets were completely efficient, QEPM would be a futile exercise. An active portfolio manager has to believe that there are inefficiencies in the market that he or she can exploit. In this section we discuss what it means for the market to be efficient and what the evidence is for some degree of market inefficiency.

2.4.1 The Efficient-Market Hypothesis

There are those who say that financial markets are efficient and those who say they are not, but what are people really talking about? In general terms, an efficient market is a market in which all information is reflected in current stock prices. For instance, in an efficient market, an investor could not earn excess returns by buying a stock on a tip that the company has a new CEO who is expected to improve operations. The price of the stock already would reflect this news and any other relevant information. In a perfectly efficient market, investors cannot earn excess returns without bearing extra risk, and portfolio managers, therefore, cannot create α.[10]

In an efficient market, stock prices move randomly. If we looked at the historical prices of any stock in an efficient market, the day-to-day changes would show no discernible pattern. When stock prices do move in some predetermined fashion, then presumably an investor can make money by trading around the pattern, which means that there is an inefficiency in the market.

Not even *near-arbitrage* opportunities exist in a perfectly efficient market. For example, if a publicly traded company traded at a price of \$100, a subsidiary in which it owned a 95% stake could not trade at \$200. If it did, an investor could buy the main company's stock at \$100 per share and short sell the subsidiary's stock at \$200 per share, making a profit when the two prices eventually aligned. When PALM was spun off from 3COM on March 1, 2000, the two companies' stock prices displayed a similar imbalance, leading some to seriously doubt the efficiency of the stock market.

Asset bubbles are also signs of inefficient markets. In early 2000, the NASDAQ was overvalued by almost any measure of valuation. Internet startups traded at market capitalizations many multiples of the market caps of established brick-and-mortar companies in similar lines of business. From March 2000 to January 2003, however, the NASDAQ fell by 71%. A $100 million portfolio invested in the new technology revolution in 2000 would have been reduced to a mere $29 million in less than three years. The internet bubble was a huge, unsustainable deviation from market efficiency.

Perhaps one of the greatest dislocations of prices from fundamentals occurred during the housing bubble of 2008, also known as the Great Recession. In the period from 2003 to 2008, easy liquidity and lack of accountability caused housing prices and the stock market to soar. In 2008, the housing market came crashing down along with the stock market. The S&P 500 declined by 38% in 2008, and the national housing price index declined by 11.88%, with some cities experiencing even larger declines in housing prices. This perplexing inefficiency may have been caused by an immensely crowded investment space and pointed to the potential inefficiencies of markets.[11]

In the 1970s, Eugene Fama, a professor at the University of Chicago, decided to establish testable criteria for market efficiency. Fama recognized that the market's level of efficiency has to do with the degree to which security prices reflect information relevant to the investment decision. The nature of investment-relevant information is a continuum that spans from the most obvious public information to the most private insider information. In Fama's criteria, market efficiency is directly related to how much of the continuum is incorporated in market prices.

Fama described three increasing levels of efficiency—*weak form, semistrong form*, and *strong form*—at which market prices reflect increasingly more of the information continuum. (Table 2.1 provides a quick overview of the three definitions of efficiency.) At the lowest level of efficiency, weak-form efficiency, *security prices reflect the information available in previous security prices*. This means that past prices of stocks give an investor no actionable information. For portfolio managers, it means that *technical analysis* does not work.

TABLE 2.1

Definitions of Market Efficiency

Type of Efficiency	Definition	Practical Implication
Weak form	Security prices reflect the information available in previous security prices.	You should not expect technical analysis to aid you in picking stocks.
Semistrong form	Security prices reflect all information that is publicly available.	You should not expect models built from macroeconomic, fundamental, analyst, or any other publicly available information to aid you in picking stocks.
Strong form	Security prices reflect all information that is publicly or privately held.	You should not expect inside information or proprietary information to aid you in picking stocks.

Technical analysis is popular with many investors, especially traders. A classic text written by John Murphy defines technical analysis as the study of market action through the application of charts and/or statistical techniques to forecast future price trends.[12] Whereas the fundamental analyst looks for the causes of market movements in the form of economic or operating data, the technical analyst is more concerned with patterns in the market movements themselves. Technicians look for patterns in data directly obtainable from securities, such as past prices, volume, and open interest. Like fundamental analysts, technicians believe that prices adjust slowly to their proper levels when new information reaches the marketplace because not all investors have equal access to information. The time it takes for information to percolate through the hierarchy of professional traders down to the general investing public creates a window of opportunity for technicians to trade on the patterns that they have identified in the data.

One of the simplest patterns identified by technicians is *momentum*. The idea is that if a stock had a positive return in the period prior to this one (e.g., last week), then there is a good chance that the stock will have a positive return in the subsequent period (e.g., this week). If this pattern holds, the market is not weak-form efficient because past prices, which indicate the past returns, help you to invest for the future. In a weak-form efficient market, current prices incorporate all the information contained in past prices.

The next level of efficiency covers more of the public-to-private continuum of information. In the semistrong form of market efficiency, *security prices reflect all information that is publicly available*. This means that quantitative portfolio managers and other investors cannot use any publicly available information to pick stocks and earn risk-adjusted excess returns. Semistrong-form efficiency is bad news for QEPM. Most quantitative portfolio managers use some form of macroeconomic or fundamental data to choose superior stocks. If markets were semistrong efficient, these methods would not work. For example, some portfolio managers buy stocks with low price-to-book (P/B) ratios on the belief that such stocks are probably undervalued. This would not work in a semistrong-form efficient market because P/B ratios are public information, so stock prices already would correctly reflect low P/B ratios. Active management succeeds only by finding niches of the market that are not semistrong-form efficient, perhaps owing to the slow diffusion of information, or by developing proprietary investment strategies, which become themselves nonpublic information. In the semistrong form of efficiency, market prices do not incorporate nonpublic information.

At the highest level of market efficiency, strong-form efficiency, *security prices reflect all information that is publicly or privately held*. In a strong-form efficient market, even private information gives an investor no consistent advantage in picking stocks that will outperform the rest. Strong-form efficiency is the hardest form of efficiency to test because testing requires a precise definition of private information, yet the boundary between public and private information fluctuates. Many years ago (November 4, 1998), the Bureau of Labor Statistics (BLS) accidentally posted its weekly unemployment report on its Web site two days before the report was supposed to be published. Some investors found the number, acted on it, and made abnormal profits. Since no one expected the report to be released until two days later, what was supposed to be released as public information ended up available ahead of time to a small group of lucky, or particularly diligent, investors. Private information often transforms into public information, but as the bureau's slip-up demonstrates, information also can pass from the private domain into an in-between space not universally accessible. The law does at least draw one bright line between public and private by prohibiting trading on *inside information*. Company insiders and those who receive information from them are not allowed to take advantage of privileged knowledge. If the law is effective in barring people from trading on this sort of private information, then the market cannot be strong-form efficient.

The entire premise of active management, including QEPM, is that there are market inefficiencies prevalent and predictable enough to exploit consistently for high returns. Semistrong-form efficiency and strong-form efficiency would make active management nearly impossible. After all, a portfolio manager cannot depend on fluke opportunities such as the BLS's early unemployment data release, much less on illegal insider stock tips. The fact that some active managers consistently outperform the market or a benchmark is enough to convince some people that the market is not wholly efficient. More rigorous evidence against strong-form and even semistrong-form efficiency comes from researchers who have documented signs of inefficiency in market prices.

2.4.2 Anomalies

Investment professionals and academics have observed patterns in historical financial data that contradict the theory of efficient markets.[13] Table 2.2 lists some of these so-called anomalies, along with references to studies on each of them.

TABLE 2.2

Well-Known Anomalies

Anomaly	Associated Factor	Research Papers
1. Value effect. Low price-to-earnings ratio stocks outperform high price-to-earnings ratio stocks. Similar results for price-to-book ratio, price-to-sales ratio, and price-to-dividend ratio.	Price-to-earnings ratio, price-to-book ratio, price-to-sales ratio, and price-to-dividend ratio.	Ball (1978), Barberis and Schleifer (2003), Basu (1977, 1983), Bhanderi (1988), Conrad et al. (2003), Daniel and Titman (1997), Cook et al. (1984), Fama and French (1992, 1993), Goodman et al. (1986), Reinganum (1981, 1988), Rosenberg et al. (1985), and Senschak et al. (1987).
2. Size effect. Small-cap stocks outperform large-cap stocks.	Market capitalization (size).	Banz (1981), Fama and French (1992, 1993), Ferson and Harvey (1999), Kim and Moon (2002), and Reinganum (1997).
3. January effect. Small-cap stocks and last year's poorly performing stocks tend to outperform in January.	Low market capitalization, low returns in the previous year.	Blume and Stambaugh (1983), Constantinides (1984), Ferris et al. (2001), Givoly (1983), Haugen and Jorion (1996), Keim (1983, 1985, 1986), Lakonishik (1984,1986), Reinganum (1983), Ritter (1988), Roll (1983), Schultz (1985), and Thaler (1987).
4. Other calender effects (not including January)	Day of the week, Halloween, weather, daylight savings, and other calender events.	Ariel (1987), Bouman and Jacobsen (1997), Cross (1973), Harris (1986a, 1986b), Hensel and Ziemba (1996), Hirshleifer and Shumway (2001), Jaffe and Westerfield (1985), Kamstra et al. (2000, 2003), Rogalski (1984), Smirlock (1986), Sullivan et al. (2001), Thaler (1987), and Wang et al. (1997).
5. Neglected firm effect. Stocks with low analyst coverage tend to have risk-adjusted excess returns.	Number of analysts following the stock.	Arbel (1985), Arbel and Strabel (1982, 1983), Arbel et al. (1983), Dowen and Bauman (1984, 1986), Beard and Sias (1997), and Merton (1987).
6. Price-to-earnings-growth	Price-to-earnings-growth	Peters (1991) and Reilly

ratio effect. There is an inverse relationship between the price-to-earnings-growth ratio of a stock and its performance.	ratio.	and Marshall (1999).
7. IPO effect. IPOs tend to underperform on a risk-adjusted basis in the first 3–5 years.	Dummy variable indicating whether the stock recently had an IPO.	Carter et al. (1998), Dharan and Ikenberry (1995), Loughran and Ritter (1995), and Carter and Ritter (1991).
8. Index change effect. The stocks to be included in the index will have significantly positive returns, while the stocks to be dropped from the index will have significantly negative returns. (Every year the Russell 1000, 2000, and 3000 are rebalanced. The S&P 500 is also rebalanced at the discretion of the S&P Index Committee.)	Dummy variable indicating whether the stock is going to be included in or dropped from an index.	Jain (1987) and Harris and Gurel (1986).
9. Momentum. Stocks that perform well during one investment period will continue to do well in the subsequent investment horizon.	Return of stock in prior investment period.	Chan et al. (1996), Chan et al. (2000), Grundy and Martin (2001), Hong and Stein (2000), Jegadeesh (1990), Jegadeesh and Titman (1993, 2001), Moskowitz and Grinblatt (1999), O'Neal (2000), Richards (1997), Rouwenhorst (1999), and Swinkels (2002).
10. Other technical effects (other than momentum). Certain technical indicators have been shown to provide excess returns.	Various technical indicators, including volume, reversals, relative strength index, Bollinger bands, moving average, and others.	Brock et al. (1992).
11. Wall Street analyst forecasts. Stocks with buy ratings have been shown to outperform, but this	Analyst recommendations on stocks.	Alexander and Ang (1998), Arnott (1985), Barron and Stuerke (1998), Bartov et al. (2000), Bauman et al.

effect has subsided in recent years. Changes in analyst ratings have been shown to be predictive of security returns, and earnings surprises in which reported earnings exceed the consensus estimate have been shown to produce excess returns.		(1995), Brown and Chen (1990, 1991), Brown and Rozeff (1978, 1980), Chopra (1998), Clement (1999), Cornell (1989), Dowen (1990), Dreman and Berry (1995), Givoly and Lakonishok (1980), Jones et al. (1984), Kim and Kim (2003), LaPorta (1996), Latane and Jones (1977), Mendhall (2004), Peters (1993), and Womack (1996).
12. Insider trading. Stocks that insiders have recently bought produce excess returns. Stocks that insiders have recently sold produce negative returns.	Net purchases or sales by insiders.	Fishe and Roby (2004), Givoly and Palmon (1985), Jaffe (1974), and Seyhun (1986, 1988).
13. Overreaction. It has been found that investors overreact to news events, bad and good. One basic result is that previous "loser" stocks tend to outperform previous "winner" stocks over a 3- to 5-year period.	Past returns of stocks conditional or unconditional on certain news events.	Chincarini and Llorente (1999), Chopra et al. (1992), Conrad and Kaul (1993), Daniel et al. (1998), Debondt and Thaler (1985, 1987), Hong and Stein (1999), Lo and MacKinlay (1990), and Veronesi (1999).
14. Stock buybacks. When a company buys back its stock, the stock subsequently has positive risk-adjusted returns.	Change in treasury stock.	Choi and Chen (1997) and Ikenberry et al. (1995).
15. Stock splits and reverse splits effect. Stock splits are followed by positive returns.	Signals of a stock split.	Desai and Jain (1997), Doran (1994), Dowen (1990), Ikenberry et al. (1996), and Ye (1999).
16. Spin-offs effect. A parent company has excess returns after the spin-off of a subsidiary.	Signals of a spin-off.	Cusatis et al. (1993) and Desai and Jain (1999).
17. Accruals. It has been	Accrual of company in prior	Sloan (1996), Bradshaw,

documented that stocks that have low accruals perform better in the subsequent investment horizon than companies with high accruals. Accrual accounting measures a company's performance by recognizing economic events regardless of when cash transactions occur.	period. Accruals are defined as revenues earned or expenses incurred that impact a company's net income on the income statement, even though cash related to the transaction has not yet changed hands. Accruals can also affect the balance sheet, as they involve non-cash assets and liabilities.	Richardson, and Sloan (2001), Lev and Nissam (2006), Ali et al. (2008), Richardson et al. (2005), Thomas and Zhang (2002), Xie (2001), Shi and Zhang (2011), Allen et al. (2010), Richardson et al. (2006), Bhorraj and Swaminathan (2008), and Liepold and Lllore (2012).
18. Low volatility. It has been documented that stocks with low historical volatility outperform stocks with high historical volatility.	Historical volatility in prior period. Different researchers have used different methods to compute the historical volatility of each stock.	Thomas and Scapiro (2009), Clarke et al. (2006), Ang et al. (2006), Blitz and Van Vliet (2007), and Baker et al. (2011),
19. Low beta. It has been documented that stocks with low historical measured beta outperform stocks with a high historical measured beta on a risk-adjusted basis.	Historical beta of stock in prior period. Different researchers use different methods to measure historical beta.	Black et al. (1972), Black (1993), Frazzini and Pederson (2014), and Baker et al. (2011),
20. Liquidity. There is research that shows that less liquid stocks provide higher average returns than more liquid stocks. Despite lots of evidence, there are also researchers who doubt that this is a true anomaly.	There are various measures of liquidity. Share turnover, which is volume of shares traded divided by shares outstanding. Also called *trading turnover* when done in dollars. Amihud liquidity, which is the return of stock for the day divided by dollar volume for the day. The Modified Amihud measure, which is the same but measures the daily return using open-to-close prices. Pastor and Stambaugh use regressions on daily stock returns to assess how volume causes reversal in stock returns due to liquidity. Invariance is also used, which is related to the standard deviation of returns and average volume.	Chincarini and Llorente (1999), Chincarini and Kim (2006), Amihud (2002), Amihud and Mendelson (1986), Pastor and Stambaugh (2003), Brown, Crocker, and Foerster (2009), Chrodia et al. (2001), Liu (2006), Acharya and Pederson (2005), Fama and French (2008), Drienki et al. (2017), Hoe et al. (2017), Li et. al (2017), Barardehi et al. (2019, 2020), Kyle and Obizhaeva (2016), and Ibbottson et al. (2013).

21. Crowding. There is evidence that stocks that are disproportionately invested in by money managers underperform stocks that are less saturated by money managers.	The measure of crowding in an investment space. There are various measures of crowding, including return-based measures and holding-based measures. An example of a holding-based measure is the percentage of holdings by active managers relative to average turnover. An example of a return-based measure is to look at how the returns or residual returns of grouped stocks are correlated over time.	Chincarini (1998, 2012, 2017), Chincarini et al. (2018), Cahan and Luo (2013), Yan (2014), Chue (2015), Zhong et al. (2016), Baltas (2019), Brown et al. (2019), and Volpati et al. (2020).
22. Profitability	The profitability of a company. There are various measures of profitability that seem to generate excess returns.	Chincarini and Kim (2006), Novy-Marx (2013), Ball et al. (2015), Wahal (2018), Linnainma and Roberts (2018), Fama and French (2015, 2018), Chen et al. (2018), Cacecki et al. (2020), Bouchaud et al. (2019), Wahal (2019), and Asness et al. (2019).

Anomalies pose a significant challenge to the theory that markets are perfectly efficient. Proponents of efficiency might dismiss certain price irregularities as one-time aberrations in a generally efficient market. Anomalies cannot be dismissed so easily because they are patterns of recurrent irregularities. An anomaly suggests that investors habitually fail to consider and correctly interpret all the information relevant to the investment decision, or that institutional barriers prevent them from acting on certain information, or that, even with all the relevant information staring them in the face, they persist in making irrational choices. There are anomalies that contradict each definition of market efficiency from weak form to strong form.

Weak-Form Anomalies

Typical tests of weak-form market efficiency try to determine whether past prices can be used to predict future prices of individual stocks. For example, researchers may test whether there is significant autocorrelation in a stock's returns. *Autocorrelation* is the statistical term for correlation between returns from period to period. If stock returns are positively autocorrelated over time, a positive return in one period suggests a positive return in the next. An investor could profit from this pattern, which technical analysts call *momentum*, by buying stocks that performed well in the last period.

Researchers have found that there is autocorrelation in stock indices over a daily, weekly, and monthly horizon. One study found that between July 1962 and December 1994, the autocorrelation in a certain stock index was about 35%. Individual securities exhibited a slightly negative autocorrelation, but this autocorrelation was economically and statistically insignificant.[14] Other studies found significant positive autocorrelation in quarterly stock returns over horizons of less than one year.[15] However, for horizons of one week or one month, individual monthly stock returns seem to be negatively autocorrelated.[16] There is also evidence that industry returns may exhibit positive autocorrelation.[17]

There is mixed evidence on other technical indicators. Traders believe that they work, but tests of technical rules do not clearly confirm that belief. The standards for testing technical rules have been inconsistent. Many practitioners fail to include transactions costs in their tests, making excess returns appear greater than they actually would have been. Tests are also usually done on closing prices, which may be the best and most complete data available but are not necessarily the prices at which trades would have been placed. In general, tests cannot easily replicate a trader's discretion in applying technical strategies. A trader might have thought that a technical rule applied during one period but not during another. Technical trading is also not necessarily comparable with portfolio management. Some trading rules might work on intraday data, but they will not serve as portfolio strategies, which are supposed to work for months or years. Taking these limitations into account, the results of the tests of technical rules tend to fall on the side of weak-form efficiency because most technical rules do not seem to work consistently over time.

Semistrong-Form Anomalies

If technical data fail to provide reliable investment strategies, what about other publicly available information? There is a host of anomalies that provide evidence that market prices do not reflect all public information and that therefore the market is not semistrong-form efficient.

One well-known anomaly is the *earnings surprise anomaly*. Stocks that report higher-than-expected earnings tend to have excess risk-adjusted returns in the weeks following the announcement. What do we mean by *higher-than-expected earnings*? Wall Street stock analysts typically forecast the earnings they expect a company to report each quarter. The average of all analysts' earnings forecasts is known as the *consensus estimate* of earnings for that company. When a company's earnings clear the hurdle of the consensus estimate, they are higher than expected. The earnings surprise anomaly violates semistrong-form efficiency because some stocks continue to earn excess returns for weeks after earnings are reported to the public. A semistrong-form efficient market would incorporate the earnings report into the stock price immediately, but the real market responds more slowly. Studies show that stocks actually provide excess returns for 13 to 26 weeks after positive earnings surprises.

Another old and well-known anomaly is the *January effect*.[18] Small-cap stocks and the previous year's poor performers tend to do very well in January, especially in the first week. There are some logistical reasons for this phenomenon. One is that tax-loss harvesting by institutional and private investors involves selling poorly performing stocks in December to generate capital losses that offset capital gains and, to a lesser extent, income taxes.[19] The December sell-off depresses those stocks to prices below their actual nontax values, so investors might repurchase them once the new year rolls around.[20] The January effect is stronger when more investors engage in this tax-sensitive strategy.

Aside from tax harvesting, mutual fund managers also sell losing stocks at year end before they report their holdings to the Securities and Exchange Commission (SEC), in compliance with the required quarterly reporting. By selling a loser stock, the mutual fund manager does not have to report it as a holding and be scrutinized by investors who notice it on the SEC report. This tactic is known as *window dressing*. In the SEC report, investors will be able to see a mutual fund's overall low return, but window dressing prevents them from seeing each of the manager's bad picks.

It is also possible that the January effect occurs because many mutual fund managers "bet the house" early in the year. The annual bonus system at most funds might give managers an incentive to take big risks in January. They might think that if they lock in good returns at the beginning of the year, they will have gained a head start toward their bonuses; at the same time, if they lose a lot early on, they will have the rest of the year to grind their way back to a decent return.[21] Whatever the reasons for January buying, it is interesting to note that it is possible to see the forces of market efficiency at work on this anomaly. The January effect is gradually shifting backward as arbitrageurs and quantitative managers try to take advantage of it by placing trades ahead of time in December.

Other calendar effects exist as well. Mondays and the onset of daylight savings are among the calendar events associated with trading patterns. The *Monday effect* is that the market tends to drop slightly on Mondays. The *daylight savings effect* is that the market tends to drop slightly on the first trading day after the beginning of daylight savings, supposedly because the disruption of traders' circadian rhythms increases their aversion to risk.

There are many anomalies associated with fundamental data, which are the data obtained from a firm's income statements, balance sheets, and cash-flow statements. Several studies have shown that a portfolio of stocks with a low P/B ratio or a low price-to-earnings (P/E) ratio will earn high risk-adjusted returns. If a simple strategy of purchasing stocks based on publicly available P/B or P/E ratios can lead to excess returns, then the market is clearly not semistrong-form efficient. Small-cap stocks also have tended to earn higher risk-adjusted returns than large-cap stocks over long periods, even though large-caps might outperform small-caps in any given time period. Is there any rationale for this behavior? Some people argue that small-cap stocks are actually inherently riskier than large-caps because there is less information on them, and standard measurements of risk do not capture this discrepancy in the amount of available information. Small-cap outperformance may, however, be an instance of the *neglected-firm effect*, in which firms with low analyst coverage or low institutional ownership tend to have higher risk-adjusted returns. Whatever the explanation for it, the fact that small-caps do especially well over the long run seems to imply that the difference between the amount of public information on small-caps and large-caps is a kind of information itself, and this sort of second-degree information is not efficiently factored into stock prices.

In recent years, some new anomalies have been documented and some old anomalies have become in vogue again. In particular, the low volatility and low beta anomalies have become popular again. These anomalies find that stocks with low measured historical volatility outperform stocks with high measured historical volatility. Similarly, stocks with low beta tend to outperform stocks with high beta. More recently, factors regarding liquidity and crowding have surfaced. *Crowding* is a recently identified risk in investing that might occur when multiple market participants begin to follow the same trade in such concentration that liquidity becomes fragile, altering the risk and return dynamics of the trade. Research has found that there are excess returns to purchasing less crowded stocks.

There are many studies that point to additional violations of semistrong-form efficiency (see Table 2.2). Some academics still believe that these sorts of anomalies are not a sufficient basis for successful portfolio strategies. Professor Richard Roll of UCLA once wrote that "[i]t's extremely difficult to profit from the slightest deviation from market efficiency."[22] Quite a few active managers, however, make a living doing exactly that.

Strong-Form Anomalies

The strong form of market efficiency is probably the hardest kind to believe in because it means that market prices already reflect *all* information, both public and private. It is also hard to test for strong-form efficiency. Some clever tests have been done, though.

One test is known as the *insider-traders test*. Company executives and other insiders have, *ceteris paribus*, a better understanding of a company's operations and financial health than an outsider does.[23] Executives are required by the SEC to report their own purchases and sales of their companies' securities. Researchers have obtained these reports and asked the following question: If an outsider purchased a stock when insiders purchased it and sold it when insiders did, would the outsider have earned risk-adjusted excess returns? The answer is oftentimes yes. Insider information is not immediately priced into stocks, as it would be if the market were strong-form efficient.[24]

In the past, researchers were able to perform a test of the profits from specialists on the exchange. Stock exchange specialists were supposed to maintain an orderly market and manage their limit-order books. Access to the limit-order book was equivalent to having private information, because it contained the prices at which prospective buyers and sellers were prepared to buy or sell a stock. One could imagine that a specialist might be able to use this knowledge to his or her advantage, and, indeed, historically, specialist profits have been extraordinary. Was this due to the contrarian nature of their trades, which must balance the order flow, or was it due to the fact that they could make use of private information? If it was the latter, the specialists' profits of the past were another sign that markets were not strong-form efficient.[25]

Behavioral Explanations for the Anomalies

Traditional theories of investor behavior have a difficult time explaining market anomalies. Staunch believers in the efficiency of the market usually argue that any strategies that yield excess returns must be a great deal riskier than other strategies. Another argument for market efficiency is that any statistical arbitrage opportunities that do exist are extremely small, not scalable, fleeting, and very costly to discover.[26] These arguments are hard to reconcile with the many studies that describe long-term strategies that produce risk-adjusted excess returns. The seemingly small eddies of inefficiency in the market sometimes create major opportunities for statistical arbitrage. The risk involved in statistical arbitrage, including the possibility of misjudging how long an anomaly will persist, does not always negate the return. It may be necessary to turn from the CAPM to a multifactor model of stock returns in order to completely understand a less-than-efficient market.

Additional explanations for the existence of anomalies can be found in a field of economics known as *behavioral finance theory*. Researchers cannot fully explain the presence of anomalies with theories that presume that investors behave rationally. Behavioral finance theory attempts to account for anomalies with a study of the effect of psychology and irrational behaviors on investment decisions. Behavioral biases, like the ones listed in Table 2.3, contribute to the kind of buying and selling that leads to mispriced securities.[27]

TABLE 2.3

Common Behavioral Biases

Name	Description	Example
Anchoring	Anchoring bias is the tendency to rely too heavily on, or to anchor to, a past reference or single piece of information when making a decision. Anchoring bias can occur, for example, when investors base their future decisions on an initial set of information they began with—as when investors make future trade allocations based on their initial wealth many years ago. This allocation is no longer the correct allocation since their wealth is much greater now.	Anchoring bias can also occur when investors find a target price for buying a stock. The stock passes this price and the investor doesn't reevaluate the target price, thus never buying a possible long-term winner—missing out on the investment due to anchoring around their initial estimate.
Ambiguity aversion	An avoidance of unfamiliar stocks and a preference for familiar ones, even if some unfamiliar stocks are *great buys*.	Most investors invest the bulk of their money in their home countries, even though diversification would require owning more foreign stocks. In defined-contribution plans, many employees place much of their money in the stock of their own companies, which is terrible for diversification. Some studies have shown that investors place more money in stocks headlined in the newspaper, possibly because of their familiarity (this could also be due to the saliency effect).
Availability bias	The tendency to judge a future event more likely if there is a memory of a similar event, even though the memory may be distorted.	A trader may think that the stock market rallies every time a war starts, even though he has only experienced this once.
Confirmation bias	The tendency to give greater weight to information that confirms one's original beliefs and less weight to information that contradicts those beliefs.	Analysts who originally recommended a stock might be more likely to upgrade it on good news than to downgrade it on bad news. Wall Street analysts often follow stocks that they like, and confirmation bias may

		explain why the majority of stock ratings by Wall Street analysts are typically quite positive (either "buys" or "strong buys").
Disposition effect	Holding on to *losers* (stocks with negative returns) too long and selling *winners* too soon. The psychological explanation is that humans fear losses more than they value gains. Holding on to a poor stock lets an investor avoid realizing paper losses and makes it easier to pretend that the stock was not a bad pick.	A foreign exchange trader one of the authors knew used to call the disposition effect, "Hope, Mope, and Dope." Basically, you make a stock purchase and, when it falls, you first hope it will go back up. Then you mope around as it continues going south. Finally, when it really goes down, you've become the dope. Although different from disposition, investors also have inertia bias, which prevents them from taking appropriate actions.
Endowment effect	The idea that people will demand more to give up an object than they would be willing to pay to acquire it.	This could lead to investors preferring stocks that they already own above and beyond the reasonable price, which leads to other attractive stocks remaining at prices below their reasonable value. Eventually, these mispricings will adjust as the inertia is overcome. Can also lead to less efficient after-tax returns by investors who avoid tax harvesting because of their reluctance to sell stocks they already have.
Escalation bias	The tendency to put more money in an investment even when the original decision to buy it was based on bad analysis, thus compounding any potential losses. Also known as *commitment bias*.	Suppose an investor purchases a stock at $40 and tells everyone what a great buy it is. The stock then falls to $30. There are two ways to look at the situation. One reaction is, "I made a poor decision." The other is, "If it's a buy at 40, it's a steal at 30!" Usually you run into the second kind of thinking. The investor therefore purchases more of the stock

		purchases more of the stock, which was a bad investment to begin with.
Herding mentality	The idea that people like to follow what others are doing. This applies to many aspects of life but also to investing.	Herding may lead to asset prices deviating from fundamental values. For example, the inelastic market hypothesis predicts that flows into an asset class can lead to a much larger move in valuation than might be justified. Herding can also lead to a phenomenon known as *crowding*. Crowding occurs when lots of investors have similar trades or de facto similar trades that are squeezing the liquidity of a space. This lack of liquidity can lead to a dramatic collapse in prices and mismeasurement of the risk of an investment.
Illusion of knowledge	The assumption that having more data on a stock automatically means having a better understanding of the stock.	This may be one reason why studies have found that investors who trade online trade more often. The massive amounts of information provided by brokerage firms to investors feed an illusion of certainty. Or take the example of the game of roulette. A player chooses a number to bet on the roulette wheel. Suppose you then give the player a piece of paper with the results of the last 1,000 spins of the roulette wheel. His number has shown up frequently in recent spins, and this makes him more confident about his bet. This is only an illusion of knowledge, though, since every outcome is independent and his chance of winning is the same no matter what.
Narrow framing	Considering individual gambles or investments in isolation from other, related investments. Also sometimes referred to	An investor might shy away from stocks that experience big short-term fluctuations even though those stocks could, if they

	as mental *accounting* or as individuals having different *mental pockets.*	are negatively correlated with the rest of the investor's portfolio, reduce the overall portfolio's long-term level of volatility.
Overconfidence	Humans tend to be overconfident. Researchers have documented that the average investor is overconfident in his own ability, causing him to trade too often, incur more transactions costs, and cut into returns. Overconfidence may come from *self-attribution bias*, which is the tendency to attribute one's successes to one's own talent but consider one's failures the fault of other people or circumstances. Overconfidence may also relate to *hindsight bias*, which is what persuades people that they knew in advance how a decision was going to turn out.	One example of overconfidence is demonstrated in many classrooms. The professor asks the class, "How many people believe they are better at driving than the average driver in this room?" Typically, more than 50% of the students will raise their hand, which is a clear impossibility. Only 50% can be better than the average in the room. Self-attribution bias probably had a hand in the internet bubble. In the late 1990s, investors believed they were genius stock pickers as they watched their stocks soar. After the bubble burst, they blamed their losses on a terrible economy or a world that had gone mad.
Probability distortion	Individuals' perceived probabilities might differ from actual objective probabilities. This is because, as psychologists observe, individuals exhibit diminishing sensitivity to actual probabilities. As individuals move away from so-called reference points, such as sure bets (i.e., probability equals 1) and sure losses (i.e., probability equals 0), they are less sensitive to changes in this probability. For example, an individual reacts to a change in probability of 1 to 0.95 much more than a reduction in probability from 0.55 to 0.50. In addition to this, psychological experiments found that individuals tend to overweight low-probability events and underweight high-probability events.	This probability distortion might lead to overestimating a bad situation. For example, from March 2 to March 23, 2020, the S&P 500 declined by 28%, only to rebound strongly throughout the remainder of the year. This may have been rational due to extreme uncertainty over the Covid-19 virus, but some of it may also have been investors' overweighting the probability of a really bad scenario versus the actual likelihood of a bad scenario. In general, this may lead to bouts of overreaction to bad news, which could present a profitable investment opportunity.

	probability events.	
Recency effect	Investors might be overly influenced by recent events.	For example, if there has been a huge drop in the stock market as in 2000 or 2008, certain investors may shy away from the market for a while. As the market slowly does better, a slow trend of these investors returning eventually adds to the momentum. This might also be called the *snake-bite effect* when the recent experiences are negative. It might also lead to positive momentum effects when recent experience is positive.
Regret aversion	Regret aversion or regret theory is related to an anticipated regret that an individual may consider when facing a decision. Individuals might anticipate regret and incorporate in their choices their desire to eliminate or reduce this possibility. Regret is a negative emotion with a powerful social and reputational component. Researchers have shown that it might explain rationally various behaviors of humans.	There are many potential examples of regret aversion in the financial markets. One example is the investor who buys an overvalued stock because he does not want to later be upset with himself for not buying a winner like Tesla or Amazon. This could push the stock even higher. Another example is investors who have recently witnessed a market downturn and are very reluctant to invest newly acquired cash; as a result, they hold off, thus missing years of gains in the market.
Representativeness	Oversimplification of investment decisions using rules of thumb. Investors try to come up with easy tests for *good* and *bad* stocks. Most of the time, rules of thumb are helpful in summarizing a lot of complicated information, but taken by themselves, they can also lead to irrational decisions.	Investors might pour money into stocks that have recently had good earnings announcements simply because that is said to be the sign of a good company. However, consideration should also be given to whether the market expected the earnings, in which case the stocks' prices should already reflect the news. This may be the cause of an overreaction of stock prices to recent earnings announcements.
Saliency	Salience refers to any aspect of	For example, investors

Saliency	Salience refers to any aspect of a stimulus that, for any of many reasons, stands out from the rest. Investors might thus give more attention to higher-stimulus items, while neglecting lower-stimulus items. Also related to *vividness* in the psychological literature.	For example, investors might be more attracted to attention-grabbing stocks (e.g., stocks in the news). This could lead to an overvaluation of stocks that are grabbing everyone's attention, leaving opportunities for less glamorous stocks.

Note: Although there are many papers on these particular behavioral biases, we try to cite just the early studies in these areas. Anchoring [Kahneman and Tversky (1981)], ambiguity aversion [Ellsberg (1961)], availability bias [Kahneman and Tversky (1973)], confirmation bias [Bacon (1620), Wason (1960), Lord et al. (1979)], disposition effect [Shefrin and Statman (1985)], endowment effect [Kahneman and Tversky (1979), Thaler (1980)], escalation bias [Staw (1976)], herding mentality [Keynes (1936), Asch (1955)], crowding [Chincarini (1998, 2012)], [Stein (2009)], Illusion of Knowledge [Oskamp (1965)], narrow framing [Kahneman and Tversky (1981)], overconfidence [Marks (1951), Irwin (1953), Oskamp (1965), Fischoff and Slovic (1978), Odean (1998)] and for the subset of hindsight bias [Miller and Ross (1975) and Fischoff (1977)], probability distortion [Preston and Barrata (1948)], recency effect [Ebbinghaus (1902), Crowder (1976), and Roediger and Crowder (1976)], regret aversion [Loones and Sugden (1982) and Bell (1982)], representativeness [Kahneman and Tversky (1973)], and saliency [Kanouse (1972), Taylor and Fiske (1975), Nisbett et al. (1977), Nisbett and Ross (1980), and Bordalo et al. (2012)]. There is disagreement on whether the endowment effect truly exists or not and whether the phenomenon reflects patterns of decision mistakes that go away with experience and careful analysis [Plott and Zeiler (2007)]. In a survey of behavioral biases that plague investment advisors in 2020, recency bias and loss aversion seemed to affect people the most [Smith (2020)].

2.4.3 Market Efficiency and QEPM

Anomalies and behavioral biases give fairly strong evidence that markets may not be much more than weak-form or only sporadically semistrong-form efficient. Summarizing the evidence, here are our top 10 reasons why markets are not perfectly efficient and why active portfolio management such as QEPM is a worthwhile endeavor:

1. Obtaining information is costly. Not everyone is able or willing to pay for information.
2. Information, even public information, travels somewhat slowly through the market.
3. Not every investor has the ability to process a large amount of information, especially quantitative information.
4. By filtering public information, some people may create what amounts to private information.
5. Some investors base their investment decisions on sentiment rather than on the logical interpretation of information.

6. Some attempts to exploit others' presumed irrationality actually creates more inefficiency.
7. Economic conditions, especially the state of technology, change all the time, and it takes time for people to adapt to these changes.
8. Transactions costs create gaps between economic models and reality.
9. Taxes cause distortions in the markets.
10. Government regulation of financial markets creates gaps between economic models and reality.

Efficiency means that the market incorporates all relevant information into security prices, and this happens only when all investors have all the information relevant to investment decisions. In reality, investors operate with varying sets of information partly because accessing it is costly. Portfolio managers have access to extensive market databases paid for by their firms. Even with the introduction of low-cost databases tailored for individual investors, the majority of them simply cannot replicate the volume and functionality of commercial databases that contain historical and up-to-the-minute data.

Since people's access and exposure to information vary, information moves through the market slowly and unevenly. Private information obviously stays within a fairly small radius, but even public information travels slowly and stops before reaching everyone. Many people rarely come in personal contact with the markets, do not keep track of market news, and never receive some information. Also, private information usually becomes public eventually, but often in a delayed and uneven fashion. This is especially the case with bad corporate news, which companies sometimes release in bits and pieces in order to avoid dropping bombs on their own stock prices.

Once information is publicly accessible, quantitative portfolio managers are in the best position to make use of it. They are alerted to news almost immediately via electronic data services, whereas average investors may not learn about it until later from the TV or newspaper headlines. Compared with other professional managers with access to data services, quantitative managers also have a great advantage because they make use of software and quantitative methods that sort through large data areas relatively quickly, and their stock-return models can be updated accordingly. The quantitative model of stock returns will warn the manager in as little time as it takes for the computer to run the program whether any news warrants selling stock.

By filtering public information with quantitative analysis and proprietary models, quantitative portfolio managers gain insights and uncover strategies that are essentially private information derived from public inputs. Since most investors cannot uncover the same things, this information is not yet priced into securities, and there is the potential for statistical arbitrage. Even if a number of portfolio managers use similar models, they still can earn excess returns as long as they are relatively few compared with the entire population of investors.

Investors' biases also create statistical arbitrage opportunities because they cause mispricings. People inevitably let irrational hopes and fears creep into investment decisions, which causes them to ignore or misjudge certain relevant information. Quantitative equity portfolio managers are in a good position to both avoid and exploit irrational decisions. Quantitative models of stock returns help the manager to take his or her own emotions out of the investment decision while also uncovering irrational price movements that are vulnerable to statistical arbitrage.

Statistical arbitrage usually helps to return security prices to their efficient levels. Sometimes, though, it actually exacerbates market inefficiencies. For example, during the internet bubble of the late 1990s, many stocks were overvalued. Amazon (AMZN) was in the same line of business as Barnes & Noble (BKS), but it traded at a much higher multiple. If someone assumed that the higher multiple was a symptom of a short-lived overvaluation, then one could have gone long BKS and short AMZN. In contrast, if one assumed that the internet bubble was going to get bigger, one could have gone long AMZN and short BKS, thereby compounding the overvaluation.[28]

The internet bubble also was a case in which the market did not appreciate or understand a change in economic conditions. Sometimes information is not priced into securities because it is not yet understood or even recognized. It takes time for even the best analysts, economists, and strategists to understand the real significance of changes in the economic environment. During the learning period, markets most likely will not factor the changes into securities prices correctly. During the late 1990s, for instance, investors' zeal over the prospects of new e-commerce business models outstripped the actual growth potential of many startups.

The process of buying and selling in the market is itself a source of inefficiency. Transactions costs divert money away from its most efficient use. While an economic model may say what prices should be in equilibrium, actual prices may be quite different because of transactions costs. These costs can take the form of *broker commissions, price impact*, and *delay*. Mispricings due to transactions costs may not be exploitable by any investors, or they may be exploitable only by those who can keep their own transactions costs low. For instance, commissions and delays prevent many small investors from trading as frequently as professional traders do and from pursuing strategies of scale that require large funds.

Taxes are another significant cost and source of inefficiency in investing. Investors are ultimately concerned with after-tax returns, so tax consequences weigh on investment decisions. The distribution of investors' tax rates, as well as any changes in that distribution as a result of changes in the tax laws, affects the relative prices of securities. If even a few large institutions buy or sell stock for tax reasons, prices can shift away from their true nontax values. At times, the price shifts create exploitable market inefficiencies.

Market inefficiencies can be the by-products of other government regulations as well. Some countries with fixed exchange rates have experienced fluctuations in which the exchange rate suddenly dropped by as much as 40% of its value in a matter of hours. The drastic rate corrections have cost foreign investors enormous amounts of money. Clearly, such fixed exchange rates did not price the currencies according to all the information available in the marketplace. Government-imposed systems also create less dramatic distortions. In the case of equities, numerous government regulations alter the flow of investment, including restrictions on short selling, rules related to tick increments (i.e., decimal pricing), and rules related to stock ownership.[29] Although the intent of the regulations is presumably to ensure some public good, they still cause distortions that may be exploited for excess returns.

These are the essential reasons why markets are not perfectly efficient, which is to the benefit of quantitative equity portfolio managers. Why haven't QEPM managers and other arbitrageurs already eliminated inefficiencies from the market? There are two simple answers to this question. The first is that inefficiencies lead not to pure arbitrage opportunities but rather to statistical arbitrage opportunities that are low but not zero risk (as stated in tenets 2 and 3). The second answer is that there are simply not enough arbitrageurs in the marketplace, with enough investing power relative to the rest of the investing public, to trade away every inefficiency. The markets remain an abundant source of potentially profitable mispricings for the active manager.

2.5 TENETS 3 AND 4: THE FUNDAMENTAL LAW, THE INFORMATION CRITERION, AND QEPM

The portfolio manager has to apply good quantitative analysis to market data to find and exploit the opportunities for excess return that are hidden in market inefficiencies. Tenets 3 and 4 of QEPM state that quantitative analysis opens up the possibility of statistical arbitrage so long as the methods and models that are used combine all the available information efficiently. These two tenets are best illustrated within the framework of the *fundamental law of active management*.[30] The fundamental law has gained popularity among portfolio managers. Through a simple formulation, it shows the portfolio manager's contribution to the portfolio management process. We know from Section 2.2.6 that a high information ratio is one of the goals of QEPM, and the fundamental law helps us to understand how to achieve it through the application of statistics and the efficient, full use of information.

The fundamental law states that the *information ratio* (*IR*) is the product of the *information coefficient* (*IC*) and the square root of *breadth* (*BR*), that is,

$$IR = IC \cdot \sqrt{BR} \tag{2.8}$$

Given the definition of the information ratio in Eq. (2.7),

$$\frac{\alpha^B}{\omega} = IC \cdot \sqrt{BR} \tag{2.9}$$

This equation shows that a higher information ratio can be achieved by increasing the information coefficient or by increasing the breadth. For quantitative managers who build stock-return models, the *IC* can be increased by finding factors that are more significant than the ones already in the model, and the *BR* can be increased by finding more factors that are relatively uncorrelated with the ones already in the model.

The fundamental law was first introduced as a way to gain insights into the quantitative portfolio management process, but its nature and use are sometimes misunderstood. We use the results of standard econometrics to accurately quantify and make clear statements about the components of the law, which will be especially useful to quantitative managers who use linear stock-return models to build their portfolios. We stress, however, that the fundamental law is for *understanding* QEPM, not for *doing* QEPM. A quantitative portfolio manager does not use the law to actually create portfolios. Rather, the law provides the manager with a useful conceptual framework for analyzing the QEPM process.

2.5.1 The Truth about the Fundamental Law

One of the essential tasks of QEPM is to predict future stock returns by estimating a model that specifies a relationship between the stock return and a list of explanatory variables (a.k.a. *factors*). Suppose that the model specifies that the return of stock i at time t, r_{it}, is a linear function of the value of K factor premiums at time t, that is, $f_{1t}, \ldots, f_{Kt}$:

$$r_{it} = \alpha_i + \beta_{i1} f_{1t} + \cdots + \beta_{iK} f_{Kt} + \epsilon_{it} \tag{2.10}$$

where α_i, $\beta_{i1}, \ldots, \beta_{iK}$ are parameters to be estimated, and ϵ_{it} is the random-error term (i.e., the deviation of the stock return from its expected value). Assume that the portfolio manager estimates the preceding equation using data from time 1 to T, that is, $t = 1, \ldots, T$.

The fundamental law assesses how well Eq. (2.10) explains the stock-return process, and it expresses the equation's goodness of fit as the product of the *number of explanatory variables* and each variable's *average contribution*. Depending on what portfolio managers do after estimating Eq. (2.10), the fundamental law may be expressed in different ways, but several truths always hold[31]:

1. IR^2 approximately equals the goodness of fit (R^2) of the return forecasting equations.[32]
2. The breadth is the number of explanatory variables in the return forecasting equations.[33]
3. IC^2 is the average contribution of each explanatory variable in increasing R^2.
4. The fundamental law decomposes the goodness of fit into the number of explanatory variables and their average contribution.
5. When the benchmark is ignored and the risk-free rate is subtracted from the portfolio returns, IR is essentially the maximum Sharpe ratio one can achieve, and the fundamental law decomposes the maximum Sharpe ratio into the number of explanatory variables and their average contribution.

2.5.2 The Information Criterion

One misconception about the fundamental law is that it applies to all active portfolio management. *The fundamental law applies only when the portfolio manager creates the optimal portfolio*. While it may be surprising, many portfolio managers unknowingly create suboptimal portfolios often because they do not use all the available information in the most efficient way (thereby violating tenet 4). When constructing the portfolio, a manager should use all the information gathered from the estimation of Eq. (2.10). This important point has not been emphasized enough by proponents of the fundamental law, so we want to state it as a general principle.

LEMMA 1 (THE INFORMATION CRITERION). *The fundamental law of active portfolio management as expressed in Eq. (2.8) is valid only if the portfolio manager combines all the available information in the most efficient way*.

An example helps to illustrate the preceding criterion. A portfolio manager may follow the analyst-revision strategy: Construct an equal-weighted portfolio of stocks whose analyst ratings improved in the last month. We can consider "inclusion in the portfolio" a variable. That is, we may define a variable β_{it} that has a value of 1 if stock i's rating improved at time t and a value of 0 otherwise. The manager estimates the following equation from historical data:

$$r_{i,t+1} = \alpha + \beta_{it} f + \epsilon_{i,t+1} \tag{2.11}$$

where $r_{i,t+1}$ is the return to stock i at time t+1, α and f are the parameters to be estimated, and $\epsilon_{i,t+1}$ is the error (the part of the return that is not explained by the model). If the value of f is positive, the equation suggests that those stocks with $\beta_{it} = 1$ will have higher future returns. Thus the portfolio manager constructs an equal-weighted portfolio of stocks for which $\beta_{it} = 1$, indicating that their ratings improved in the last month.

In doing this, however, the portfolio manager is not using all the available information. Equation (2.11) not only says that those stocks with $\beta_{it} = 1$ have a higher expected return, but it also identifies which stocks have higher risk (volatility). Stocks with higher volatility should have smaller weights in the eventual portfolio. Since the portfolio manager did not use this piece of information, the information criterion is not satisfied.

We believe that some of the current industry practices do not pass the information criterion. We will encounter such examples in the following chapters, and we will discuss exactly what aspects of the practices violate the information criterion in each case. The bottom line is that if the portfolio construction strategy does not satisfy the information criterion, then there must be a better way to construct the portfolio.[34]

2.5.3 Information Loss

The loss that results from not using all information can be quantified easily within the framework of the fundamental law. The fundamental law suggests that the contribution of the portfolio manager (and, by implication, of the information that the portfolio manager uses to increase the portfolio's return) is summarized by the information ratio. Similarly, the lack of contribution of the portfolio manager is also reflected in the information ratio. By comparing the information ratio that the portfolio manager could have achieved using all available information with the information ratio that he or she actually obtained using only a subset of the available information, we can quantify the amount of information he or she lost, known simply as the *information loss* (IL). Thus

$$IL = \text{maximum } IR - \text{actual } IR \tag{2.12}$$

To the extent that the information ratio is similar to the maximum Sharpe ratio, the information ratio can be understood as the reward-to-risk ratio. If the information ratio is 0.5, this means that taking on 10% more risk will result in 5% extra return. On the flip side, the information loss shows the reduction in the reward-to-risk ratio. If the information loss is 0.1, then the portfolio manager is missing out on 1% of potential extra return for every 10% of risk he or she takes.

2.6 TENETS 5, 6, AND 7: STATISTICAL ISSUES IN QEPM

Tenets 5, 6, and 7 of QEPM concern issues arising from the application of statistical techniques to the portfolio-choice problem. In this section we discuss the three issues that a portfolio manager cannot neglect: *data mining, parameter stability*, and *parameter uncertainty*.

2.6.1 Data Mining

One of the biggest challenges for quantitative portfolio managers is to avoid the problem of data mining. Data mining violates tenet 5, which requires that quantitative models be based on sound economic theory. In some quantitative analyses, the use of data mining is quite obvious, but in other cases it is harder to detect.

Data mining is the practice of running regressions of historical stock returns on so many combinations of factors that one is virtually guaranteed of finding a model or handful of factors that seem significantly correlated with stock returns but are in fact not particularly meaningful. Suppose that there are 99 potential factors $f_1, \ldots, f_{99}$ that may explain a stock's return. Let $f_{1t}, \ldots, f_{99t}$ and r_t be the values of the factors and the stock's return for month t. Suppose that we observe the value of these factors and the stock's return for 100 months, and then we regress the stock return on all 99 factors. That is, we estimate the following equation:

$$r_t = \alpha + \beta_1 f_{1t} + \cdots + \beta_{99} f_{99t} + \epsilon_t \qquad t = 1, \ldots, 100 \qquad (2.13)$$

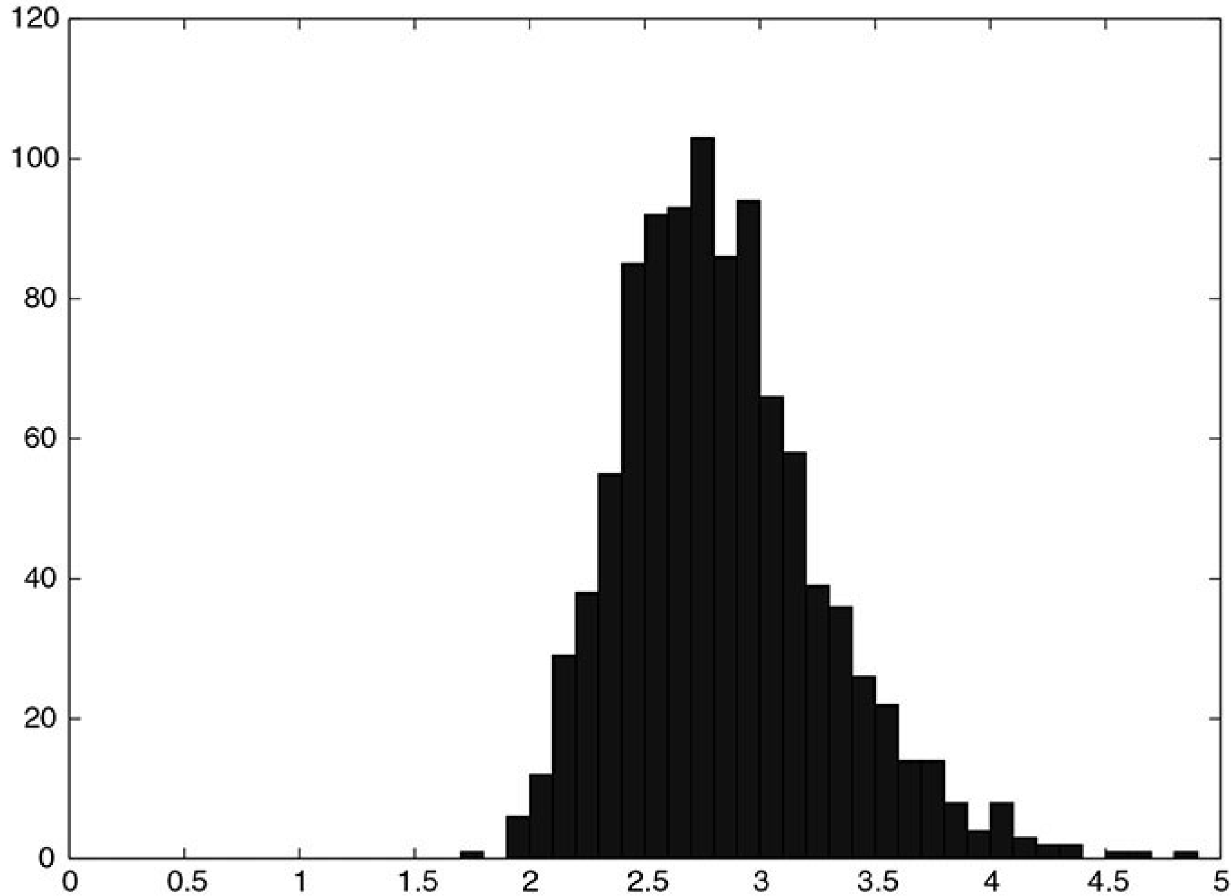
What will we find out from the estimation? In this case, the goodness of fit (R^2) of the regression will be 100% because there will be no error in the regression (i.e., $\epsilon_t = 0$). With this equation, we superficially "explain" the stock return completely.[35] In truth, though, the R^2 of 100% simply reflects the fact that we included too many variables in the regression. From this regression, we cannot make any statistical inference about whether the model is valid or not. Even if we had made up the values of factors for the 100 months using a random-number generator, the regression still would have assigned coefficients so that R^2 would equal exactly 100%. In fact, when there are 100 observations and more than 99 explanatory variables, the R^2 always will be 100%. Basic statistics tells us not to include 99 variables when there are 100 observations.

Sometimes the problem is not so obvious. Suppose that we do a so-called stepwise regression. That is, we deliberately find the most significant variable out of 100 variables. Can we make statistical inferences from such a regression? No. Imagine again that we made up the value of factors using a random-number generator. If we deliberately look for the most significant of the 100 variables, we are guaranteed to get a significant variable because we have such a large pool of variables from which to choose. We set up the regression so that it would generate a significant variable, knowing in advance that whatever variable we eventually selected would be significant. The statistical significance of the variable is therefore no real indication that the variable explains stock returns well.

To illustrate the concept, Fig. 2.1 shows the frequency distribution of the absolute value of *t*-statistics when we deliberately choose the most significant variable out of 100 randomly generated variables. For each simulation, 100 observations of the dependent variable and 100 explanatory variables were generated from the standard normal distribution independently. Given 100 explanatory variables, we selected the most significant explanatory variable by running regressions of the dependent variable on the selected explanatory variable and a constant. The simulation was repeated 1,000 times.

FIGURE 2.1

Frequency distribution of the absolute values of *t*-statistics from 1,000 simulations.

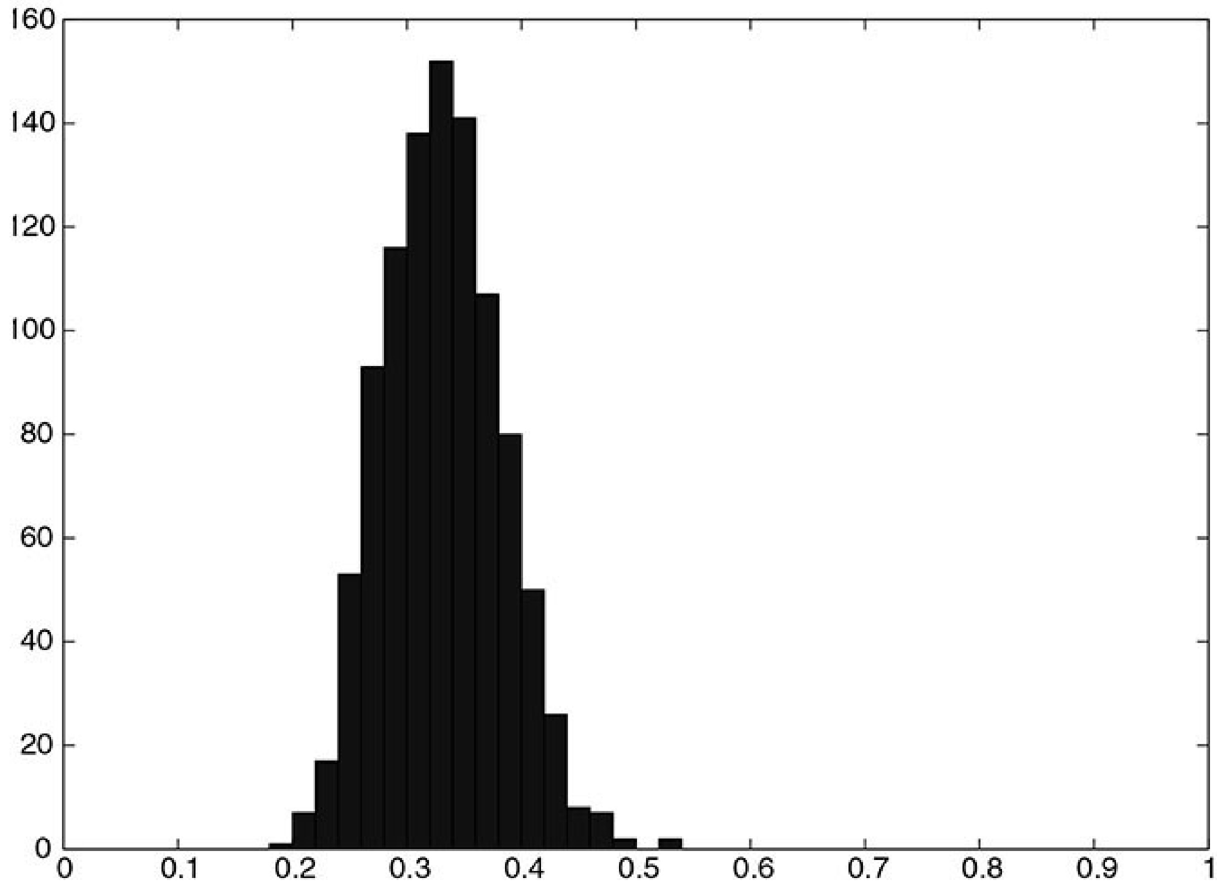


As can be seen from the figure, in most cases the absolute value of the *t*-statistic is greater than 2. The standard interpretation of this high *t*-statistic is that the selected variable is significant. But we know from the design of the simulation that the selected explanatory variable does not have anything to do with the dependent variable. The R^2 of the regression will be high simply because the *t*-statistic is high, so the R^2 also will be meaningless. For each simulation just described, we selected the 10 most significant explanatory variables and ran a regression of the dependent variable on the selected variables and a constant. The frequency distribution of the resulting R^2 is plotted in Fig. 2.2. R^2 is mostly around 30% and 40%, quite high values, but again, the high R^2 does not suggest true statistical significance.

The outcomes of stepwise regressions are, in a sense, rigged, and this is not the kind of research a portfolio manager ought to perform.

FIGURE 2.2

Frequency distribution of R^2 from 1,000 simulations.



Even avoiding stepwise regression in one's own analysis, statistical conclusions still may be tainted indirectly by data mining. Even when an individual researcher does not select variables deliberately, the community of researchers may be doing a stepwise regression collectively. For example, in the 1970s, an MBA student in the business school of the University of Chicago found that the size (i.e., market capitalization) of a stock explained its return. Let's say that the student came to this conclusion through valid statistical methods. We know, however, that there must have been thousands of other MBA students (not to mention hundreds of academics and practitioners) who have tried to explain stock returns with other variables. The fact that one student's conclusion about stock size gained widespread acceptance, whereas the conclusions of other studies did not, suggests that the community of financial researchers as a whole probably has been engaging in a kind of collective data mining referred to as *data snooping*. One therefore cannot accept in full faith the finding that the size of the stock is a significant explanatory variable.

We cannot completely avoid data mining or data snooping when we test a factor unless we use data that have never before been used to test that particular factor. When we test a factor with the same data that many other people have already used to test it, our conclusions may be influenced by their conclusions. For instance, authors of numerous investment textbooks have written about tests that they and others performed using U.S. stock data from the S&P Compustat database. In fact, many of these tests analyzed the same factors with the same stock data over the same time period. The P/E ratio in particular was one of the factors studied quite extensively. Thus, when we read in some textbooks that the P/E ratio explains stock returns well, we know, even before actually running any of our own regressions, that the P/E ratio will be a good explanatory variable for much of the available data on U.S. stock returns. We cannot make any statistical inference from our own regression of returns on the P/E ratio unless we use data that have not already been extensively tested.

These problems suggest that careful empirical study is not enough to avoid the data-mining problem. The best way to generate meaningful statistics is to follow tenet 5 and make sure that the model is based on sound economic theory and common sense. Empirical evidence alone is not a sufficient basis for good practical decisions. A model will work only if there is a good reason to think that its factors explain stock returns and that the relationship between those factors and stock prices has not been exploited already by other investors.

In addition to letting theory guide the portfolio manager and using out-of-sample data, there are other methods to mitigate the data-mining problem.[36] In particular, portfolio managers, guided by the past research that has been done on factor anomalies, should attempt to control for data snooping through one of two methods: using statistical techniques like a Bayesian adjustment about a factor or investment strategy or adjusting the *p*-values or *t*-statistics of the significance of tests by the number of tests they are performing—and even the number of tests that may have occurred prior to their own testing and that might have influenced their beliefs. In the case of the Bayesian methods, they should adjust the significance of the empirical tests by their a priori belief about the importance of the factor (that is, theory leading). In the case of the p-value or t-statistic adjustments, the methods increase the critical values by which a factor should be accepted due to the repeated testing of many factors.[37]

2.6.2 Parameter Stability

To make forecasts based on past data, we have to assume that history repeats itself. If the historical average of the stock return is around 1%, then we may assume that the stock return will be around 1% next month. If the standard deviation of the stock return is historically around 10%, then we may expect it to be around 10% next month.

Unfortunately, the constant flux of the market often interrupts the historical patterns that we might count on continuing. CEOs, employees, products, market conditions, and regulations change. As companies change, the properties of stocks also change. When the properties of stocks change, the parameters of stock-return models change. Thus, if we estimate β from the CAPM for, say, General Electric, then we have to ask ourselves, "Is the β of General Electric going to be the same next month? Next quarter? Next year?" If we estimate a more complicated model, we have to determine whether the estimation will be stable over time.

Consideration of parameter stability plays an important role in determining data sample size. For example, if we believe that β is generally not constant for more than five years, we should not estimate it from a sample that includes data covering more than five years. Or knowing that Daimler and Chrysler merged, we should not create a sample that includes both premerger data and postmerger data. Understanding parameter stability is crucial to QEPM. In Chapter 16 we will discuss some simple ways to ensure parameter stability.

In the last decade, the threat to factor return stability has increased, and the risks associated with quantitative investing have also increased. The reason is partly due to the phenomenon of crowding.[38] Crowding is a recently identified risk in investing that can occur when multiple market participants begin to follow the same trade in such concentration that liquidity becomes fragile, altering the risk and return dynamics of the trade. That is, as portfolio managers and investors discover a new factor, the popularity of the strategy can actually reduce the future performance of the strategy. When quantitative strategies or factor strategies become public through publication or through the launch of an investment product, the future risk-adjusted returns of that strategy decrease by a large magnitude. Thus, the measured performance might not be a stable estimate of the future performance.[39] In order to assess future strategy performance, portfolio managers should make sure historical backtesting is realistic (i.e., consider trading costs), be aware of crowding, and be aware of how the nature of the factors themselves is changing over time.

2.6.3 Parameter Uncertainty

Having extra information never hurts and is often helpful. However, extra information should not necessarily alter the portfolio composition. A portfolio manager may be able to construct a portfolio that is expected to outperform the benchmark. Does this mean that he or she should automatically choose the active portfolio instead of an indexed one? No. The active portfolio may have too much risk compared with the gain in expected return. Whether the risk is “too much” depends on an individual portfolio manager’s attitude, but the amount of risk can be checked in principle.

Even if the active portfolio does not look too risky on its surface, it could have a high degree of parameter uncertainty, which is a component of risk. Standard statistics overlooks this important point. Parameter uncertainty exists in all statistical estimations of stock-return models because all estimations contain some degree of estimation error, as measured by the standard error. Mean-variance optimizations therefore should take standard error into account as one aspect of a portfolio’s risk. When the standard error is considered, it may become apparent that a supposedly superior active portfolio’s risk-adjusted return is, in fact, no better than the benchmark’s.

The standard error, which measures the error of the estimation, depends on two things: the variance of the error in the model and the variance of the explanatory variables.[40] The variance of the error shows how precise the model is. If the model is 100% precise, the error always will be zero. The variance of the explanatory variables shows how informative the data are. If the variables do not change at all, their variance is zero, and the standard error will be infinite. Therefore, to reduce the standard error, the model and the explanatory variables should be selected carefully.

In general, the portfolio manager should consider not only the estimated value but also the standard error of the estimates. If the standard error is large, then that should be considered part of the portfolio's risk. This is one of the fundamental implications of Bayesian econometrics for portfolio selection. Bayesian econometrics is gaining acceptance by industry practitioners, and we will discuss the Bayesian approach in Chapter 14.

2.7 CONCLUSION

We started this chapter with a discussion of α and the information ratio, the concepts of risk-adjusted excess return, because the goal of QEPM is to create portfolios with high levels of return for risk. We then looked in detail at the seven tenets of QEPM that all quantitative equity portfolio managers should follow. The seven tenets deal with concepts of market efficiency, the fundamental law of active management, and statistical issues. As a form of active management, QEPM is based on the belief that markets, despite functioning efficiently in general, contain many patterns of inefficiency that open up opportunities for statistical arbitrage. An understanding of the information criterion and the concept of information loss from the fundamental law helps portfolio managers to exploit these opportunities to the fullest. Successful QEPM also depends on managers being vigilant about data mining, parameter uncertainty, and parameter stability.

Now that we have a firm grasp of QEPM concepts, we turn to the key piece of QEPM, the quantitative model of stock returns, which brings together theory and observation into one tool for selecting stocks for the portfolio.

QUESTIONS

2.1. (a) What are the three types of α used in QEPM, and how are they different?
(b) What is another type of α commonly used by the general investing public, and why is it less relevant to QEPM?

2.2. (a) When does $\alpha^B = \alpha^{\text{CAPM}}$?

(b) Can α^{MF} ever be the same as the other types of α?

2.3. (a) What distinguishes ex-ante α from ex-post α?

(b) Ideally, what relationship would a portfolio manager want between them?

2.4. (a) Define the information ratio.

(b) Why is it important in QEPM?

2.5. Name the seven tenets of QEPM.

2.6. Explain the difference between a pure arbitrage and a statistical arbitrage.

2.7. Name the three types of market efficiency.

2.8. Define weak-form market efficiency.

2.9. Define semistrong-form market efficiency.

2.10. Define strong-form market efficiency.

2.11. Give an example of a type of quantitative analysis that would not work for each of the types of market efficiency.

2.12. Name three documented anomalies. Can you think of a theoretical or behavioral justification for these anomalies? If so, explain.

2.13. Name the three anomalies that are most likely to be subject to data mining or data snooping. Explain.

2.14. What is ambiguity aversion? Give an investment example.

2.15. What is the disposition effect?

2.16. It has been documented that trading volume decreases during bear markets. Can you explain this phenomenon with common behavioral biases?

2.17. Give three reasons why practitioners of QEPM might believe that markets are inefficient.

2.18. A special case of the fundamental law is related to the idea that independent forecasts with informational value can improve the power of the overall forecasts. Suppose that you and your QEPM department decided to forecast the equity markets every month. The choices are market up, market down, or market flat. You collected the forecasts of every participant and then constructed the QEPM department's consensus forecast as follows. The aggregate forecast is the majority. Thus, if there are more than $n/2$ "long" signals, the combined forecast goes "long"; if there are fewer than $n/2$, the combined forecast goes "short"; and if there are exactly $n/2$, the combined forecast is determined by a coin flip. Suppose furthermore that you make the assumption that everyone's opinions are independent of each other.

(a) Under what conditions would the QEPM department's consensus forecast be better than any individual forecast?

(b) Suppose that everyone had the same probability of being right, p. Consider the following individual probabilities: 0.5, 0.53, 0.60, 0.65, 0.70, and 0.75. What would be the corresponding probabilities of the QEPM department's consensus being right, assuming that there are 22 participants in the consensus?

2.19. The fundamental law makes a number of approximations. One of them is

$$\frac{R^2}{12\,R^2} \approx R^2$$

(a) Calculate the error from the preceding approximation when the value of R^2 is 10%, 20%, 50%, and 80%.

(b) Express the approximation error as a function of R^2. Is the approximation error bounded?

(c) We have claimed in this chapter that IR^2 is approximately equal to R^2. However, while the value of R^2 is bounded ($0 \leq R^2 \leq 1$), the value of IR^2 is not. Can you explain this discrepancy?

2.20. That IR^2 is approximately R^2 can be verified by a very simple situation. Suppose that we estimated the following equation for stock *XYZ*:

$$r_{t+1} = \alpha + \beta_1 f_{1t} + \cdots + \beta_K f_{Kt} + \epsilon_{t+1} \quad t = 1, \ldots, T-1$$

where $f_{1t}, \ldots, f_{Kt}$ are forecasting factors. Using this equation, we predict the return for $T+1$.

(a) What is the expected return of stock *XYZ* for $T+1$? What is the standard deviation of stock return for $T+1$? Express the Sharpe ratio of stock *XYZ* for $T+1$ in terms of data and estimates.

(b) Express R^2 and $R^2/(1-R^2)$ in terms of data and estimates.

(c) Show that the squared Sharpe ratio of stock *XYZ* can be approximated by R^2. When would the approximation create a large error?

(d) Show that IR^2 can be approximated by R^2 if the portfolio is made only of stock *XYZ* and the benchmark is the cash return.

2.21. Suppose that a portfolio manager predicts the return of MSFT and GE from earnings forecasts of 10 analysts. Let r_{MSFT} and r_{GE} be the return of MSFT and GE and $E_{\mathrm{MSFT},1}, \ldots, E_{\mathrm{MSFT},10}$ and $E_{\mathrm{GE},1}, \ldots, E_{\mathrm{GE},10}$ be the forecasts of 10 analysts.

(a) Using the historical data, the portfolio manager may estimate the following equations:

$$r_{\mathrm{MSFT},t} = \alpha + f_1 E_{\mathrm{MSFT},1,t} + \cdots + f_{10} E_{\mathrm{MSFT},10,t} + \epsilon_{\mathrm{MSFT},t}$$

$$r_{\mathrm{GE},t} = \alpha + f_1 E_{\mathrm{GE},1,t} + \cdots + f_{10} E_{\mathrm{GE},10,t} + \epsilon_{\mathrm{GE},t}$$

If the portfolio manager predicts return based on this model, what would be the breadth?

(b) Quite often in reality the portfolio manager has access only to the consensus forecasts (i.e., the average earnings forecasts). Then the portfolio's return prediction may be based on the following equations:

$$r_{\mathrm{MSFT},t} = \alpha + f \frac{1}{10} \sum_{j=1}^{10} E_{\mathrm{MSFT},j,t} + e_{\mathrm{MSFT},t}$$

$$r_{\mathrm{GE},t} = \alpha + f \frac{1}{10} \sum_{j=1}^{10} E_{\mathrm{GE},j,t} + e_{\mathrm{GE},t}$$

What would be the breadth?

(c) An alternative definition of the breadth is the number of "distinct signals." If we adopt this alternative definition of the breadth, do your answers to (a) and (b) change?

(d) One disadvantage of defining the breadth as the number of distinct signals is that the two preceding models will have the same breadth. Discuss why this is a dis-advantage.

2.22. Suppose that a portfolio manager predicts the return of MSFT and GE from earnings forecasts of 10 analysts. Five analysts provide earnings forecasts for MSFT, and five analysts provide earnings forecasts for GE. Let r_{MSFT} and r_{GE} be the return of MSFT and GE and $E_{\text{MSFT},1}, \ldots, E_{\text{MSFT},5}$ and $E_{\text{GE},6}, \ldots, E_{\text{GE},10}$ be the forecasts. Using the consensus earnings forecasts (i.e., average forecasts), the portfolio manager may estimate the following equations:

$$r_{\text{MSFT},t} = \alpha + f\frac{1}{5}\sum_{j=1}^{5} E_{\text{MSFT},j,t} + e_{\text{MSFT},t}$$

$$r_{\text{GE},t} = \alpha + f\frac{1}{5}\sum_{j=6}^{10} E_{\text{GE},j,t} + e_{\text{GE},t}$$

(a) If the portfolio manager predicts return based on this model, what would be the breadth?

(b) If the portfolio manager applies the models to GM stock as well as MSFT and GE, would your answer to (a) change?

(c) If we define the breadth as the number of "distinct signals," would your answer to (b) change? Explain why it is not practical to define the breadth as the number of distinct signals.

2.23. The information loss is defined as the difference between the maximum information ratio and the actual information ratio.

(a) What does it mean to have an information loss of 0.1?

(b) What is the value of the information loss when the information criterion is satisfied?

(c) What is the minimum possible value of the information ratio? Is there a maximum possible value of the information ratio?

2.24. A portfolio manager believes that the expected returns of stocks *A, B, C, D*, and *E* are as follows:

$$E(r_A) = 10\% \quad E(r_B) = 20\% \quad E(r_C) = 30\%$$
$$E(r_D) = 40\% \quad E(r_E) = 50\%$$

She also found out that the risk (standard deviation) of each stock is 25% and that the five stocks are independent from one another. After examining this information, the portfolio manager created an equal-weighted portfolio of stocks *C, D*, and *E*.

(a) Is the information criterion satisfied?

(b) Calculate the expected return and the standard deviation of the portfolio.

(c) What is the best portfolio one can create out of five stocks? Calculate the expected return and the standard deviation of the best portfolio.

(d) Calculate the information loss.

2.25. (a) What are data mining and data snooping?

(b) Is it possible for a quantitative portfolio manager to avoid them? Explain.

2.26. What does it mean for parameters to change? What might cause parameters to change? Why might a quantitative portfolio manager be worried about parameter stability?

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1 In practice, many quantitative managers simply refer to benchmark α as α because ultimately they measure their success or failure against a benchmark. The portfolio manager does typically increase his or her residual risk when pursuing α. Theoretically, a portfolio manager could increase his or her α with zero incremental risk, but practically this is rare.

2 CAPM α is also sometimes referred to as *Jensen's* α.

3 The reader can think of these returns in these regressions as the returns minus the risk-free rate.

4 Note that this usage of residual is somewhat different from the standard usage in econometrics. In econometrics, the residual refers to the realization of ϵ and does not include α. Unfortunately, the use by practitioners does not always coincide with the use by academics.

5 For those unfamiliar with the CAPM, see a brief discussion of the fundamental models of stock return in Appendix B at www.ludwigbc.com under QEPM Exclusive Content.

6 Suppose that we postulate the following models of stock returns: $r_P = \alpha' + \beta' r_B + \epsilon'$, $r_P = \alpha'' + \beta'' r_M + \epsilon''$, and $r_B = \alpha + \beta r_M + \epsilon$. By substitution of variables, we see that $r_P = \alpha' + \beta'\alpha + \beta'\beta r_M + \beta'\epsilon + \epsilon'$. From this relationship we can see that $\alpha'' = \alpha' + \beta'\alpha$. Thus, for a positive β', $\alpha'' < \alpha'$ when $\alpha < 0$, and vice versa.

7 For those unfamiliar with the APT, see a brief discussion of the three fundamental models of stock returns in Appendix B at www.ludwigbc.com under QEPM Exclusive Content.

8 In golf, there is a saying that goes, "Drive for show, putt for dough." In QEPM, it would be "Ex-ante α for show, ex-post α for dough."

[9] Parameter uncertainty depends, among other things, on sample size. If the sample size is small, the parameter uncertainty will be large. By properly accounting for the parameter uncertainty, portfolio managers can avoid the common mistake of exaggerating the estimation from small sample sizes.

[10] If the market were perfectly efficient, would it have been possible for Warren Buffett to earn an average annual return of 20% between 1965 and 2020, beating the S&P 500 by 9.8% per year? (*Source*: Berkshire Hathaway 2021 Annual Report, Chairman's Letter to Shareholders, p. 2.) Even in a perfectly efficient market, a few investors would indeed earn very high returns because there would still be some outliers in the distribution of returns. In a truly efficient market, however, Buffett's returns would have come at the price of extra risk rather than generating α. Also, his performance would have been simply good fortune rather than an ability to find companies selling below their intrinsic values.

[11] See Chincarini (2012) for a detailed account of crowding and the financial crisis of 2008.

[12] See Murphy (1999).

[13] A test of the efficient-market theory requires an asset pricing model (e.g., CAPM) for the purpose of calculating risk-adjusted returns. Thus the rejection of efficient-market theory, or the discovery of an anomaly, may indicate the failure of the chosen asset pricing model rather than the failure of the efficient-market theory. This ambiguity regarding the nature of an anomaly is known as the *joint hypothesis problem*. In this chapter, for the purpose of simple exposition, we describe an anomaly as representing the failure of efficient-market theory.

[14] See Lo, MacKinlay, and Campbell (1997), p. 66.

[15] See Conrad and Kaul (1988) and Jegadeesh and Titman (1993, 2001).

[16] See Jegadeesh (1990). Also see the list in Table 2.2.

[17] See Moskowitz and Grinblatt (1999).

[18] The January effect is often attributed to Sidney Wachtel [see Wachtel (1942)].

[19] The law currently only allows up to $3,000 to be used to offset income taxes.

[20] There is a wash-sale rule that prohibits repurchasing before 30 days after the sale, so that adds some delay to the repurchasing by sellers.

[21] In general, some portfolio managers may take big bets regardless of the month because they figure that the worst that will happen to them personally is to miss a bonus. There are no income penalties for poor performance, generally. Of course, after too many years of poor returns, a manager is likely to lose his or her job, not just his or her bonus. For more information on mutual fund incentives at other times of the year, see Brown et al. (1996), Busse (2001), and Taylor (2003).

[22] Let us mention that Professor Roll used to have a money management shop with the late Professor Stephen Ross of MIT. In addition, Fuller-Thaler Asset Management, named after Nobel prize–winning behavioral economist Richard Thaler, manages around $11 billion to exploit behavorial biases.

[23] As Gekko says in the movie *Wall Street*, "If you're not inside, you're way outside."

[24] It is illegal to trade on "material, nonpublic" information. However, to a certain degree, insiders have greater information about a company than outsiders. Some of this information can be traded on legally, leading them to profit from the trades they place on their companies' stock. According to the SEC's Rule 10b5–1, an executive's trade is not considered "insider trading" if

a detailed contract planning the trade was established before the executive gained the "material, nonpublic" information in question and the contract was executed exactly as written.

25 This test was known as the *specialist's* test, but it is no longer useful because the structure of securities markets has dramatically changed and specialists are now known as *designated market makers* (DMMs). Electronic limit-order books and high-frequency trading have dramatically reduced the importance of specialists on the exchange.

26 See Ross (2002).

27 Some investors believe anomalies exist due to institutional constraints. That is, institutions may have constraints on their ability to buy or sell that create arbitrage opportunities for less constrained investors.

28 At the end of 1999, Amazon was trading at a large multiple to Barnes & Noble. A short position in AMZN and a long position in BKS would have made a return of 74% on the AMZN position and a 28% return on the BKS position as the price of Amazon collapsed while Barnes & Noble's price rose. In the long run, however, it turned out that Amazon was much more than a bookseller, and an investment in Amazon turned out to be spectacular. From 2000 to 2019, a long position in Amazon yielded around 7,500%, while Barnes & Noble lost about 14%. This brings to mind another conundrum: is efficiency intrinsically related to the investment horizon?

29 For example, Regulation T of the Federal Reserve permits borrowing of stocks up to only 50% of their investment. Thus, borrowing, in reality, is not unlimited. Markowitz (2005) has reiterated that in the absence of unlimited borrowing and lending, the market portfolio will not be efficient.

30 See Grinold and Kahn (1995).

31 Since the mathematics involved is rather heavy, we have placed the derivations of these truths in Appendix 2A, which can be found at www.ludwigbc.com under QEPM Exclusive Content.

32 One may notice that the range of values R^2 can take is bounded, whereas that is not the case for IR^2. This discrepancy results from a number of approximations introduced in the fundamental law. If IR^2 has a high value, the approximations that the fundamental law introduces create a huge approximation error.

33 In theory, the breadth can be much larger than the number of explanatory variables if one counts the number of distinctive signals. We show in Appendix 2A, which can be found at www.ludwigbc.com under QEPM Exclusive Content, that this way of determining the breadth has limited practicality.

34 See Chincarini and Kim (2007).

35 We have 100 unknown parameters and 100 observations. Thus 100 unknown parameters are determined from a system of 100 equations. As long as the 100 equations are not linearly dependent, there will be a unique solution, and ϵ_t will not have any role.

36 Since the publication of the first edition of this book in 2006, there has been an explosion of factor research and a movement to address the data-mining problem that we emphasized in the first edition of this book. See Arnott et al. (2019), Harvey et al. (2016), and Harvey (2017).

37 These methods include the Bonferroni test, the Hommel test, the Holm test, the Hochberg test, the Dubey-Armitage-Parnar test, and the Tukey-Ciminera-Heyse test. See Sankoh et al. (1997), Hochberg (1988), Holm (1979), Hommel (1988), Armitage and Parmar (1986), Dubey (1985), Shi et al. (2012), and Tukey et al. (1985). The simplest method, the Bonferroni method, was named after Italian mathematician Carlo Emilio Bonferroni. A simple example may help.

Suppose a researcher wants to test some data on whether 160 factors are significant with 60 months of data. The typical 5% error threshold for each factor would imply a *t*-statistic of 2 or greater. A Bonferroni adjustment would require a *t*-statistic of 3.83 for each factor. This assumes that factors are independent. One can adjust Bonferroni for correlations in a multiple of ways. One of the simplest would reduce the critical *t*-statistic for each factor to 3.66 with an average correlation of 0.4 amongst factors. See Appendix 4B for more information on these techniques.

[38] See the work of Chincarini (2012), as well as Chincarini (1998, 2017), Chincarini et al. (2018), Cahan and Luo (2013), Yan (2014), Chue (2015), Zhong et al. (2016), Baltas (2019), Brown et al. (2019), and Volpati et al. (2020). For the problems related to copycat trading amongst quantitative funds, see Chincarini (1998), Rothman (2007, 2008), Goldman Sachs (2007a, 2007b), and Khandani and Lo (2008). There are also more references for crowding in some of the presentations at https://ludwigbc.com/presentations/slides/.

[39] See Arnott et al. (2016), Calluzzo et al. (2019), McLean and Pontiff (2016), Jacobs and Muller (2020), and Hou et al. (2020).

[40] Readers who are not familiar with these concepts may consult Appendix C at www.ludwigbc.com under QEPM Exclusive Content.

CHAPTER 3

Basic QEPM Models

The greatest work is inside man.

—Pope John Paul II

3.1 INTRODUCTION

The central, unifying element of quantitative equity portfolio management (QEPM) is the quantitative model that relates stock movements to other market data. Quantitative equity portfolio managers create such models to predict stock returns and volatility, and these predictions, in turn, form the basis for selecting stocks for the portfolio.

Some readers may wonder whether it is necessary to know how to make quantitative models from scratch when there are so many excellent commercial software packages with prepackaged models. When a portfolio manager relies completely on commercial software, he or she may not be able to use all the information available to him or her. Different models make use of different types of information, and any given software program likely ignores some relevant information. If the portfolio manager tries to be "creative" and combine his or her own calculations with a prepackaged model, the resulting hybrid most likely will violate the information criterion. Some dependence on commercial software may be unavoidable,[1] but with an understanding of the modeling process, the manager will know how to get the most out of the software, what to do with it, and what not to do with it.

We begin this chapter by discussing the two basic models of QEPM. As we discuss these models, we consider how they fit into the entire construction of the portfolio, from picking factors to determining the portfolio weights. It turns out that the basic models of QEPM share many properties and produce similar portfolios in certain circumstances. We explain the equivalence of the models in Section 3.3. Portfolio managers often combine one of the basic models with some ad hoc model of stock returns. In Section 3.5 we discuss how these combinations are attempted and why they are not advisable in light of the information criterion. Section 3.6 weighs the basic models' strengths and weaknesses and discusses reasons for using one rather than the other.

3.2 BASIC QEPM MODELS AND PORTFOLIO CONSTRUCTION PROCEDURES

The central idea of modern financial economics is that the average return of a stock is the payoff to the shareholder for taking on risk. Factor models express this risk-reward relationship. *Factors* are explanatory variables that represent different types of risk. A factor model shows that the average stock return is proportional to the stock's exposure to the risk that the factor represents (the *factor exposure*) and to the payoff for each unit of exposure to the risk (the *factor premium*).

There are two generic factor models in QEPM that are used to determine how stock returns and risks vary with factors. They are the *fundamental factor model* and the *economic factor model*. The models take their names from the types of factors typically associated with them. The fundamental factor model uses fundamental factors, which are stock characteristics such as the P/E ratio and market capitalization. The economic factor model was developed originally for macroeconomic variables such as gross domestic product (GDP) and inflation, but it is general enough to handle other types of factors as well. While the models are distinguished in name by the types of factors that go into them, it is important to understand that, in addition, *the fundamental factor model and the economic factor model employ different techniques for modeling stock returns*.

Both the fundamental factor model and the economic factor model are based on the principle that the average stock return is determined by the product of the *factor premium* and the *factor exposure*. The factor premium measures how much investors are willing to pay for each factor, whereas the factor exposure measures how sensitive the stock return is to a factor. The factor premium and factor exposure operate differently in the fundamental and economic factor models. Factor exposures are directly observable for the fundamental factor model but not directly observable for the economic factor model. For the fundamental factor model, the exposures of fundamental factors can be read straight from companies' financial statements and other data sources. For the economic factor model, each stock's exposure to economic (or other) factors is not observable and instead must be estimated from the historical relationship between stock returns and factor premiums.

The factor premium functions in almost the opposite manner in the two basic models. In principle, the factor premium is not observable and must be estimated. This is the case in the fundamental factor model, for which the factor premium must be estimated from the historical relationship between stock returns and the factor exposures. In the economic factor model, however, the factor premium can be determined up to a proportionality without a statistical estimation in certain cases. In other cases, it is determined by constructing *zero-investment portfolios* or through a mathematical method called *principal-component analysis*. We will explain these methods a little later.

Table 3.1 summarizes the different ways that the basic QEPM models estimate the expected returns and risks of stocks. The specific steps involved in constructing a portfolio depend partly on which model is used. In the remainder of this section we summarize the procedures of portfolio construction from A to Z, beginning with the factor choice and ending with the assignment of portfolio weights. The procedures outlined in this overview will be discussed in detail in subsequent chapters.

TABLE 3.1

How Expected Return and Risk Are Determined in Basic QEPM Models

	Fundamental Factor Model	Economic Factor Model
Factor exposure (β)	Directly observed	From time-series regression
Factor premium (f)	From cross-sectional regression	Directly observed*
Expected return	Factor exposure × factor premium	Factor exposure × factor premium
Risk	Factor exposure × risk of factor premium + specific risk	Factor exposure × risk of factor premium + specific risk

*Up to a proportionality. Depending on the factors used, the factor premium may be determined by constructing zero-investment portfolios or by principal-component analysis.

3.2.1 Factor Choice

The first step toward building a quantitative portfolio is choosing factors that seem to drive stock returns. Good factors are really the secret sauce of QEPM. All quantitative portfolio managers are at least reasonably proficient at using the various types of quantitative models. What distinguishes one manager from another is the particular set of factors that he or she uses in his or her model. Managers use a wide variety of factors to explain stock returns, but not all factors can be used in all models. The fundamental factor model restricts factor choice somewhat because it typically takes only fundamental factors. The economic factor model is more flexible, allowing for both economic factors and all fundamental factors. We catalog many factors and discuss how to choose them in Chapter 4, so for the rest of this chapter's discussion on portfolio construction we shall assume that we have already completed the first step of portfolio construction and chosen K factors to represent the behavior of stock returns.

3.2.2 The Data Decision

The data set for the model is determined with a type of factor model and a set of factors in mind. A data set has two dimensions: the *cross-sectional dimension* and the *time-series dimension*. The cross-sectional dimension defines a data set by the characteristics of the stocks it includes. One may decide to gather data on stocks in a specific industry, stocks in the Standard and Poor's (S&P) 500, or all stocks traded on the New York Stock Exchange (NYSE). The cross-sectional dimension of data matters because the attributes of the data set will affect the attributes of the ultimate portfolio and because the number of stocks included in the data set will affect the ease of estimation. The time-series dimension concerns the *periodicity* of the data (i.e., the time intervals at which data points were recorded) and the time *period* of the entire data set. The data could have been recorded daily, weekly, monthly, quarterly, or annually. The entire set of data could cover a few years or just one. The periodicity and the period of the data may affect parameter stability and parameter uncertainty.[2]

Given the model and the data set, the portfolio manager may have to make a few more decisions. If the portfolio manager measures his or her performance against a benchmark, which is typically the case, there is the question of whether to include the benchmark itself as a factor in the model. Also, the portfolio manager should decide whether to use the gross return or the logarithm of the return and whether to use the return in excess of a risk-free rate.

3.2.3 Factor Exposure

Once the portfolio manager has chosen the factors and defined the data set, it is possible to determine the factor exposure and factor premium. We begin with the factor exposure, also referred to as *factor loading*. In the fundamental factor model, determining the factor exposure is fairly straightforward. For example, if the factor is the P/B ratio, the factor exposure of stock i is the latest observed value of the P/B ratio for stock i. (The factor also could be something more complicated, such as an average of the P/B ratio or a forecast of the P/B ratio.)

We will now introduce some mathematical symbols to represent the factor exposure. Suppose that there are K factors chosen for the model. We denote the K factor exposures of stock i by $\beta_{i1},\ldots, \beta_{iK}$. If the first factor in the model is the P/B ratio, then β_{i1} is the P/B ratio of stock i.[3]

In the economic factor model, because the factor exposure is not directly observable, the factor premiums must be determined first (we explain how in the next section). Then the factor exposure can be estimated from the relationship between returns and factor premiums. We denote the factor premium of K factors as $f_1,\ldots,f_K$. Given the factor premium, the following equation for the return of stock i, r_i, can be estimated:

$$r_i = \alpha_i + \beta_{i1}f_1 + \cdots + \beta_{iK}f_K + \epsilon_i \tag{3.1}$$

where $\beta_{i1},\ldots,\beta_{iK}$ are the factor exposures of stock i, whereas α_i is the constant term of the equation. The last term, ϵ_i, is the error, which reflects the random nature of returns. The equation is typically estimated by a time-series regression using observations made at various time periods. That is, the portfolio manager takes factor premiums, which are the variables of interest that affect stock returns, and regresses stock i's returns at every time interval in the data period on the corresponding factor premiums. The estimate from this regression becomes the factor exposure. For example, the factor premium f_j might be real GDP growth. The factor exposure β_{ij} found by estimating the regression shows how sensitive stock returns are to real GDP growth.

3.2.4 Factor Premium

The quantitative portfolio manager also must know the factor premium, which is essentially the premium that the market places on exposure to whatever risk a factor represents. If we know that a stock has a high exposure to the P/B ratio factor, for instance, we need to know what kind of return the P/B ratio provides. These two pieces of information together will allow us to predict the return of the stock.

In the fundamental factor model, the factor premium is estimated from the historical relationship between the stock return and the factor exposure. The equations representing the fundamental and economic factor models are identical to Eq. (3.1), but whereas in the economic factor model the factor exposures $\beta_{i1},\ldots,\beta_{iK}$ need to be estimated, in the fundamental factor model the factor premiums $f_1,\ldots,f_K$ must be estimated. The factor premium for the fundamental factor model is estimated by means of either cross-sectional regression, which involves using observations for various stocks at a single point in time, or panel regression, which entails using observations for various stocks at many points in time.

In the economic factor model, the factor premium is determined first and in various ways depending on the nature of factors. For macroeconomic factors, the values of the variables themselves are taken as factor premiums. For example, if the inflation rate is 3% this month, then the premium for the inflation factor is 3%. This value is not exactly the premium; it does not mean that investors are willing to pay exactly 3% for a unit of stock exposure to inflation. However, the exact premium is proportional to the inflation rate, so for the purpose of forecasting the stock return, the rate itself can be called the factor premium. More generally, the factor premium is calculated by constructing zero-investment portfolios. A *zero-investment portfolio* is a theoretical construct in which the hypothetical investor does not have to invest any capital. For example, if an investor shorts $100 of one stock, theoretically he or she can use the $100 in short-sale receipts to buy $100 of another stock. In the real world, margin requirements prevent such a neat transfer of funds, but for theoretical exercises, it is helpful to imagine such a zero-investment portfolio. We can determine the factor premium by calculating the return on a zero-investment portfolio of investments that exhibit the factor in question. Suppose that size is one of the factors in a model. A zero-investment portfolio could be created for the size factor by going long a small-cap subportfolio and shorting an equivalent amount of a large-cap subportfolio. The factor premium on size would be the difference between the return of the small-cap subportfolio and the return of the large-cap subportfolio.

3.2.5 Expected Return

Whichever factor model is used, estimation of the model provides important information about a stock's expected returns. The estimation essentially allows us to determine the expected returns of stocks based on their factor exposures and the factor premiums.

3.2.6 Risk

The estimation of the model also provides important information about stock risks. The estimation essentially decomposes the risk of stocks into two components: nondiversifiable risk and diversifi- able risk. *Nondiversifiable risk* is the primary concern for investors because it comes from a stock's exposure to risks in the market that cannot be removed from the portfolio. Nondiversifiable risk is represented by $\alpha_i + \beta_{i1} f_1 + \ldots + \beta_{iK} f_K$ in Eq. (3.1). *Diversifiable risk*, on the other hand, can be removed from the portfolio by diversifying holdings. It is captured by ϵ_i. Diversifiable risk is often called the *stock-specific risk*. By calculating the variance or the standard deviation of each component of risk, we can estimate the stock's total risk.

3.2.7 Forecasting

Expected stock returns are simply the product of the factor premiums and the factor exposures. Once the stocks' risk levels have been determined, the portfolio manager has gathered all the information needed to construct a portfolio. But he or she does not have all the information to construct an optimal portfolio. The factor premiums and factor exposures in the model are determined from past data. The relationships among return, exposure, and premium are likely to change in the future. The portfolio manager wants to know *future* values, not historical values. In all likelihood, the factor exposure will not change in the immediate future, but the factor premium *is* likely to change in a very short amount of time. Thus it is usually necessary to forecast the factor premium. We explain how to forecast it in Chapter 8.

3.2.8 Security Weighting

With the estimates of stock returns and risks based on the new forecasts, the portfolio manager can use optimization techniques to select the stocks for the portfolio and assign them their respective weights. The weighting of stocks in the portfolio can be set to maximize the portfolio's overall return, minimize its risk, and satisfy other constraints such as diversification requirements and specific investment style requirements. Alternatively, the portfolio manager may decide to minimize the tracking error between the portfolio and the benchmark. This issue is explored fully in Chapter 9.

3.3 THE EQUIVALENCE OF THE BASIC MODELS

The fundamental factor model and the economic factor model are built on the same principle: The expected stock return can be expressed as the product of the factor exposure and the factor premium. Therefore, it is not surprising that the two models produce identical portfolios when certain assumptions are satisfied. The equivalence of the factor models can be established starting with the following proposition:

LEMMA 1 (THE EQUIVALENCE OF FACTOR MODELS) *The fundamental factor model and the economic factor model are equivalent to each other if the expected stock return is a linear function of the fundamental factor exposure.*

If the fundamental factor model is a correct model, then the economic factor model is also a correct one.[4] The proof of this claim proceeds in the following way: First, we assume that expected stock returns are a linear function of fundamental factor premiums, as described by the fundamental factor model. Then, under this assumption, we show that the expected stock return can be expressed as a linear function of the factor exposure, as described by the economic factor model.

Let us first express the stock return using the fundamental factor model. If there are K factors, then the return of stock i, r_i, is

$$r_i = c_i + \pi_1 x_{i1} + \cdots + \pi_K x_{iK} + \omega_i \tag{3.2}$$

where $x_{i1},\ldots, x_{iK}$ are the factor exposures of stock i, $\pi_1,\ldots, \pi_K$ are the factor premiums, and c_i is the constant term of the equation. For this section only, we have deliberately changed the notation to distinguish the fundamental factor model from the economic factor model.

What happens if we use the same K factors but in the economic factor model? Does the economic factor model correctly describe stock returns? The answer is yes. All we need to show is that the expected stock return can be expressed as a linear function of the factor premium, as described by the economic factor model. We provide the proof for the case in which the factor exposures are independent of one another.[5] In the economic factor model, the factor premium is determined by constructing the zero-investment portfolio. To determine the premium of factor k, one could distribute all stocks into two or three groups based on the value x_{ik}. Suppose that each stock is assigned to one of three groups: a high group $(\text{if } x_{ik} > \overline{x}_k)$, a low group (if $(\text{if } x_{ik} < \underline{x}_k)$, and a middle group (with the cutoff values $\overline{x}_k$ and $\underline{x}_k$). The factor premium f_k is the expected return to the zero-investment position that puts \$1 into the high group and shorts \$1 in the low group,[6] that is,

$$f_k = E(r \mid x_k > \overline{x}_k) - E(r \mid x_k < \underline{x}_k) \tag{3.3}$$

If factor exposures are independent, then, using Eq. (3.2), we can rewrite the equation as

$$f_k = \pi_k d_k \tag{3.4}$$

where d_k is a constant defined as

$$d_k \equiv [E(x_k \mid x_k > \overline{x}_k) - E(x_k \mid x_k < \underline{x}_k)] \tag{3.5}$$

Using Eq. (3.4), we can rewrite Eq. (3.2) in the following way:

$$r_i = c_i + \left(\frac{f_1}{d_1}\right)x_{i1} + \cdots + \left(\frac{f_K}{d_K}\right)x_{iK} + \omega_i \tag{3.6}$$

This equation shows that the expected stock return can be written as a linear function of the factor premiums. In fact, it shows the exact relationship between the fundamental factor model and the economic factor model. The constant term in the economic factor model (α_i) is identical to the constant term in the fundamental factor model (c_i). The factor exposure in the economic factor model (β_{ik}) is proportional to the factor exposure of the fundamental factor model (x_{ik}/d_k).

For practitioners, the equivalence of the factor models means that to the extent that the fundamental factor model is a correct model, it does not really matter whether one use the fundamental factor model or the economic factor model. The two models produce essentially identical results. The only catch is that if the fundamental factor model is not correct, then it is best to use the economic factor model because it has a stronger theoretical basis than the fundamental factor model does.

3.4 THE SCREENING AND RANKING OF STOCKS WITH THE Z-SCORE

Many portfolio managers screen and rank stocks using *Z-scores*. Sometimes stock screens and rankings supplement the use of a factor model in determining the mix of stocks for the portfolio. More often they are used as alternatives to the factor model. Since screening and ranking are widespread practices among managers, we discuss them more fully in Chapter 5.

In the context of stock screening and ranking, a Z-score is a stock's standardized exposure to a fundamental factor. To calculate the Z-scores for the stocks in some investment universe, we need to know the factor exposures of every stock. Then, for each stock, we subtract the universe's mean factor exposure from the stock's individual factor exposure and divide that difference by the standard deviation of factor exposures for the universe. What comes out of this standardization is a set of Z-scores, factor exposures with a mean of 0 and a standard deviation of 1.

For example, suppose that we want to calculate Z-scores for exposure to the P/B ratio. If $\beta_1, \ldots, \beta_N$ are the P/B ratios of stocks 1, …, N, then the Z-score of stock i is defined as

$$z_i = \frac{\beta_i - \mu}{\sigma} \tag{3.7}$$

where μ and σ are the average and standard deviation of $\beta_1, \ldots, \beta_N$. The standardization allows us to interpret the Z-score in the following way: If z_i is 2, then the P/B ratio of stock i is 2 standard deviations away from the average.

Given the Z-score of one factor—or, more commonly, of many factors—portfolio managers can develop a number of screening and ranking strategies. We look at a simple example in the next section and more realistic examples in Chapter 5.

3.5 HYBRIDS OF THE MODELS AND THE INFORMATION CRITERION

Practitioners often try to add extra inputs to the basic QEPM models by combining them with ad hoc models. The resulting hybrid models are sometimes reasonable descriptions of stock returns, but in many cases they have a number of undesirable effects. The most critical problem with hybrids is that they frequently violate the information criterion.

Recall that in order to uphold tenet 4 of QEPM and satisfy the information criterion, a portfolio manager must use all available information in the most efficient way. A hybrid model combining two models that are based on different information may violate the information criterion by combining the two original sets of information inefficiently. If a hybrid model is based on two models that use the same information, the hybrid may violate the information criterion by incorrectly "double counting" the information. Consider the following hypothetical situation: Adam, a portfolio manager, creates a fundamental factor model but decides to combine it with a Z-score analysis. Specifically, Adam calculates the expected return based on the Z-score method and adds this expected return to the fundamental factor model as a constant. Adam uses portfolio management software that implements the fundamental factor model automatically and allows the user to add a constant [i.e., α in Eq. (3.1)].

Is Adam using all the available information in the most efficient way? Not if he created the Z-score model and the fundamental factor model from the same data. The addition of the Z-score analysis not only presents no gain, but it in fact also introduces distortion in the model, which will lead to a less-than-optimal portfolio. Even if Adam used different sets of information to create the Z-score model and the fundamental factor model, the two sets of information are still not being combined in the most efficient way. In either case, the hybrid does not satisfy the information criterion. With data on his models and portfolio, we can find out exactly how inefficient Adam's hybrid is by calculating the information loss. The following numerical example illustrates the problem with patching together two models.

3.5.1 The Setup

Imagine that there are only two stocks in the world, stock A and stock B. Suppose that current stock returns r are determined by the price-to-earnings (P/E) ratios at the end of the previous period. Specifically, we assume[7]

$$r_{A,t+1} = 0.1\left(\frac{P}{E}\right)_{A,t} + \epsilon_{A,t+1} \quad \epsilon_{A,t+1} \sim N(0,20) \tag{3.8}$$

$$r_{B,t+1} = 0.1\left(\frac{P}{E}\right)_{B,t} + \epsilon_{B,t+1} \quad \epsilon_{B,t+1} \sim N(0,10) \tag{3.9}$$

where ϵ is the random component of the stock return and is assumed to have a normal distribution with a variance of 10 or 20.[8] We also assume that the covariance between the two random errors is zero. Suppose that for the last period T, the P/E ratio of stock A is 20 and of stock B is 10. Therefore, the average P/E ratio is 15 [=(20 + 10)/2], the variance is 25 {= [(20 − 15)^2 + (10 − 15)^2]/2}, and the standard deviation is $5\ (=\sqrt{25})$. Suppose further that the average and the variance are constant over time and across stocks. Table 3.2 summarizes this assumption and the corresponding calculations of the Z-score.

TABLE 3.2

Price-to-Earnings Ratio and the Z-Score

	Stock A	Stock B	Mean	Variance	SD
$\left(\frac{P}{E}\right)_T$	20	10	15	25	5
Z_T	1	−1	0	1	1

3.5.2 The Z-Score Model

The Z-score is the normalized factor exposure. To calculate one stock's Z-score, we subtract the mean factor exposure of all stocks from the individual stock's factor exposure and then divide that difference by the cross-sectional standard deviation of the factor exposures of all stocks. The Z-score for stocks A and B are reported in Table 3.2. Once the Z-score is calculated, it is possible to estimate the following equation to predict the expected stock return:

$$r_{i,t+1} = a_i + bz_{it} + \ v_{i,t+1} \quad i= A,B;\ t = 1, \ldots, T - 1 \qquad (3.10)$$

where z_{it} is the Z-score of stock i at time t, a_i is a constant, and $v_{i,t+1}$ is the error. From the numbers given in Table 3.2, the values of a_A, a_B, and b are as follows[9]:

$$\begin{aligned} b &= \frac{1}{2}\frac{C(r_A, z_A)}{V(z_A)} + \frac{1}{2}\frac{C(r_B, z_B)}{V(z_B)} \\ &= \frac{1}{2}\sqrt{V\left[\left(\frac{P}{E}\right)_A\right]}\frac{C\left[r_A, \left(\frac{P}{E}\right)_A\right]}{V\left[\left(\frac{P}{E}\right)_A\right]} + \frac{1}{2}\sqrt{V\left[\left(\frac{P}{E}\right)_B\right]}\frac{C\left[r_B, \left(\frac{P}{E}\right)_B\right]}{V\left[\left(\frac{P}{E}\right)_B\right]} \qquad (3.11) \\ &= \frac{1}{2}\cdot 5\cdot 0.1 + \frac{1}{2}\cdot 5\cdot 0.1 = 0.5 \end{aligned}$$

$$\begin{aligned} a_A &= E(r_A) - bE(z_A) \\ &= 1.5 - 0.5 \cdot 0 = 1.5 \end{aligned} \qquad (3.12)$$

and

$$\begin{aligned} a_B &= E(r_B) - bE(z_B) \\ &= 1.5 - 0.5 \cdot 0 = 1.5 \end{aligned} \qquad (3.13)$$

Given the time T value of the Z-score, the preceding estimates imply the following expected returns for stock A and stock B:

$$E(r_{A,T+1}) = a_A + bz_{A,T} = 1.5 + 0.5 \cdot 1 = 2 \qquad (3.14)$$

$$E(r_{B,T+1}) = a_B + bz_{B,T} = 1.5 + 0.5 \cdot (-1) = 1 \qquad (3.15)$$

Note that the expected returns of stock A and stock B are correct. The Z-score model itself does not create any problem with the efficient use of information. The problem arises when the Z-score model is combined incorrectly with other models, as illustrated below.

3.5.3 A Hybrid of the Z-Score Model and a Fundamental Factor Model

Adam first estimated the fundamental factor model, i.e., Eq. (3.1). Then he did the Z-score analysis. Now he creates his hybrid model by setting the term α in the fundamental factor model according to the expected return from the Z-score model. Specifically he adjusts α of stock A to be 1 and α of stock B to be −1 so that the sum of the α's remains equal to 0. For time $T + 1$, he incorrectly modifies Eqs. (3.8) and (3.9) into the following:

$$r_{A,T+1} = 1 + 0.1\left(\frac{P}{E}\right)_{A,T} + \epsilon_{A,T+1} \quad \epsilon_{A,T+1} \sim N(0,20) \tag{3.16}$$

$$r_{B,T+1} = -1 + 0.1\left(\frac{P}{E}\right)_{B,T} + \epsilon_{B,T+1} \quad \epsilon_{B,T+1} \sim N(0,10) \tag{3.17}$$

Given time T's P/E ratio, Adam is going to construct a portfolio at time $T + 1$ based on the following numbers:

$$E(r_{A,T+1}) = 1+0.1 \cdot 20 = 3 \tag{3.18}$$

$$E(r_{B,T+1}) = -1+0.1 \cdot 10 = 0 \tag{3.19}$$

$$V(r_{A,T+1}) = V(\epsilon_{A,T+1}) = 20 \tag{3.20}$$

$$V(r_{B,T+1}) = V(\epsilon_{B,T+1}) = 10 \tag{3.21}$$

$$C(r_{A,T+1}, r_{B,T+1}) = C(\epsilon_{A,T+1}, \epsilon_{B,T+1}) = 0 \tag{3.22}$$

If the weight of stock A in the portfolio is w, then the expected return and the variance of the portfolio (according to Adam's calculations) are

$$\mu_p = 3w + 0(1 - w) \tag{3.23}$$

$$\sigma_P^2 = 20w^2 + 2 \cdot 0 \cdot w(1 - w) + 10(1 - w)^2 \tag{3.24}$$

To construct his optimal tangent portfolio, Adam finds the value of w that maximizes μ_P/σ_P. This value is $w = 1$. Thus 100% of Adam's portfolio will go into stock A and 0% into stock B.

3.5.4 Information Loss

Adam thinks that he has created the optimal portfolio, but he is actually not achieving the maximum Sharpe ratio because his estimates of the expected return are not correct. The information loss reveals exactly how much potential Sharpe ratio Adam is losing by combining two models. Let us first determine the maximum Sharpe ratio Adam could have achieved if he had not combined the Z-score analysis with the fundamental factor model. By not combining the models, we obtain the true expected returns of stock A and stock B:

$$E(r_{A,T+1}) = 2 \tag{3.25}$$

$$E(r_{B,T+1}) = 1 \tag{3.26}$$

We should emphasize the fact that the loss of potential returns does not arise from choosing one model over the other. Adam would have found the correct expected return whether he used the Z-score model or the fundamental factor model. It was the combination of the two that muddied the analysis. Given the correct expected return of the individual stocks, the expected return and variance of Adam's current portfolio (invested 100% in stock A and 0% in stock B) are

$$\mu_p = 2w + (1 - w) = 2 \tag{3.27}$$

$$\sigma_P^2 = 20w^2 + 10(1 - w)^2 = 20 \tag{3.28}$$

Thus Adam's portfolio's actual Sharpe ratio (*SR*) (ignoring the risk-free rate) is

$$\text{Actual } SR = \frac{\mu_P}{\sqrt{\sigma_P^2}} = 0.4472 \tag{3.29}$$

Adam's current portfolio weights are not optimal. The maximum Sharpe ratio is achieved when $w = 0.5$. When $w = 0.5$, the expected return and the variance of the portfolio are

$$\mu_p = 2w + (1 - w) = 1.5 \tag{3.30}$$

$$\sigma_P^2 = 20w^2 + 10(1 - w)^2 = 7.5 \tag{3.31}$$

If the portfolio had been constructed optimally, Adam could have attained a maximum Sharpe ratio of

$$\text{Maximum } SR = \frac{\mu_P}{\sqrt{\sigma_P^2}} = 0.5477 \tag{3.32}$$

Given the actual Sharpe ratio and the maximum Sharpe ratio, the information loss (*IL*) is the difference between the two:

$$IL = \text{maximum } SR - \text{actual } SR = 0.1005 \tag{3.33}$$

The information loss of 0.1005 means that when Adam takes 1% extra risk, his reward for taking that extra risk is 0.1% lower than it could have been. Since Adam is currently taking 4.5% risk, he earns 0.45% less than he could have. He loses 0.45%, or 45 basis points, in expected return by incorrectly combining two models. The moral of the story is clear. The best thing to do is to choose one model and stick with it. Combining basic models is hard to justify in terms of the information criterion. Another practical implication of this example is that there are benefits to building one's own model at the outset. Many managers want to model information in a specific way, but prepackaged programs are not flexible to modifications. It does no good to tack an ad hoc model onto the one that the software generates, so designing an entire custom model may be the best course of action.

3.6 CHOOSING THE RIGHT MODEL

It is clearly best to choose just one model of stock returns in order to create the portfolio, but which one of the two general factor models is the right one to use in a given situation? As we noted earlier, the economic factor model handles a different range of factors and operates differently from the fundamental factor model. Here we discuss some criteria for choosing one type of model over the other (see Table 3.3 for a summary).

TABLE 3.3

Criteria for Selecting the Right Model

Criterion	Fundamental Factor Model	Economic Factor Model
Theoretical basis		Better
Factor accommodation		Better
Ease of implementation	Better	
Data requirement	Better	
Intuitive appeal	Depends	Depends

3.6.1 Consistency with Economic Theory

Tenet 5 of QEPM states that quantitative models should be based on sound economic theory. The economic factor model upholds tenet 5 better than the fundamental factor does. Economic theory suggests that high expected returns are justified only as the payoff for bearing extra risk. Small-cap and value stocks might produce extra returns not because there is anything especially good about being small cap and value but because being small cap and value increases the stocks' risk. If this were the case, a fundamental factor model using those two variables would not give us meaningful information about the sources of risk and how they affect stock prices. It would be better to use a model that gets to the underlying risk of the stocks directly rather than looking at size and value. An economic factor model can measure the sensitivity of stocks to various economic risk factors.

Of course, if the relationship between the stock return and the fundamental factor exposure is linear, then the fundamental factor model is equivalent to the economic factor model, as we showed earlier. If the relationship is not linear, however, the fundamental factor model not only lacks theoretical backing, but it also lacks econo-metric justification because the cross-sectional regression becomes misspecified. In other words, we are estimating the wrong equation.[10]

3.6.2 Ability to Combine Different Types of Factors

Not all factors can be used in both the economic and fundamental factor models. Both models can handle fundamental factors, such as the P/B ratio, the P/E ratio, and size. They also can use technical factors, such as momentum and trading volume. There are some factors, however, that only the economic factor model can handle. These include macroeconomic factors, such as GDP and inflation. If a portfolio manager wants to use a mix of all types of factors, the economic factor model is the necessary choice.

3.6.3 Ease of Implementation

In terms of ease of implementation, however, the fundamental factor model is preferable to the economic factor model. For the fundamental factor model, only factor premiums need to be estimated. For the economic factor models, factor exposures of individual stocks need to be estimated, and forecasts of factor premiums are usually needed as well. A manager should remember, though, that the ease of implementing the fundamental factor model comes at the cost of restrictions on the types of factors that can be used.

3.6.4 Data Requirement

The fundamental factor model can be estimated without a large amount of historical data.[11] On the other hand, to estimate the economic factor model, portfolio managers have to gather a relatively large period of historical returns because the estimation of factor exposures requires a time-series analysis of returns.

In terms of the cross-sectional dimension, the data requirement for the economic factor model may be lower than the data requirement for the fundamental factor model. The economic factor model is estimated for each stock separately, and in the extreme case, it may be estimated for just one stock. To estimate the fundamental factor model, however, one needs a significant number of stocks (typically a few hundred).

However, this difference should not be exaggerated. When the economic factor model is estimated for a small number of stocks, the estimation error in the factor exposure may remain significant *after* the portfolio is constructed. If the portfolio is composed of a large number of stocks, the estimation error in the factor exposure of one stock is likely to be canceled out by the estimation error of another stock. This sort of canceling out is not likely to happen in a portfolio composed of very few stocks. Thus, in practice, the economic factor model also requires data on many stocks in order to be useful in the portfolio construction process.[12]

3.6.5 Intuitive Appeal

The intuitive appeal of a model goes beyond whether it can be explained with economic theory. The best model is the one that makes common sense and is easy enough to explain in plain language. Portfolio managers always should understand the models they are using. Even an excellent model is dangerous in uneducated hands. In terms of gaining familiarity and becoming comfortable with quantitative models, reading this book carefully is a step in the right direction.

3.7 CONCLUSION

In this chapter we introduced the basic models that quantitative equity portfolio managers use to predict stock returns. The main models of QEPM are the fundamental factor model and the economic factor model. The fundamental factor model is used primarily to explain stock returns with stock fundamentals, whereas the economic factor model can be used to explain returns with almost any type of factor. The fundamental and economic factor models work somewhat differently but follow a common formula: the average stock return equals the product of factor exposures and factor premiums. We demonstrated that the two types of models produce identical results under certain conditions, but we also looked at reasons for choosing one type over the other.

Many managers use stock screening and ranking methods in addition to or instead of factor models. Although the Z-score analysis involved in screening and ranking stocks works fine by itself to estimate stock returns, it is best not to combine it with factor models. The resulting hybrid models fail the information criterion, and the resulting portfolios miss out on potential returns.

In this chapter we described how quantitative models fit into the seven steps of portfolio construction (factor choice, data preparation, estimation of factor exposures, estimation of factor premiums, determination of risk, forecasting of factor premiums, and, finally, security weighting). We will examine the fundamental and economic factor models in greater detail in Chapters 6 and 7, again in the context of the overall portfolio construction process, which is the subject of Part II of this book.

QUESTIONS

3.1. What are the two most frequently used models of stock return in QEPM?
  (a) How do you obtain the factor exposure for each model?
  (b) How do you obtain the factor premium for each model?
  (c) How do you obtain expected returns for each stock from each model?
  (d) How do you measure the risk of each stock with each model?

3.2. What are the steps in the general portfolio construction process of QEPM? Describe each one briefly.

3.3. What is the difference between a cross-sectional data set and a time-series data set?

3.4. Below is a series of statements by a quantitative portfolio manager. Decide whether each statement relates more to a fundamental factor model or to an economic factor model.
  (a) The average P/B ratio of our portfolio is very high.
  (b) Our portfolio will be quite resilient if oil prices rise to our negative exposures.

(c) Our philosophy at the Financial Artists Hedge Fund is that economics drives stock returns; thus we use economic data to predict individual stock returns.
(d) The factor exposure of every stock is known and publicly available.
(e) Analyst forecasts drive stock returns.
(f) Our investment department is very good at predicting presidential elections. We choose stocks accordingly.

3.5. Macroeconomic variables cannot be used in stock return models because they are the same for all stocks. True or false. Explain.

3.6. How do you obtain the factor premium for an economic factor model and a fundamental factor model? Give an example of each.

3.7. How useful is the annual inflation rate of the United States as a factor premium?

3.8. Why is forecasting factor premiums necessary? Why doesn't this same logic apply to factor exposures? Please answer with respect to both types of QEPM stock return models.

3.9. In this chapter we proved the equivalence of factor models under the assumption that the factor exposures are indepen dent of one another. Prove the equivalence of factor models without this assumption.

3.10. In this chapter we proved that if the fundamental factor model is correct, then the economic factor model is also correct. However, the reverse is not true. Consider the following example. Suppose that the expected stock return is proportional to the squared firm size and that the stock return can be written as

$$r_i = \alpha + \pi x^2_i + \epsilon_i$$

where x_i is the size of stock i.

(a) Show that it is possible that the economic factor model is correct; i.e., the expected stock return is a linear function of the factor premium f defined as

$$f = E(r \mid x > \mu_x) - E(r \mid x < \mu_x)$$

where μ is the average of the firm size.

(b) Explain why the fundamental factor model may not be consistent with the preceding setup.

(c) Given this one-way relationship, which model would you prefer in practice?

3.11. What is a common standardization that portfolio managers use before screening stocks?

3.12. Why might it be dangerous for a quantitative equity portfolio manager to combine the Z-score model with a commercial risk model to build an optimized portfolio?

3.13. In the example of Section 3.5 we assumed that the true return-generating process is

$$r_{A,t+1} = 0.1\left(\frac{P}{E}\right)_{A,t} + \epsilon_{A,t+1} \quad \epsilon_{A,t+1} \sim N(0, 20)$$

$$r_{B,t+1} = 0.1\left(\frac{P}{E}\right)_{B,t} + \epsilon_{B,t+1} \quad \epsilon_{B,t+1} \sim N(0, 10)$$

That is, we assumed that α is zero. Suppose that α is not zero. Instead, assume that the true return-generating process is

$$r_{A,t+1} = 1 + 0.1\left(\frac{P}{E}\right)_{A,t} + \epsilon_{A,t+1} \quad \epsilon_{A,t+1} \sim N(0, 20)$$

$$r_{B,t+1} = 0.1\left(\frac{P}{E}\right)_{B,t} + \epsilon_{B,t+1} \quad \epsilon_{B,t+1} \sim N(0, 10)$$

(a) If Adam estimates the Z-score model, what would be the estimated expected return of stock A and stock B? Are the estimates correct?

(b) If Adam uses the estimated expected return from the Z-score model as α in the fundamental factor model, what would be the estimated expected returns of stock A and stock B? Are the estimates correct?

(c) Explain why the information criterion is not satisfied regardless of whether the true α is zero or not.

(d) Calculate the information loss.

3.14. A portfolio manager estimated the expected stock return from a Z-score model of the price-to-earnings ratio and used it as α in the fundamental factor model, as discussed in Section 3.5. This time, however, instead of estimating the fundamental factor model with the price-to-earnings ratio, the portfolio manager estimated the fundamental factor model with the size. Thus the portfolio manager is not double counting any information. Is the information criterion satisfied now?

3.15. Below we list some criteria for selecting a stock return model. Indicate which factor model (fundamental or economic) best meets each criterion.

(a) Theoretical reasoning

(b) Factor accommodation

(c) Ease of implementation

(d) Data requirements

(e) Intuitive appeal

3.16. Economic and fundamental factors are both used in practice.

(a) Which type of factor is more theoretically justified?

(b) Give one reason why this is so.

(c) Eugene Fama might be heard saying the following: "I still would rather not use the type of factor in part (a) because it does not perform well out of sample." What does he mean? Is he right?

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[1] One advantage of commercial software packages is that the companies that produce them tend to do a lot of data cleaning, which makes them a good source of polished data.

[2] We will discuss these issues further in Chapters 6 and 7.

[3] The convention in statistics is to use Greek letters for unobservable quantities and Latin letters for observable quantities. In the economic factor model, the factor exposure is unobservable, so it makes sense to use Greek letters for factor exposure. In the fundamental factor model, although the factor exposure is observable, it is still represented by the Greek letter β so that the notation is consistent in both types of factor models.

[4] The reverse relationship cannot be established in general. That is, even if the economic factor model is a correct description of reality, the fundamental factor model still may not be correct.

[5] This proof can be modified easily for the more general case.

[6] Note that the expectation is taken across stocks. It is the cross-sectional average of the expected stock return.

[7] In reality, we never know the true return-generating process, so the example may seem unrealistic. The fact that we are assuming certainty in the return-generating process, though, does not affect the lesson of the example that combining two models creates distortion.

[8] When we typically estimate these factor models, we use information at time t to predict returns at time $t + 1$. This is to avoid any look-ahead bias. Sometimes in this book, for convenience, we write the equation with time subscript t everywhere. It should still be interpreted as using information at time t to predict returns at time $t + 1$. Sometimes we will refer to this as having beginning-of-month or equivalently end-of-the-previous-month factor exposures with this month's returns.

[9] The formula can be obtained by demeaning (i.e., subtracting the mean from) both sides of the equation and applying the standard ordinary least squares (OLS) formula.

[10] Another assumption typically adopted in implementing the fundamental factor model is that the factor premium is stable for some time. Similarly, in implementing the economic factor model, the assumption is often that the factor exposure is stable for some time. Whether these assumptions make sense or not can be another criterion for choosing a model. However, as we explain in Chapter 8 on forecasting, neither of these assumptions has to be adopted.

[11] To estimate the variance of each stock return requires the variance of the error terms of each stock and the variance-covariance matrix of the factor premiums. It is possible to obtain the variance-covariance matrix of the factor premiums from (the estimation error of) a single cross-sectional regression. Thus a historical time series may improve the estimation but is not necessary. However, to be able to estimate the variance of the error of each stock, we do need some historical data because in a cross-sectional regression the variance of the error is assumed to be the same across stocks. The importance of estimating the error, however, is somewhat less than other estimations. Thus the fundamental factor model requires some historical data to be able to estimate the variance of individual stock returns, which is ultimately required to build optimized portfolios. However, the data required are still less than what are required in the economic factor model.

[12] It also means that since the fundamental factor model is affected by the errors of factor returns and not of factor exposures, the fundamental factor model's estimation errors will be similar regardless of the diversification of a portfolio. Thus one might believe that fundamental factor models would work better for very undiversified or concentrated portfolios.

PART II

Portfolio Construction and Maintenance

The next eight chapters are a user's manual for QEPM. We will lay out the nuts and bolts of constructing and maintaining the optimal portfolio for the manager's objectives. What is the job of a portfolio manager, after all, but to assemble and run a portfolio of securities that consistently outperforms the market? Intuitive and purely qualitative investment strategies can be hit or miss. QEPM helps the manager to identify the strategies that are most likely to earn high returns consistently. It is also a systematic, concrete way to make daily decisions about the portfolio.

The factor model of stock returns, the central element of QEPM, subjects investment strategies to the rigors of the scientific method. In so doing it fulfills principles embodied in several of the seven tenets of QEPM. Ideas for the model must be formed as hypotheses based on sound economic theory (satisfying tenet 5); statistical analyses used in creating the model identify which hypotheses reflect persistent, stable patterns (satisfying tenet 6); and the resulting model combines all the available information in the most efficient way (satisfying tenet 4).

QEPM guides the manager in both the initial selection of securities and in the ongoing work of maintaining the optimal mix of holdings. With QEPM, the manager even can maximize the portfolio return after transactions costs and taxes, whose corrosive effects on the accumulation of portfolio wealth are sometimes not fully appreciated by personal investors or professional managers.

CHAPTER 4

Factors and Factor Choice

Not everything that can be counted counts, and not everything that counts can be counted.

—Albert Einstein

4.1 INTRODUCTION

Factors are the ingredients of quantitative equity portfolio management (QEPM) models. Just as high-quality ingredients make for excellent cuisine, carefully selected factors can, in the right combination, create models for outperformance. Factors come in many varieties: fundamental, technical, economic, and alternative. How does a portfolio manager select the right ones for a model? QEPM tenets 4 and 5 advise the manager to build quantitative models that reflect sound economic theories and persistent, stable patterns. Good factors therefore exhibit relationships with stock returns that not only are stable and persistent but also can be explained by economic theory. As we discussed in Chapter 2, the likelihood of finding high-quality factors depends on how one goes about looking for them. Data mining produces factors that appear to be highly correlated with stock returns, but the relationships between those factors and returns are only superficial. The way to find good factors is to apply solid, well-reasoned quantitative techniques to the search. This chapter describes the main types of factors and some methods of choosing them.

4.2 FUNDAMENTAL FACTORS

A *factor* is any variable that can predict stock returns. Perhaps the most obvious influence on stock performance is the financial condition of the firm. *Fundamental* factors describe a firm's financial condition. The most common fundamental factors are ratios calculated directly from the income statement, balance sheet, and statement of cash flows. Innumerable financial ratios and combinations of financial statement variables potentially could be constructed from the financial statements and used to forecast stock returns. Since many of these possible fundamental factors are correlated with each other, we can boil the list down to some essential ones that every portfolio manager should know.

We group the fundamental factors into seven subcategories: valuation factors, size factors, operational efficiency factors, operating profitability factors, solvency factors, financial risk factors, and corporate activity factors. Valuation factors, such as the price-to-book (P/B) ratio and the price-to-earnings (P/E) ratio, attempt to measure whether stocks are relatively cheap or expensive. Size factors, such as market capitalization, attempt to classify companies by their size and measure whether size has effects on stock return behavior. Operating efficiency factors, such as inventory turnover, and operating profitability factors, such as gross profit margin, tell us how well management is running the company. Solvency factors attempt to measure a company's ability to meet future short-term obligations. Indicators in this subcategory include the current ratio and the cash ratio. Financial risk factors measure financial health in ratios such as debt-to-equity and interest coverage. Finally, corporate activity measures factors that are related to corporate executive decisions or do not necessarily fall into any of the other categories. Tables 4A.1 through 4A.7 in Appendix 4A provide complete lists of the fundamental factors for QEPM models.

4.2.1 Valuation Factors

Table 4.1 shows the values of popular valuation factors for selected stocks. Legions of managers have based investment strategies on the P/E ratio, and, as we saw in Chapter 2, many studies have documented a P/E ratio effect.[1] Low P/E ratio stocks tend to outperform high P/E ratio stocks on a risk-adjusted basis.[2] One possible explanation for this effect is that the market generally overreacts to bad news, leaving the prices of some stocks excessively deflated. Portfolio managers can purchase the "unpopular" stocks at bargain prices and eventually earn outsized returns as the rest of the market realizes that the stocks are underpriced. Since the P/E ratio is such a popular factor, it is important not to fall into the trap of including it in a model by default. There always should be a clear reason for including it based on sound economic theory. When considering P/E ratio or any other factor to add to a model, it is also important to anticipate how it will fit with the rest of the factors in the model.

TABLE 4.1

Valuation Factors for the Top 10 Standard and Poor's (S&P) 500 Companies

Ticker	Dividend Yield (DY)	Price-to- Book Ratio (P/B)	Price-to-Cash-Flow Ratio (P/CF)	Price-to-Earnings Ratio (P/E)	Price-to-Earnings-to-Growth Ratio (PEG)	Price-to-Earnings-to-Growth-plus-Yield Ratio (PEGY)	Price-to-Sales Ratio (P/S)
AAPL	0.61	34.53	27.96	39.30	3.24	3.08	8.22
MSFT	0.92	13.63	25.40	35.41	2.45	2.30	11.43
AMZN	0.00	19.74	29.56	94.04	2.59	2.59	4.70
GOOGL	0.00	5.19	19.43	30.94	1.87	1.87	6.43
TSLA	0.00	41.73	153.81	1,203.07	2.95	2.95	23.74
FB	0.00	5.58	19.43	25.98	1.57	1.57	8.31
BRK.B	0.00	1.31	13.16	15.17	N/A	N/A	1.95
JNJ	2.47	6.43	19.19	24.39	5.66	3.60	5.12
WMT	1.48	5.42	12.35	22.79	3.38	2.77	0.75
JPM	2.86	1.61	12.40	15.18	N/A	N/A	2.91

Note: DY is in percentage terms. Ticker symbols are for the following companies: Apple Inc. (AAPL), Microsoft Corporation (MSFT), Amazon (AMZN), Alphabet Inc. or Google (GOOGL), Tesla Inc. (TSLA), Facebook Inc. (FB), Berkshire Hathaway (BRK.B), Johnson & Johnson (JNJ), Walmart Inc. (WMT), and J.P. Morgan Chase & Co. (JPM). Some of these companies have multiple share listings that would have made the top 10 largest companies of the S&P 500 at the end of 2020; however, we excluded these so as to have a distinct set of different companies. PEG ratios were computed using one-year growth rates if positive. Data are for end of 2020.

The P/E ratios of the companies listed in Table 4.1 vary substantially. Tesla, one of the world's leading electric auto companies, carries a P/E ratio of approximately 1,203, whereas megabank J.P. Morgan has a mere 15.18 P/E ratio.[3] This quick cross-industry comparison does not tell us the whole story, though. A company's P/E ratio—and other financial metrics—needs to be viewed in the context of the company's industry. In 2020, J.P. Morgan's P/E ratio was roughly in line with a banking-industry multiple of about 17 times. Tesla's P/E ratio of 1,203, on the other hand, was very high compared with the automobile industry's average of about 317. Sometimes, though, a stock's P/E ratio may exceed the overall market P/E ratio but fall a bit below its industry's average P/E ratio. PayPal, the digital payment company, provides a prime example of undervaluation within its industry. Paypal's P/E ratio of 87 (as of December 2020) is above the current 78.70 P/E ratio of the S&P 500. Yet PayPal falls short of the average P/E ratio of 274 that the information technology sector commands as a whole.

Dividend yield (DY), P/B ratio, price-to-sales (P/S) ratio, price-to-cash-flow (P/CF) ratio, and price-to-EBITDA (P/EBITDA) ratio all attempt to capture the value of a stock. These variables are highly correlated with each other, so a manager can pick one according to his or her preference.[4] Some portfolio managers prefer to use P/B ratio rather than P/E ratio because P/E ratio cannot be calculated for firms with negative earnings.[5] Some portfolio managers also prefer the P/B ratio because of its prevalence in academic studies. Some quantitative portfolio managers instead use the P/CF ratio because they view cash-flow measures as less susceptible to manipulation by a company's management. All these ratios have been linked to the same trend, though. It has been documented that companies with high dividend yields and low P/B, P/S, P/CF, and P/EBITDA ratios tend to outperform stocks for which the reverse is true. There are many potential explanations for this trend. One explanation is that stocks with low values of these variables have experienced a serious price decline, which could lead many investors to shy away from them or demand a higher risk premium for holding them.

Stocks with high P/B ratios generally are referred to as *growth stocks*, and stocks with low P/B ratios are referred to as *value stocks*. In 2020, retail behemoth Walmart (WMT) had a P/B ratio of 5.42, whereas technology giant Apple had a P/B ratio of 34.53. Judging from the P/B ratios, Apple is more of a growth stock and Walmart more of a value play. As with the P/E ratio, though, we need to put the stocks in industry context to get the full story. In comparison with the 2020 retailing industry P/B ratio of about 13.21 or the subindustry of general merchandise stores' P/B ratio of 5.92, Walmart is undervalued. On the other hand, if a P/B ratio is too low, it could be evidence that management has failed to generate sufficient returns to its shareholders and that the stock price has slid in response. P/S and P/CF ratios reveal similar information and also may help to expose incidences of manipulation of earnings because sales and cash-flow numbers are generally more transparent than book values. While Walmart's P/B ratio was well below Apple's P/B ratio, its P/CF ratio, at 12.35, was closer to Apple's 27.96 P/CF ratio.

The price-to-earnings-to-growth (PEG) ratio is another valuation measure typically used to identify good growth companies. The idea behind the PEG ratio is that a high P/E ratio is justified if a company's earnings are expected to grow a lot in the next few years. From the P/E ratio alone, a stock might appear overvalued, but if the stock has a low PEG ratio, the high P/E ratio could be reasonable given the company's expected earnings growth. Consider data on Microsoft (MSFT), one of the world's leaders in software and technology, and the giant drug manufacturer Johnson & Johnson (JNJ). In 2020, Microsoft had a P/E ratio of 35, and Johnson & Johnson had a P/E ratio of about 24. From the P/E ratios, we might hasten to conclude that Johnson & Johnson was undervalued relative to Microsoft. However, as of December 2020, given the median analyst-forecasted earnings-per-share (EPS) growth rates of 14.45% and 4.31% for Microsoft and Johnson & Johnson, respectively, we can calculate the PEG ratio. Microsoft's PEG ratio is 2.45, and Johnson & Johnson's is 5.66. Owing to Microsoft's superior earnings growth outlook, Johnson & Johnson now appears to be slightly overvalued with respect to Microsoft. Microsoft's greater rate of forecasted earnings growth more than compensates for its higher price multiple. Although we used the stock analysts' consensus forecasts of earnings growth to compute the PEG ratio, some investors prefer to use historical earnings growth instead since they are unaffected by overly optimistic analyst forecasts.

The P/E ratio can be adjusted one step further by dividing it by the sum of the company's expected growth rate and its dividend yield. The resulting ratio, the price-to-earnings-to-growth-plus-yield (PEGY) ratio, indicates how attractive the stock's valuation is, given both its expected growth rate and its dividend yield.[6] For example, Apple and Walmart have PEG ratios that are roughly equivalent at 3.24 and 3.38, respectively. However, when we take into account Apple's dividend yield of 0.61% and Walmart's 2.77% yield, we end up with different PEGY ratios of 3.08 and 2.77, respectively. A relatively low PEGY ratio makes Walmart look more attractive than Apple on a pure valuation basis.

The dividend yield also can serve as an invaluable aid in determining a company's maturity and growth prospects. Stable, mature businesses tend to generate sufficient cash flow and offer relatively high dividend yields. Many of these companies find that they have limited opportunities to undertake highly lucrative growth projects, so they return discretionary cash to shareholders in the form of dividends rather than squandering it on low-return investments. Of the 10 companies in Table 4.1, J.P. Morgan Chase and Johnson & Johnson, the two oldest and most mature businesses on the list, have the highest dividend yields, 2.86% and 2.47% respectively. On the opposite end of the spectrum, Amazon, Google, Facebook, and Tesla do not even issue a dividend and instead reinvest all their earnings to fund future growth. Observations of the dividend yield will provide insights into a company's prospects for growth, whether organic or through mergers and acquisitions.

4.2.2 Size Factors

Table 4.2 shows the values of size factors for selected stocks. The market capitalization of a company, which is usually called its *size*, is another popular fundamental factor. It has been documented that small-cap stocks outperform large-cap stocks in the long run. Although this may not be the case during a single investment period, there are several reasons why it might occur over the long term. Some people believe small-cap companies pose greater risks than large, established companies and that they therefore merit an additional risk premium. The information story is that small-cap stocks receive less analyst coverage or less attention in general, so it takes longer for information about them to diffuse through the market. Ambiguity aversion might also explain why investors require a risk premium on small-cap companies, which tend not to be household names. Other size factors include the common equity of a firm and the total assets of a firm.

TABLE 4.2

Size Factors for the Top 10 S&P 500 Companies

Ticker	Common Equity (CE)	Market Capitalization (SIZE)	Total Assets (TA)
AAPL	65.34	2,255.97	323.89
MSFT	123.39	1,681.61	301.00
AMZN	82.77	1,634.17	282.18
GOOGL	212.92	1,104.81	299.24
TSLA	16.03	668.91	45.69
FB	117.73	656.67	146.44
BRK.B	415.15	543.61	829.95
JNJ	64.47	414.31	170.69
WMT	75.31	407.84	237.38
JPM	241.05	387.34	3,246.07

Note: Data are for end of 2020. For the company names, see the note to Table 4.1. Numbers are in billions.

4.2.3 Operating Efficiency Factors

In Table 4.3 we list the values of popular operating efficiency factors for selected stocks. Operating efficiency factors attempt to describe how efficiently a firm operates over both the short and the long term. An analysis of short-term efficiency would involve the inventory turnover (IT) ratio. The IT ratio is the cost of goods sold divided by average inventory, which represents the rate at which products move from inventory to sale. A high IT ratio generally is a good sign that a company's products are selling at a fast clip. Accounting standards and inventory conditions affect this ratio, so in making comparisons between companies, it is important to adjust the ratios for accounting discrepancies and to focus on companies in similar industries. For a test of a company's long-term efficiency, the total asset turnover (TAT) ratio, which is net sales divided by average total assets, attempts to measure how much revenue the company gets out of its assets. A very high ratio suggests that the company is making the most of its assets. Equity turnover (ET) and fixed-asset turnover (FAT) ratios show how well the company converts those particular classes of assets into sales revenue.

TABLE 4.3

Operating Efficiency Factors for the Top 10 S&P 500 Companies

Ticker	Equity Turnover (ET)	Fixed Asset Turnover (FAT)	Inventory Turnover (IT)	Total Asset Turnover (TAT)
AAPL	3.52	6.64	38.82	0.85
MSFT	1.28	2.85	12.84	0.49
AMZN	5.00	3.09	8.67	1.23
GOOGL	0.84	1.98	59.42	0.57
TSLA	2.37	1.35	5.07	0.62
FB	0.75	1.71	N/A	0.54
BRK.B	0.68	1.58	10.55	0.34
JNJ	1.32	4.51	2.18	0.47
WMT	6.85	4.36	9.32	2.28
JPM	0.50	5.14	4.09	0.04

Note: Data are for end of 2020. For the company names, see the note to Table 4.1.

Although a high IT ratio generally is a good thing, keep in mind that extremely rapid turnover increases the risk of failing to meet customer demand owing to a shortage of inventory on hand. Of course, this is often preferable to an extremely sluggish turnover, which means that the company either cannot sell its products or it ties up too much money by overordering supplies (which will hurt profits). The "normal" turnover for a particular line of business depends on marketplace competition and demands and industry practices and requirements. Taking a look at the consumer staples sector, of which Walmart is considered part, we find an average industry turnover rate of about 6.17 times. Walmart's IT ratio is about 9.32, so it seems to be selling its products well by the standards of its industry. The trend in turnover also matters as much as the most recent numbers. It is an encouraging sign that Walmart has maintained an average inventory turnover of approximately 8.10 times (also greater than the industry average) since 2016 and has shown an increase in turnover from 8.78 times a year ago. Walmart's turnover ratio is, by casual observation, strong for its industry and stable, which suggests that the company manages its supply chain well.[7]

The ET, FAT, and TAT ratios, like the IT ratio, vary substantially across industries. Table 4.4 lists the ET, FAT, and TAT ratios of S&P 500 companies in the semiconductor industry. We have calculated the average turnover ratios for the industry as a whole as well. Nvidia's ET ratio of 1.08 and its TAT ratio of 0.52 are lower than the industry levels of 1.50 and 0.60, respectively. These TAT ratios of roughly 1.0 or less are relatively normal for capital-intensive industries such as the semiconductor industry. We would expect higher levels of turnover for industries that maintain large levels of inventory and current assets, such as retail. The FAT ratio varies widely among the semiconductor companies as well, with Nvidia operating at 5.58 times turnover, Micron Technology at 0.72, and Applied Materials at nearly 10 times turnover.

TABLE 4.4

Semiconductor Industry in 2020

Ticker	Equity Turnover (ET)	Fixed-Asset Turnover (FAT)	Total Asset Turnover (TAT)
NVDA	1.08	5.58	0.52
INTC	1.05	1.38	0.54
AVGO	1.03	8.02	0.29
QCOM	3.96	5.99	0.61
TXN	1.59	4.00	0.75
AMD	2.86	11.78	1.23
MU	0.57	0.72	0.40
AMAT	1.84	9.87	0.77
LRCX	2.13	9.75	0.73
ADI	0.47	4.24	0.26
KLAC	2.18	9.55	0.64
MCHP	0.97	5.53	0.31
XLNX	1.15	7.18	0.53
SWKS	0.81	2.56	0.66
MXIM	1.30	3.71	0.62
TER	1.74	7.49	0.88
QRVO	0.81	2.71	0.45
Industry average	1.50	5.89	0.60

Note: Ticker symbols are for the following companies: Nvidia Corp. (NVDA), Intel Corp. (INTC), Broadcom Inc. (AVGO), Qualcomm Inc. (QCOM), Texas Instruments Inc. (TXN), Advanced Micro Devices Inc. (AMD), Micron Technology, Inc. (MU), Applied Materials Inc. (AMAT), Lam Research Corporation (LRCX), Analog Devices Inc. (ADI), KLA Corporation (KLAC), Microchip Technology Inc. (MCHP), and Xilinx Inc. (XLNX), Skyworks Solutions Inc. (SWKS), Maxim Integrated Products Inc. (MXIM), Teradyne Inc. (TER), and Qorvo, Inc. (QRVO).

4.2.4 Operating Profitability Factors

In Table 4.5 we list the values of popular operating profitability factors for selected stocks. One or more of these operating profitability factors (or transformations of these factors) appears in most QEPM models. After all, highly profitable companies are the ones to own. All the variables in the table are standard measures of profitability. Quantitative managers look not only for profits but also for some percentage growth in profits. They typically look at year-on-year growth in the gross profit margin (GPM), the operating profit margin (OPM), or the net profit margin (NPM).[8] Other common profitability factors include the return on total capital (ROTC) and the return on common equity (ROCE). Portfolio managers compare these ratios both within and across industries.

TABLE 4.5

Operating Profitability Factors for the Top 10 S&P 500 Companies

Ticker	Gross Profit Margin (GPM)	Net Profit Margin (NPM)	Operating Profit Margin (OPM)	Return on Assets (ROA)	Return on Common Equity (ROCE)	Return on Total Capital (ROTC)
AAPL	42.26	20.91	28.17	20.47	87.87	35.33
MSFT	76.75	32.29	46.64	18.65	38.49	27.18
AMZN	47.06	4.99	12.70	7.23	20.99	11.40
GOOGL	61.31	20.80	28.00	11.64	16.77	14.49
TSLA	29.89	1.97	14.87	4.28	3.47	6.10
FB	89.23	32.01	44.60	19.64	21.47	22.31
BRK.B	25.73	12.84	25.73	7.41	8.63	11.67
JNJ	74.65	21.01	33.80	11.85	26.35	19.78
WMT	26.71	3.30	6.65	10.51	23.76	16.55
JPM	73.54	19.17	43.87	1.53	9.93	5.95

Note: Data are for end of 2020. For the company names, see the note to Table 4.1. Numbers are in percent.

Let’s take a look at Microsoft’s profitability factors. Microsoft has a GPM of 76.75%, an OPM of 46.64%, and an NPM of 32.29%. It is always important to examine these three ratios along several dimensions: within an industry, in relation to the market as a whole, and over time. Each dimension will provide different insights. At the moment, we shall focus on the trend over time. Table 4.6 shows that Microsoft had a very high GPM of 77% in 2020 and that the margin has been increasing since 2016. The OPM simultaneously had risen from about 34% in 2016 to 47% in 2020. The 14% increase in OPM is larger than the 7% increase in GPM. Since operating profit is simply gross profit less sales, general, and administrative expenses, it appears that Microsoft has been able to control sales, general, and administrative expenses somewhat better than the cost of goods sold.

TABLE 4.6

Microsoft Trend Analysis